Panacea-BOCAF On-Line University

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Over View

The material published here to ensure dissemination and help spread the word to help Paul Pantone please consider this, always seek a personal license etc before using any of this. The GEET School will be stating some time in 2010, please contact Panacea for any interest.

The Panacea GEET Converted Lawn Mower

By our experiences, ***we know that a car can run with 5% of oil and 95% of water.*** – http://quanthomme.free.fr/ -English translated URL

GEET is an acronym for Global Environmental Energy Technology**.** GEET is a registered trademark of Global Environmental Energy Technologies. The GEET multi-Fuels Processor is patented technology (US005794601A1) by the inventor Paul Pantone. Research on this type of ***fuel reformer*** was started by Jean Chambrin and others around the world. Chambrin's is patented under patent numbers WO8204096 & WO8203249.

Source

There is a vast dissemination of replications using GEET variants in the country of France. More than 100 successful replications of this technology have been reported**.** The French started making GEET variants based on the free GEET plans which first surfaced in 1998. A French Farmer started mixing concepts of “water doping” with the GEET concept (Giller Pantone method).Water doping allows for a better combustion, a decrease in fuel consumption and decrease in pollution and is not new. As far back as 1901, a French Engineer, Mr. Clerget discovered it, it was used in 1942 by army air force, then in Formula 1 during the 80’s and still now in some competitions in Car Rally.

These GEET variants have also made it into commercial kits called the “SPAD/Retrokit/Nano by HYPNOW and recently to the Ecopra kit. More information below.

French Newspaper articles

DÉCOUVRIR ET EXPÉRIMENTER ...

... un système anti-pollution efficace : le réacteur Pantone

Monsieur Pantone, l'inventeur, est américain. En 1998, après une quinzaine d'années d'expérimentation, il trouve les dimensions adéquates qui miniaturisent une raffinerie et il adapte son système pour l'intégrer dans une automobile.
Son but est ambitieux : réduire drastiquement la pollution de nos véhicules.

Le réacteur Pantone est un réacteur endothermique. Son nom indique qu'il absorbe de la chaleur pour produire une réaction : la chaleur utilisée est celle perdue habituellement par le pot d'échappement et la réaction produite est un « cracking » c'est-à-dire une décomposition des molécules d'hydrocarbures en éléments plus petits, qui vont donc brûler plus facilement dans le moteur.

Les contrôles anti-pollution sont unanimes, le procédé fonctionne : dix fois moins de rejet de CO (monoxyde de carbone) et deux fois moins de CO_2 (dioxyde de carbone).

Il est possible d'adjoindre au carburant une proportion d'eau non négligeable avant de le faire passer dans ce réacteur. Certains expérimentateurs constatent que des injections d'eau via le procédé dans leur moteur diesel entraînent des baisses de consommation de carburant jusqu'à 50%.

Depuis que Monsieur Pantone a publié, gratuitement sur Internet, les plans de son réacteur, pas moins de 592 000 visiteurs francophones ont lu les pages dédiées à l'expérimentation libre de ce procédé.

Sur le site « quanthomme » les plans de base et comptes rendus d'expérimentations sont disponibles. Ils sont destinés aux amateurs qui veulent adapter le procédé sur une tondeuse à gazon. Les pages de M. David et de J.L. Naudin aideront les néophytes à s'initier.


Tondeuse équipée

Le site « econologie » de C. Martz met à disposition un forum de discussion qui regorge de conseils pratiques, d'informations et d'analyses sur le procédé. Ce site permet aux internautes de se rencontrer dans leur région afin de partager leurs expériences.

Une réalisation concrète permet de mieux comprendre l'intérêt particulier de cette invention :
Exemple d'un tracteur Massey Fergusson de 95 CV : sa consommation est passée de 21 litres à 16 litres de fioul/heure et 3,5 litres d'eau, le gain de puissance est estimé à 30 CV (une vitesse de labour de gagnée). Aucune fumée n'est visible dès que le réacteur a atteint la bonne température.
Un an plus tard les résultats continuent encore de s'améliorer : le tracteur ne consomme plus que 5 litres de fioul/heure et 10 litres d'eau, et il a toujours la même puissance au travail...


Tracteur équipé

Si ce procédé n'est pas ce que l'on pourrait appeler un moteur à eau, les avantages écologiques de cette invention sont indéniables et méritent d'être promus.

Pour plus d'informations, n'hésitez pas à visiter les liens Internet : www.quanthomme.org et www.econologie.com

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Les tracteurs passent a[u] banc d'essai à La Brillan[ne]

L'initiative est dûe à la Fédération Départementale des Cuma. Les agriculteurs ont notam[ment ...] un optimiseur compact de performances destiné à être utilisé sur les moteurs diesel e[...]

Avec un mélange hydrogène et eau

On n'a pas obtenu les résultats espérés

A suivre...

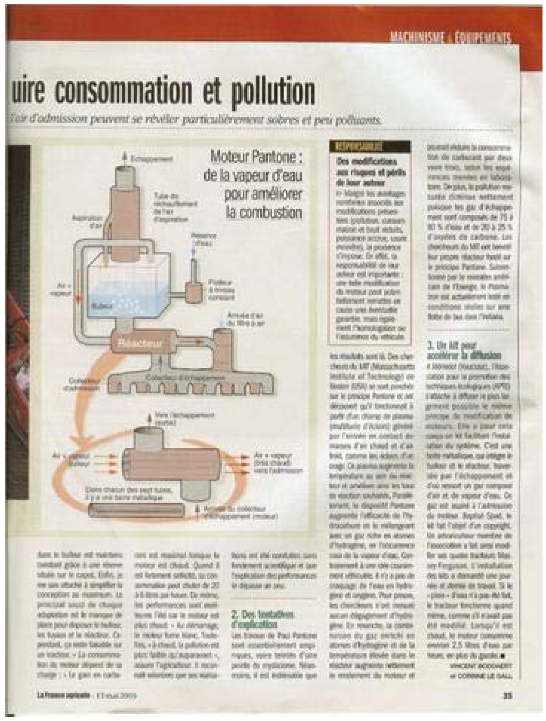

Les essais ont eu lieu en présence de nombreux agriculteurs et des techniciens de[...] mentale des Cuma de la région.

Un kit compact économiseur de [...]


Déception : l'optimiseur SPAD n'a pas donné pa[s ...] rés.

Comment ça marche ?

Cette explication ne correspond pas aux montages faits sur les tracteurs présentés qui sont des spad, eux mêmes issus du Système G, où il n'y a que de l'eau.



MACHINISME & ÉQUIPEMENTS

uire consommation et pollution

l'air d'admission peuvent se révéler particulièrement sobres et peu polluants.

Moteur Pantone : de la vapeur d'eau pour améliorer la combustion

La France agricole · 13 mai 2005

35

Ouest-France du 11 septembre 2005

Une voiture à l'eau et au colza

La vieille R25 de Jacques consomme très peu d'essence…

Pour faire rouler sa voiture, Jacques, agriculteur dans les Côtes-d'Armor, utilise de l'eau, un peu d'essence ou parfois de l'huile de colza. Et selon lui, sa vieille R25 avance bien : **« L'autre jour je suis parti de chez moi pour aller à Rennes. Environ 100 kilomètres. J'ai roulé à 110 km/h. Avec l'aller et retour, j'ai consommé 5 litres de carburant seulement et à peine un demi-litre d'eau. J'utilise un moteur Pantone, du nom de son inventeur. Avec deux ou trois tuyaux en inox, un bulleur d'eau dans le moteur et un pot d'échappement spécial, le tour est joué. Bon, il faut être un peu bricoleur. Mais ça marche et tout le monde le sait. Même les multinationales du pétrole. Mais chut… il ne faut pas le dire. Il faut d'abord gagner plein de dollars. »**

Hausse du prix de l'essence oblige, Jacques et sa voiture ont été les vedettes hier de la foire bio de Mûr-de-Bretagne dans les Côtes-d'Armor. Beaucoup de visiteurs étaient surpris de découvrir ce système : **« qui existe pourtant depuis 20 ans,** note Jacques. **Un système peu coûteux. Imaginez, on peut faire rouler une voiture avec du colza, c'est 0,32 € le litre, plus de l'eau. Autre avantage : le moteur Pantone ne pollue pas. »**

Une affaire en or ? Jacques tempère : **« Le moteur Pantone fonctionne sur des voitures qui ont plus de dix ans. Les véhicules récents ne peuvent pas accueillir ce système car elles sont bourrées d'électronique. Maintenant, il faut convaincre les compagnies pétrolières de développer le Pantone ou autre. Mais ce n'est pas gagné. »**

Jacques, agriculteur dans les Côtes d'Armor, fait rouler sa voiture avec de l'eau, un peu d'essence ou parfois de l'huile de colza.

Vincent JARNIGON.

[Link to French articles](#) Above you can see many French examples of the GEET technology being used in **tractors, generators, <u>cars,</u> lawnmowers, cranes and many others**! **The technology is even being used in a helicopter. More applications are possible**.

Reference

Tractor converted by tr24jdeere2 – GEET "SPAD" system is pictured in the center

The GEET converted Generator -By: oregonherbal

Jacques Diesel Renault

Ady' Stone's Very cool lawn mower ☺

Quote- *The French speaking people have engaged in a large battle to develop the use of the GEET of the Pantones, and with the help of J.L. Naudin, the website's managers of QUANTHOMME.com, Mr M***.** *DAVID's replication and tests plus along with Mr Martz engineering's study,* ***<u>there are now hundreds of vehicles that have been modified in France since 5 years, maybe thousands</u>****.*

Especially in the agricultural communities, these people are adapting their tractors and machines ever since one of them found a simplified version of the GEET concept, that offers enormous advantages, like ***reducing the fuel consumption by factor 2 to 5, and eliminating 95% of the exhaust fumes*** *(when testing the exhaust end with a white textile it stayed white!), and without a major modification on the vehicle*.-end quote- Reference**.**

You Tube French GEET news reports

Lorries, Boats, Big Generators have also been successfully converted to save fuel and cut piollution.

Reference

The GEET can also upgrade all existing geothermal house power systems, for example backup generators that are typically installed in solar systems, even solar self sustaining gen-sets as are shown here.

Solar self generating generator which still could be enhanced further

by conversion to GEET technology

In France the following city council adopted the GEET technology and achieved a 36% improvement in fuel economy and reduced their pollution by 80%!.

The French city council.

The following is a document which details the success and construction details and has was translated for Panacea by Jules Tresor

City council adaptation of the technology.

Unknown to all farmers and the mining industry is that there is related technology that was created exclusively from OPEN source collaboration on the GEET. This technology was inspired by the FREE GEET Plans. The first system is called the "SPAD" and is now upgraded to the "retro kit". Kits are available from a French company.

All these systems are robust and work on existing engines, have easy settings and are a reversible installation.

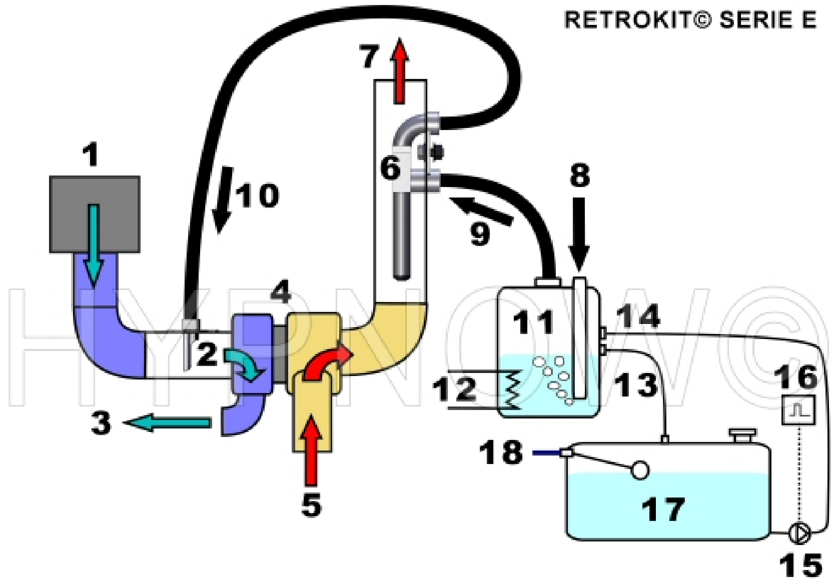
Fuel savings, pollution reduction Global Fuel Economy Initiative- **hypnow.fr**

Hypnow Retrokit Nano

View the Video here

Below is a tractor fitted with the SPAD system for saving fuel and reducing emissions. With the right tuning, savings from 30% to 60% are possible by using this technology. Emissions can be reduced by almost as much.

SPAD system fitted - Video



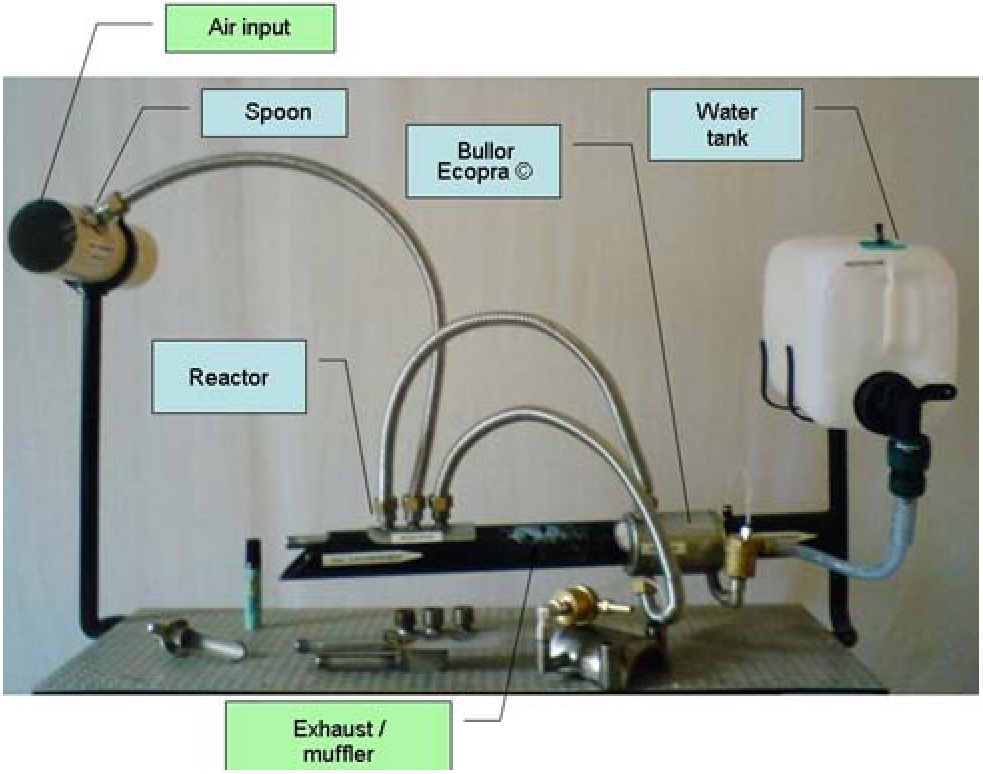
Retro kit system

Panacea's Nano Kit

Panacea Experimented with using this on a Diesel engine and mixed it with [Hydroxy booster technology].

Panacea's Nano/Hydroxy Install

More replication details have been included below. Another variant based on the GEET method has been done by a French company [ECOPRA](). This technology like the "retrokit" and "SPAD" has been given away as open source information. Kits are available.



The ECOPRA's fuel saving and pollution cutting kit.

The ECOPRA kit is a system inspired by the "Giller Pantone" GEET version and uses what is called "water doping". This process is entirely mechanical, without water your engine returns to a normal operation. This system improves combustion and reduces the unburned hydrocarbons. A reduction of 50% to 80% has been proven in gas analysis tests. The engine noise is reduced and the fuel economy is significantly improved.

Kit installed on a Mercedes 308 D -**Video**

Gains of up to 50% have been field tested on tractors. **These modifications are completely removable and reusable. If the water runs out, your engine simply returns to a normal operation**. This technology can fit vehicles from 50 to 250HP. This technology can also be fitted to the following brands of machinery.

CASE
#CATERPILLAR
CLASS
DEUTZ
FENDT
FIAT
FORD
IH
IMT
JOHN DEERE
KUBOTA
LAMBORGHINI
LANDINI
MASSEY FERGUSSON
MC CORMIK
NEW HOLLAND
RENAULT
SOMECA
VALMET
VALTRA

ZETOR

Here is a screen shot showing the Ecopra system being the first to achieve endorsement by an insurance company for installation.

Cabinet COUVE
Assurances & Placements
67 rue Jean Jaurès
B.P 154
42704 FIRMINY - Cedex
Tel : 04 77 40 53 53
Fax : 04 77 40 53 59
E-Mail : [illegible]

Orias: 07020591/07022121.

SARL ECOPRAC
RUE SAINT MARTIN
BP 13
43160 LA CHAISE DIEU

Objet : Kit Ecopra

Le jeudi 2 octobre 2008,

Madame, Monsieur,

Nous vous confirmons par la présente notre accord pour garantir par contrat AGF NOA, les véhicules automobiles équipés d'un Kit Ecopra (modèle 2008) sans restriction et sans surcoût.

Pour cela j'invite vos utilisateurs à nous faire une demande de garanties à travers notre site Internet www.agf.fr/couve en précisant que le véhicule est équipé d'un Kit Ecopra.

Nous vous prions de croire, Madame, Monsieur, en nos sentiments distingués et dévoués.

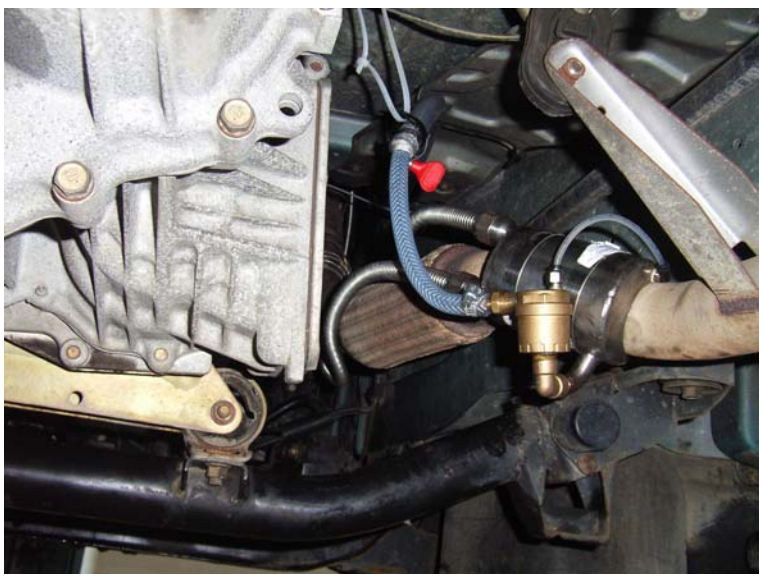
Philippe COUVE

Cabinet COUVE
67, rue Jean Jaurès - BP 154
42704 FIRMINY
Tél. 04 77 40 53 53

The Ecopra kit is approved by an insurance company

Panacea has installed this Ecopra kit onto a 1997 Mitsubishi Delica turbo diesel.

Panacea Ecopra kit



Panacea's Ecopra kit fitted.

Further replication details have been included below. **Governments must be presented with this technology in order to push for city council, businesses', cars, machinery etc to all meet the emission standards possible from the implementation of the GEET technology. Also this must be done TO SAVE THE TAX PAYER MONEY ON FUEL. Currently due to the GEET being unknown by insurance companies and manufactures, the fitting of these modifications MAY void the manufactures warranty**.

The GEET modifications will make the engine last longer and can reduce the emissions by up to 90%. This is why we need to create public pressure for governments to create

endorsements, subsidies and carbon credits for this technology to help the retrofitting and cut the carbon foot print.

Despite the GEET being proven technology and available for many years, other western countries do not have this fuel reforming emission cutting technology in place and faculties are still unaware of its power management process. This is due to the inventor Paul Pantone and related GEET groups encountering "interference" and or suppression - Reference.

In 2002, Pantone was found guilty of securities fraud for selling shares under false pretenses, cheating investors out of as much as $25K each. **HOWEVER, there are many inconsistencies going on with his case to suggest there is something other than fair justice prevailing**.

Paul Pantone has been held against his will and it is reasoned that his human rights were being violated due to neglect of needed medical treatment. Reports given to the nonprofit organization state that Paul Pantone has been wrongfully imprisoned and tortured for three years in the Utah State Hospital. He is a victim of civil rights violations legal criminality and medical mal-practice and sabotage. He tells about his experience trying to gain his freedom from the Utah Justice System.

Current Event: Inventor Wrongfully Imprisoned for over 3 Years Part 1

Current Event: Inventor Wrongfully Imprisoned for over 3 Years Part 2

Current Event: Inventor Wrongfully Imprisoned for over 3 Years Part 3

Paul Pantone Interview, late January 2009: No help for USH patients

Wrongful Incarceration- Paul was incarcerated on dubious charges in January of 2006, and kept incarcerated far longer than he was supposed to be. Thanks to donations from supporters there is now enough money to hire an attorney for Paul, and as of MAY 2009 he has been finally let out.

However that is only the first step. Paul is also need of financial assistance to get back on his feet. Hasn't he suffered enough? Paul still needs our help with his recovery. He needs some blood tests, x-rays, braces, and healing herbs and other natural remedies that he has located but lacks the funds to get.

If you can make a donation, please go to http://www.geet.nl and click the PayPal 'Support Paul' donate button. It will help Paul recover quickly so he can focus his attention to developing technologies which benefit all mankind and the Earth.

Despite all this, it is not all Paul's fault his company failed and what he says about how to make a simple fuel reformer is something people should pay attention to. Despite his legal problems, the technology works and many similar patents have been in the hands of auto and oil companies since the 1970s. Some of the 1st catalytic converters made were for intake gases but the auto makers would not use them. By doing simple search for "Steam Reforming" and/or "fuel reformer" one can see that the technology works well and gives a significant increase in fuel economy long before Paul made a similar device. Many others also have made fuel reformers and they work well. The reforming can be done with plasma, a catalyst or both and releases hydrogen from both the fuel and the water steam.

People must not focus on Paul's plight and ignore the facts that the reforming of fuel with water plus vapors works great to make hydrogen rich gases which when added to an engine intake give much better fuel economy.

The GEET technology is understood to be a self-inducing plasma generator or a plasma reactor with an endothermic reaction –reference. The GEET system uses a special "reactor" vessel heated by a modified Diesel or Internal Combustion Engine's exhaust (and electrically and magnetically "boosted" to create a plasma effect) that allows virtually any organic waste to be broken down and converted in usable fuel.

This is then mixed with small-percentage amounts of petroleum-based oils of any type (including waste motor oil) to efficiently fuel Diesel or internal combustion engines. The result is very high "mileage" and very low emissions from the engine (because the special Reactor captures and reuses unspent fuel; with amazingly efficient results). All by using existing-design engines without too much modification needed. Fun anecdotal stories have been reported about using banana peels, orange rinds, catsup and mustard. Nearly anything organic to fuel an engine abound when talking about GEET.

This technology has taken some criticism over the last few years, and the inventor was even once imprisoned for problems associated with his investors. Yet there is clear proof that the GEET concept works and would be an excellent alternative to fossil fuels alone; especially for the agricultural sector which has large amounts of organic waste cheaply available.

Greatly helped through the efforts of Panacea-BOCAF, there have been over 100 successful reproductions of GEET ; many of them done in France which seems to have embraced the technology more than other countries."

A U.S. patent was issued to Pantone for a "Fuel pre- treated apparatus and method" on 18 August 1998. **Independent reports by replicators confirm that the GEET can triple fuel efficiency and cut pollution by up to 90% by simply transferring exhaust heat to the fuel intake**. Pantone explains that the instantaneous pressure fluctuations in the exhaust help to create a vacuum that, when combined with the heat, creates micro-magnetic forces. This produces plasma that dissociates the hydrogen from the oxygen in the carburetor.

Initially the GEET is applied as a dynamic fuel-exhaust recycling device that can be fitted to an engine, between the air intake and the exhaust. **Many variants and improvements have since been patented by engineers in France**. Currently the open source Vortex Heat Exchanger group located at: http://tech.groups.yahoo.com/group/VortexHeatExchanger/ is working on a Hambrin/Pantone/Martz reproduction version.

The GEET became popular when Paul and his wife released a version as free internet plans. These plans we only meant to give an idea of the process, and not a demonstration of the efficiency one can reach with the GEET. Also these plans are only suitable for small engines, and not the auto version. Here is a video of Paul Pantone explaining the version of the free internet plans.

Paul's Comments about the FREE GEET PLANS

Due to what most consider the unlawful incarceration of Paul Pantone, currently the only access one has to the official version of the plans can be found through the Tesla Tech web site. What one must keep in mind is that in these plans it is stated that the size of the reactor rod used in the GEET is specific to the alternative fuel used, (for example water or waste oil) the size of the engine and what application you're running. They ask you to call the GEET center to find out this information; and that this is stated as being done to protect them from theft. **One cannot ring the GEET institute as Paul is not there**. Regardless there seems to be enough clues in there to get started and there are more listed in this document.

The Full GEET (Deluxe) plans as listed on the Tesla Tech site

Quote- *The Deluxe GEET Plans will show you how to make GEET Gas to run your entire home... heating, refrigeration, stoves, furnace, and all of your power needs. They will* *also allow you to retrofit up to TEN cars, and provide you with one year of free updates. GEET technology is the independent homeowners dream come true. Through the use of GEET technology, a lot of waste materials become valuable fuel... even the lowly septic tank becomes a fuel source! This means significant savings over conventional technology.*

For gas engines using gasoline as the primary fuel, you will be able to use "junk" fuels in conjunction to stretch out fuel supplies. However, heat is one of the main elements in the GEET reaction process and until the reactor is warmed up, the engine is running on whatever fuel you feed it. You must start the engine on something that it will run on, like gasoline, propane, or even GEET Gas from a previous run (storage tank and pump setup). Once warmed up, you can switch over to the junk (alternate) fuel.

Small engines can be retrofit for as little as $20.00 to $30.00. Automobiles can be retrofit for as little as $75.00. Depending on your tools, skill level, and how much you need to farm out, small engines average $50.00 to $75.00, and automobiles average $200.00 to $1,000.00. Take the first step to convert your homestead to GEET-End.

The only Tesla Tech large engine plans we have seen are from 1998. There are to date 2008 no newer ones. The basic difference with them and the free plans is:

The large engine plans show using a larger reactor but not a lot larger. Paul Pantone used to give phone support saying what size tubes worked best for different size engines. I got the impression it was important to get the size correct but the whole reactor is still never really big even on really big engines. They also show using a modified carburetor to get a rich fuel vapors mix through the reactor, else using fuel injection. I think a bubbler was not recommended for autos.

Then a large part of the large engine plans was about the Air Management Valve. Some reports we have read state it didn't really work well and people who got it working well had come up with other designs to get the correct mix of reactor gases and fresh air while also maintaining a good vacuum to the reactor. Since Paul Pantone can no longer provide phone or online support that used to come with those plans we don't think these plans are worth the money at this time.

Please note that the GEET is not just a vaporizer, a properly working GEET is a "fuel reformer".

This Video shows the so called GeetGas that is left over in the GEET Reactor after the engine was stopped. We do not know if anyone has analyzed this gas to see what it consists of, using a gas spectrometer, gas chromatograph or some such sophisticated equipment.

http://www.youtube.com/watch?v=zD6UE6pr-Rg

In France they have many variants of the GEET systems, including the SPAD system (which is included below however is more of a vaporizer system). The SPAD system as now evolved to the “retrofit” system which uses a reactor (more info below). Many people have demonstrated how to do it wrong and they document their vaporizers (not fuel reformers) on youtube. Build it correctly and it works as a fuel reformer. Some evidence it is working correctly:

Exhaust is almost as cool as the air going in, indicating there is an endothermic chemical reaction taking place inside that absorbs heat from the exhaust. There is higher oxygen content in the output gases as measured by people who test their units at an exhaust inspection station.

The input no longer requires much fresh air added with the reactor output gases and in fact a normal amount of fresh air will stop it from working.

There are many fuel reformer designs. This can be seen by doing a Google search of "fuel reformer" or "steam reforming" or "Plasma fuel reformer" or "catalytic fuel reformer". It is understood that both the fuel and the water get cracked.

Auto makers, oil companies and universities together own hundreds of fuel reformer patents specifically for use on board a vehicle for providing "Hydrogen Enhanced Combustion". But for decades auto makers refuse to use them. In 1974 NASA engineers reported in a paper to the SAE, Society of Automotive Engineers, the benefits of using an on board fuel reformer. In 1975 an in car demonstration study was done and reported to SAE proving the benefits of hydrogen enhanced combustion. This can be found by a Google search:

SAE reformer 2007-24-0078

SAE reformer 1972 "boston car"

SAE reformer 1999-01-2927

SAE reformer 981920

There was a surge in the number of patents filed for various types around that time frame but we never actually see them in cars. Many of the patents include detailed records of the testing in real world conditions proving the efficiency of the on board fuel reformer.

Years ago MIT spent millions proving that on board fuel reformers would give us all better fuel economy and cleaner air. They did long term in bus in and in car proofs. They teamed with the very large auto parts supplier Arvin Mentor to do the final step of putting them in production cars and trucks. It was then reported that allegedly "One Equity Partners" bought out Arvin Mentor's division that did the entire final work to get fuel reformers in all our vehicles. OEP created a new company and that company dropped the fuel reformer from their product line, not because it did not work but because it did work. Reference.

*The microplasmatron fuel converter (plasmatron, winner of the 1999 Discover Award for Technological Innovation) is a device that would be used on a vehicle to transform gasoline or other hydrocarbons into hydrogen rich gas. The plasmatron uses an electrically conducting gas (a plasma) to accelerate reactions that generate hydrogen rich gas. The hydrogen-gas, which is a high quality fuel, is then used as a fuel in the engine resulting in greatly reducing pollution-*Source

MIT Plasmatron-Source

The MIT style plasma fuel reformer or a catalytic fuel reformer can be connected to a car and get MUCH better fuel economy. The basic difference is Paul Pantone made reformers with self generating plasma and MIT's is powered plasma. Both types perform fuel reforming and release hydrogen. The engine burns much cleaner and much more efficiently.

Reports given to us state that The next day after Paul went to jail allegedly MIT and Drexel and many other company's fuel reformers were still working, still capable of giving us all much better fuel economy. Paul's works also when built correctly.

Note also The PICC Pre Ignition Catalytic Converter is a GEET reactor. Dennis Lee developed the PICC product further from the original GEET product. Reports given to us state that he was supposed to pay a license fee to Paul Pantone's GEET (Global Environmental Energy Technology) company and was supposed to put the GEET logo on the product but Dennis Lee says that he developed the product so much more that it is his design now and he refused to pay any fees to GEET and Paul Pantone took legal action against Dennis Lee.

For 100 years people have been doing fuel reforming using steam and fuel vapors through a catalyst or plasma. It's called steam reforming or fuel reforming. It works just fine using the engine's waste heat along with a catalyst or a plasma for "cracking" fuel and water molecules which supplies lighter molecules to the engine and gives better fuel economy and cleaner air.

The extra oxygen is released from the water in a steam reforming reaction. Normally in most fuel reformers, the oxygen combines with carbon but Paul's was able to output the extra oxygen. Google search: steam reforming or fuel reforming. Between 1983 and now, big oil has taken whatever steps necessary to insure this technology does not get installed in cars. Automakers and oil companies own many fuel reformer patents for designs specifically for use in cars, but they won't use them.

Google search:

Steam reforming

Catalytic fuel reformer - patent GB129963 (1920 catalytic fuel reformer)

Plasma fuel reformer - Patent US4066043 (Nippon 1978 spark (plasma) reformer)

Hydrogen generator "waste heat" 1974 waste heat recovery hydrogen

1974 "On-board hydrogen generator for a partial hydrogen injection internal combustion engine"

1975 "Feasibility Demonstration of a Road Vehicle Fueled with Hydrogen Enriched Gasoline."

Hydrogen, catalytic heat exchange apparatus

Hydrogen, endothermic catalytic heat exchange apparatus waste heat, hydrogen, endothermic catalytic heat exchange apparatus – Patent Numbers - US5794601 and US4214867

Gliding arc fuel reformer

Gasoline fuel reforming

MIT plasma fuel reformer

Despite the GEET using a transmutation and plasma process (being a relatively new field of science), the GEET fuel processor is a combination of very basic scientific principles which fall within most of the normal rules and laws of thermodynamics. But **some of the 70 simultaneous phenomenon's are not found in those books**, since it is the *combination of events*, which is the body of this discovery- Reference.

The Answers to global warming are contained in the GEET technology, in the following French new report, a French resident applied the GEET technology to his car; the French news report conducted a dyno and emission test on the car. Their findings showed that **the CO2 levels with the GEET technology** were as low as 0.1%**, without the GEET it registered 8.6%.** Other Nox and hydrocarbon reductions were as significant! Plus the car gained up to 20% better fuel economy (more is possible).

The Emissions test report which was done by French news

THERE IS NO OTHER MODIFICATION IN THE WORLD WHICH CAN BE APPLIED TO EXISITNG TECHNOLOGY THAT CAN CUT CO2 AS WELL AS THE APPLICATION OF THE GEET TECHNOLOGY.

A scene from the news report - Here a car is fitted with the GEET system and undergoing emission testing.

The above emission test illustrates what our world would be like with the GEET or any modification based on the GEET. We would be closer to eradicating cancers from hydrocarbons and stopping CO2 emissions. HOWEVER the GEET is unknown to the majority of the general public. This method is not taught at any mainstream faculty, So far Panacea's engineers are the only ones intent on training others as a public service.

French engineer filling his reaction chamber with water for the GEET fuel processor

A Transcription from the above French news report:

3:52-Journalist: This gesture that looks usual is simply revolutionary!

3:57-Kevin: Here I fill my tank with water, to run my car with it, to save some fuel.

4:06-Journalist: You run with water ?

4:07-Kevin: Yes, I run my car with water and diesel.

4:09-J: With the enthusiasm and energy of his 24 years old, Kevin tinker on his old Renault diesel, a system that injects water vapor with the fuel.

4:20-J: The processes is simple, it's composed of a tank, called bubbler and a reactor.

4:24-J: A tube with a rod inside that we introduce in the exhaust pipe.

4:30-K: The vapor will be produced in the bubbler.

4:33-K: The vapor will come and slip between the rod and the tube that is heated by the exhaust gases, and at the exit of this reactor we will send this gaseous mix coming from the water vapor, in the air intake.

4:45-J: It's the Pantone motor, named after its inventor, an American Engineer. **Since 1998, this schematics are available on internet and in the ecologist movement.** The genius of Kevin, self taught in mechanic, is to have replicated it, and it works.

5:04-K: So the system is runing, the engine runs better, we feel like the engine is better tuned. We need to push less on the gas pedal to climb a big hill, especially for the old vehicles.

5:15-J: We measured in a technical centre, the gain obtained in term of pollution. It's spectacular !

5:22-J: For the CO2 emissions, we go from 8,6 to 0,1 with the Pantone system. The results are very satisfying as well for the Nitrogen Oxydes and the Unburned Hydrocarbons.

5:32-J: Demonstration with the smoke opacity test. On the right side without transformation of the engine, on the left with the Pantone system.

5:41-J: As for the fuel saving, it's about 20%, and that's just tinkering.

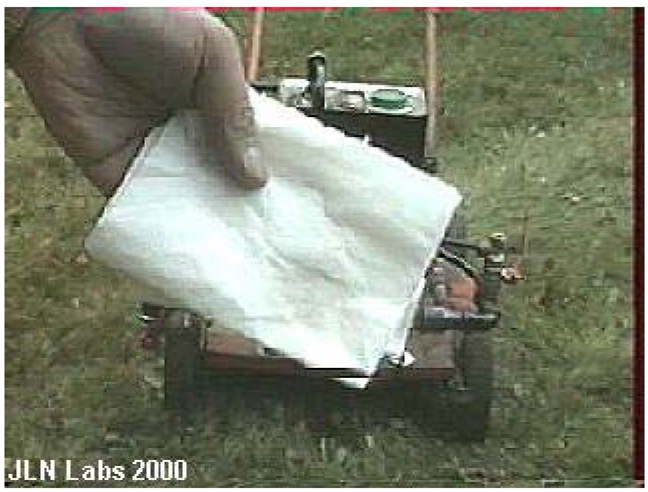
5:47-J: **Then we can ask ourselves why the car manufacturers are still not interested.**

5:53-Screen: The efficiency of the Pantone reactor under the day light. -END

Even if a low cost electric car ever really reached the market place, it would still take the existing infrastructure an average of 5 years to change over all their technology to the new ones. The proposed fuel standards in as far as 2012 can be reached TODAY. The GEET is the lowest cost modification in the world which can be implemented to existing vehicles to cut their carbon foot print. Please **READ that again.**

Another French experimenter J-L Naudin retrofitted a lawn mower with the GEET Multi-Fuels Processor. Not only was he able to run it on 75% water and 25% gasoline he was also able to practically eliminate the pollution.

The JLN' s lawnmower runs with 75% of water and 25% gasoline



<u>Links to French videos validating the technology</u>

http://www.youtube.com/watch?v=El0tIGcwpcM

http://www.youtube.com/watch?v=gT_69ra2PB0

http://www.youtube.com/watch?v=HJQc7Et7xQc

http://www.youtube.com/watch?v=f-lNWi3JuVA

http://www.youtube.com/watch?v=gv0AjiPj34E

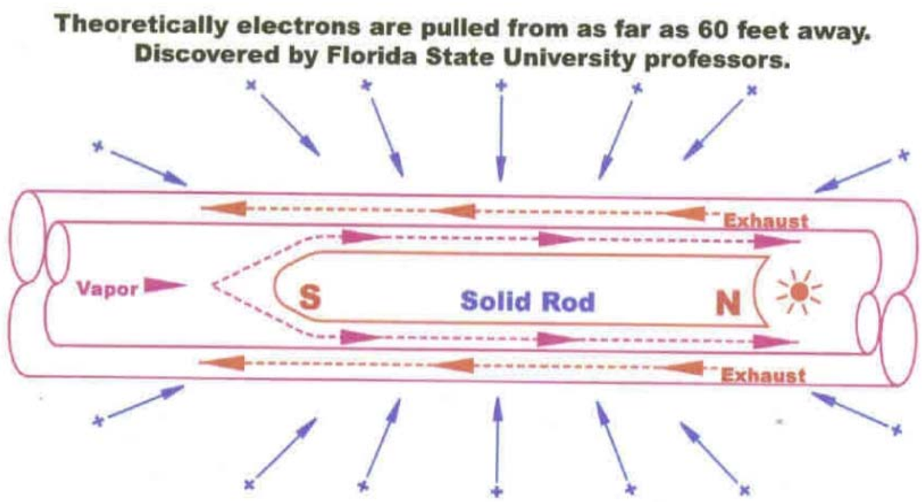
http://au.youtube.com/user/worldwideeagles

User- http://www.youtube.com/user/worldwideeagles

At this stage the city council and other businesses etc are advised that all cars, machinery etc meet or exceed the emission Standards. Also with the manufactures warranty, the fitting of any modifications is likely to void factory warranty. These modifications will make the engine last longer and can reduce the emissions by 90%. This is why we need to create public pressure for governments to create endorsements, subsidies and carbon credits for the technology.

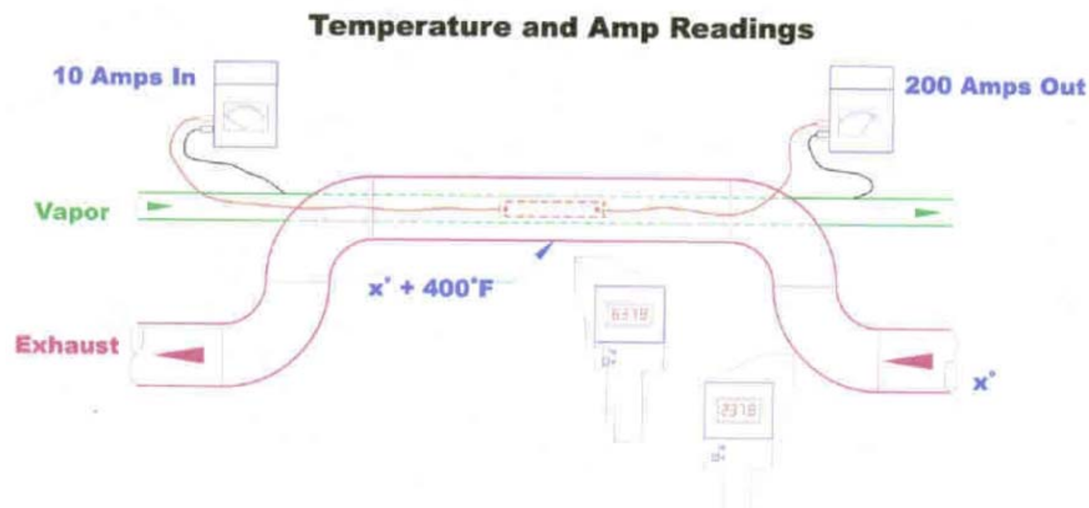
Given the efficiency reports by the GEET replicators, this technology is an invaluable power management process which the mainstream faculties must benefit from. **As an emission cutting device and power savings device alone, the GEET technology justifies (and needs) law for its mandatory implementation**.

The GEET more than deserves further research and development into its operation, specifically towards its ability to transmute and theoretically rip EXTRA electrons out of the air.



This transmutation may be the reason why more energy was reported to come out then was put in by the user, as this would create an open system effect where the device has a COP (co efficiency of performance) of more than one. Open systems (like a solar panel or windmill or heat pump) are able to extract additional energy out of the environment and add it to the users input to equal the total output. The reactor can make such a strong electrostatic field that it attracts electrical charges from a good distance and when it works very well it is making a strong magnetic field also. Empirical reports given to us by open source engineers state that the GEET reactor definitely creates a good current flow when it is running. Engineers have measured current with clamp around current meters than can read both AC or DC current.

In fact if you cannot measure any current or see a change in magnetic fields while it is running then it is an indication that the GEET fuel reformer is not working.



Also engineers are begging to experiment with using the GEET technology with Hydroxy technology. Specifically the following open source engineer has had some very encouraging results.

Web site

You tube channel

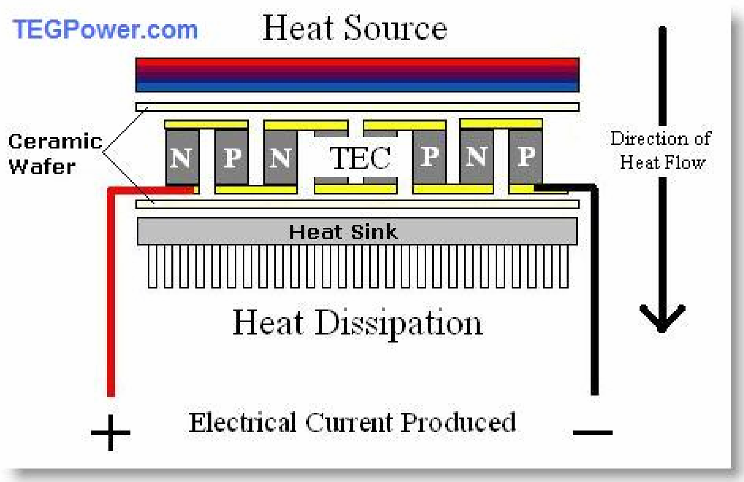
Further, interesting "closed loop" configurations have already been done showing a lot of merit and promise in the technology.

Lawn mower modified for GEET operation

Web site - http://www.thejoeball.com/

Video- http://www.youtube.com/watch?v=MKY5aH0Fp7E

Recently a thermo electric effect has been identified by open source engineers. This effect could also extract extra power from the GEET's normal operation as the heat exchange is energy produced waiting to be harnessed. A thermo electric effect can be described by the following:



Source

This can be placed on the exhaust hose or the reaction rod or both for extra energy extraction.

The Nonprofit organization Panacea-BOCAF intends to support open source engineers working with the GEET and other suppressed clean energy technologies. These engineers require grants, resources, faculty recognition and security. All this can be created in Panacea's proposed granted research and development center. For those able to help this effort, please Contact us. **Also Paul pantone is in need of help, anybody who can help please contact us immediately. Please consult** this web site **to see how you can help Paul.**

Panacea has completed a video production containing background information on the case of Paul Pantone and what applications are possible**.**

Panacea-BOCAF Re loaded GEET production

Description

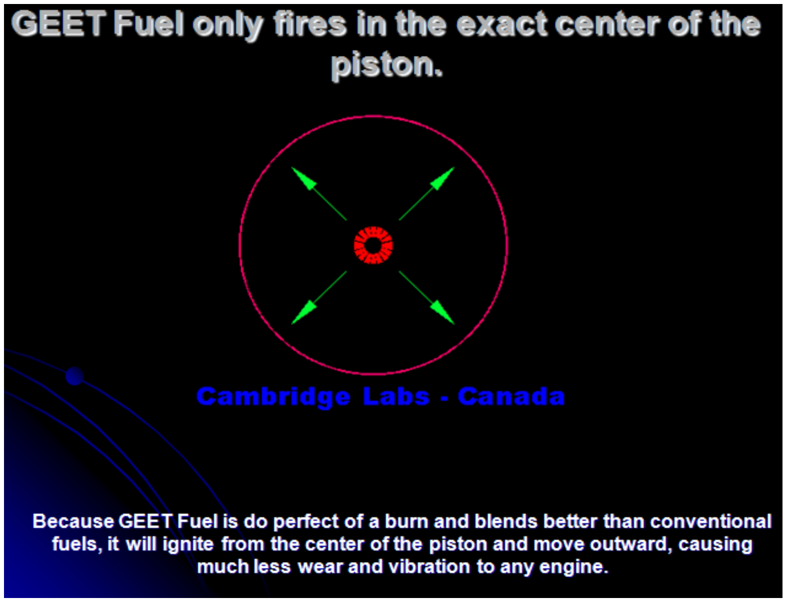
Inventor Paul Pantone pictured next to the GEET.

Energy losses occur in an internal combustion engine (ICE) due to the incomplete conversion of combustion energy (chemical energy) into mechanical energy. The overall engine efficiency is about 25% for a diesel cycle, and even lower for a gasoline engine. The ICE conversion from the air / fuel mixture of chemical energy into useful mechanical energy **wastes around 75% of potential energy**. This wasted energy results in **harmful emissions** and is expelled as a mixture of gases and **heat losses** evacuated through the engines exhaust.

Paul Pantone's GEET multi fuel processor recovers these heat losses into a form that can be directly transformed into mechanical energy by the same engine. The GEET recovers the lost heat to transform the air and water vapor/fuel mix incoming into the engine into a combustible usable mixture. **This device allows a significant reduction of pollution of almost 85% generated by the gas exhaust compared to a conventional engine-** Reference**.**

With a GEET, whatever "fuel" you put in at one end comes out the other as a hydrogen rich vapor(when working correctly) and is not troubled as much with flame front speed and timing issues as "normal" fuel's. The problem with "normal" fuel is it can't burn fast enough to be converted to mechanical energy at higher engine speeds; hence, it has to be ignited earlier in the cycle as the engine speed increases. To enable the use of poor quality fuels, such as gasoline, the ignition timing need's to advance automatically with engine speed.

Hence the GEET is practical as a vacuum advance ignition system! It is also predicted that Diesel engines running on the reformed GEET Gas produced by the reactor rod could fire at exactly top at dead center, but more testing is needed to find out.



The invention could be called a new type of carburetor with a miniature refinery built in. This invention can be fitted to 2 or 4 stroke engines, cars, scooter and even diesel generators and cut their emission to virtually nothing- Reference**.** This GEET allows for the use of any type of hydrocarbon fuels like crude oils, methanol, gasoline, plus various solvents, kerosene, bio-diesel to be **mixed with water for a usable fuel-**Reference. The GEET is a dynamic fuel-exhaust recycling device that can be fitted to an engine, between the air intake and the exhaust.

Also at 1:32 minutes into the Paul Pantone Plasma Reactor Motor youtube video Paul Pantone says the reactor gives off slight radiation when running that is not Alpha, Beta or Gamma radiation.

Devices Fitted Successfully:

http://64.233.179.104/translate_c?hl=...

http://translate.google.com/translate...

http://translate.google.com/translate...

Description of Device Operation: Plasma Fuel Reforming with PMC (Processing Multi-Carbons). The bubbler is a tank containing a mixture of water and hydrocarbons (gasoline, diesel, kerosene, crude oils and others derived from hydrocarbons...).

Plasma technically is made of gases that are ionized sufficiently to be electrically conductive. But they can be electrically conductive even though they are not energized or excited enough to emit visible light. On the other hand, Paul Pantone said that in his glass reactor(see in the faculty section below) , he saw sparks along the rod and a small glow area behind the end of the rod.

The hot gas flow coming from the exhaust of the engine circulates by the outside part of the reactor with a strong kinetic energy, that contributes to bring up to very high temperature the steel rod (being used as heat accumulator) contained in the pyrolytic chamber. The gases cross the engine and penetrate then in the bubbler containing the water/hydrocarbon mixture. The vapor of the mixture is strongly aspired by the vacuum created by the engine intake and is pushed by the pressure coming from the exhaust. The kinetic energy of the vapor is increased considerably by the reduction of the diameter in the pyrolytic chamber (by Venturi effect). The combined effect of the high temperature and the increase of the kinetic energy produce a thermo chemical decomposition (molecular break down) of the water/hydrocarbon mixture.

The endothermic reactor forms an Electro-Plasma-Chemical unit (EPC) and it is now possible to create a high-output fuel coming from the decomposition of the water contained in the water/hydrocarbon mixture. This fact is confirmed by the presence of oxygen gas (O_2) in great amount measured in the exhaust.

Replication

Note please read through the "Naresh Group files" listed below courtesy of the vortex heat exchanger group. Naresh has provided a complete reference including beginner's instructions to research material and more. This is a 170mb file which has every ting need to get started and to understand the complete fuel reforming process. PLEAE READ THROUGH THESE FILES BEFORE STARTING A GEET. The file is zipped and contains an off line viewer for files.

Note – if using gasoline (petrol) you may find that portions of the gasoline boil away leaving the heavier less volatile portions behind. To solve this problem you may need a metal bubbler inside another metal container with exhaust gases going through it to heat the bubbler. That is how Paul Pantone's "Old Blue" is designed. Search You-tube videos and you will see he has an aluminum container inside an aluminum container. –It is recommended you use waste or heavier oils.

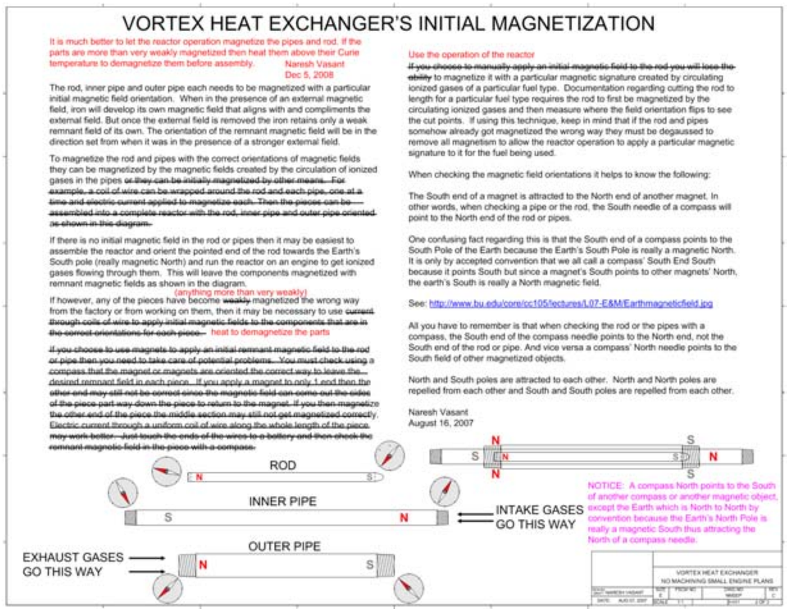
Leo Umila Supplemental GEET system

You can also use the GEET as a supplemental fuel system, the following comes courtesy of Leo Umila who retrofitted a double GEET chamber and used the Gases as a "booster". He reports 3 X the fuel economy.

Leo Umila supplemental GEET

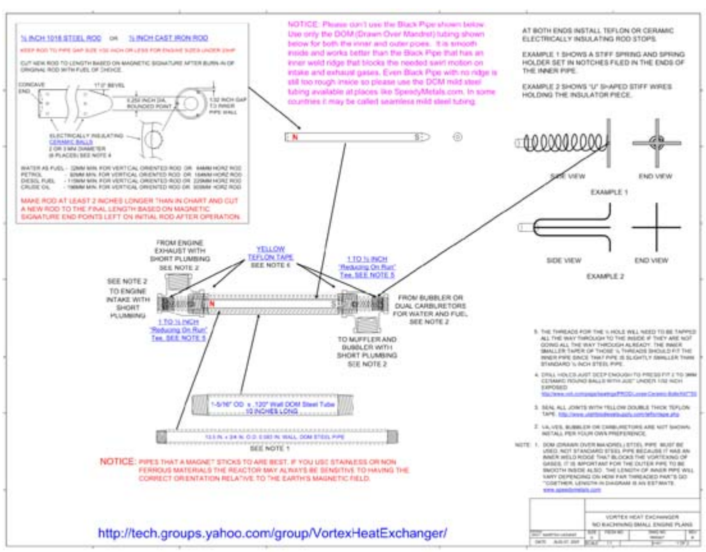
The following files come from the vortex heat exchanger group who are so far the most advanced and active in the GEET technology. The following example by john has very good overview covering the similar concepts employed by the group.

GEET Reactor magnetic's



[File -GEET Reactor Magnetics]

Vortex Heat Exchanger Small engine plans revision 2



[Down load - Vortex heat exchanger Small engine plans revision 2]

Others

Background - I (Naresh) have been trying to determine what sizes are best based on information gathered from others who have built various size GEET reactors. Also, David Pantone had previously mentioned that the reactor should be smaller than the "Free Internet Plans" version when the engine is smaller than 20hp. Joe Ball had also built a vertical reactor with only a 1/8 or 1/4 inch NPT inner pipe and it worked well. Note that a 1/8 NPT pipe is larger than 1/8th of an inch. "Pipe" sizes are not actual sizes but "Tube" sizes are. See his video: http://www.youtube.com/watch?v=MKY5aH0Fp7E

Based on that, I started making diagrams for smaller reactors for smaller engines.

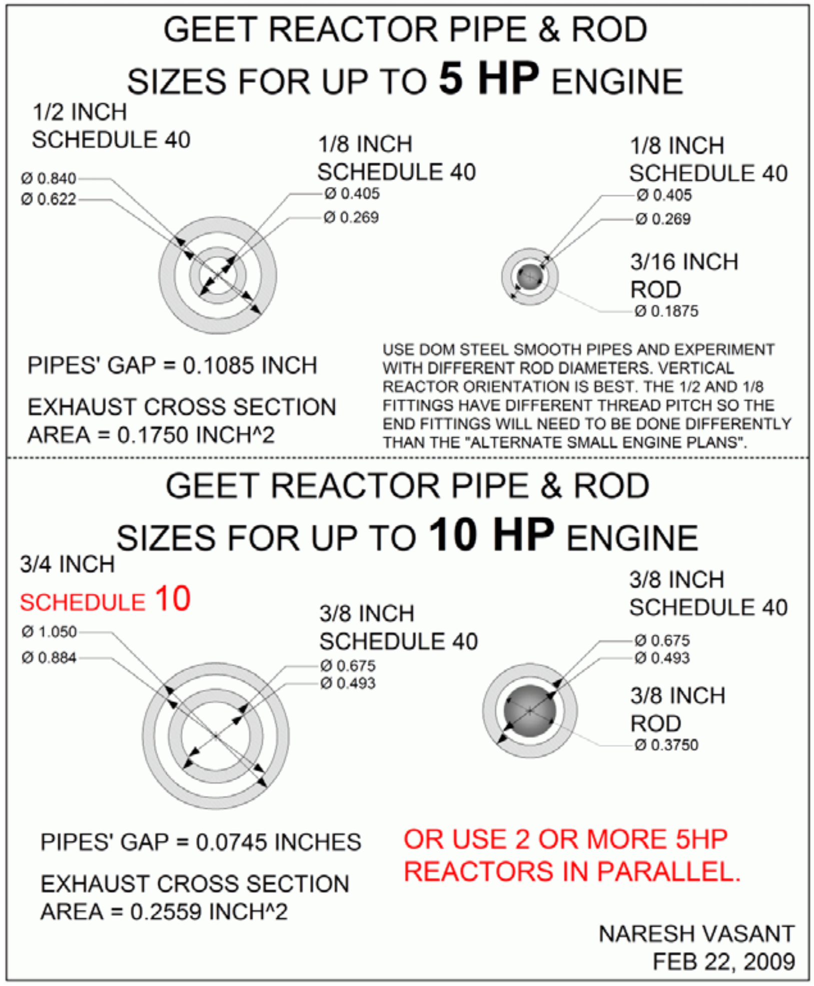
0.1 To 5hp GEET Reactor Plans

10 to 15hp REACTOR PIPE SIZES

20 to 25hp GEET Reactor Plans

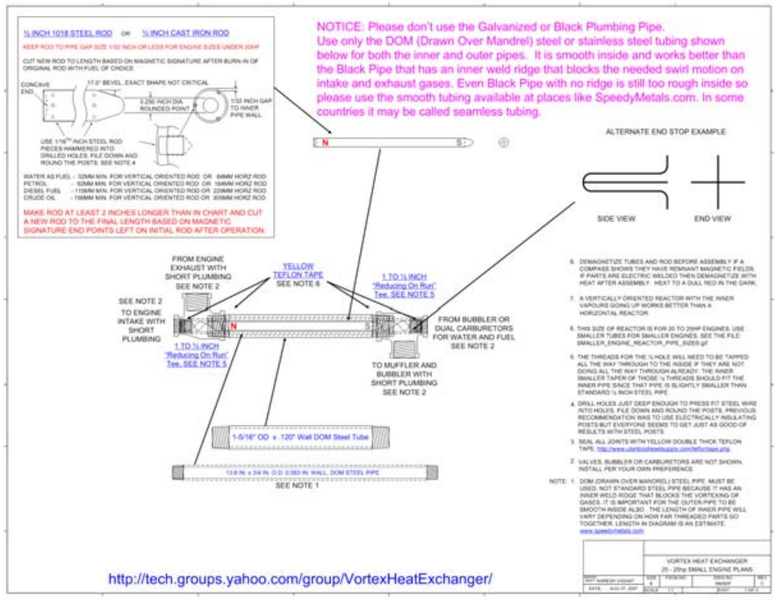
Only the 20_to_25hp_GEET_Reactor_Plans.gif reactor has been built by several people and those parts are sure to work together. Its' 3/4 inch inner tube can be threaded the same as a 1/2 inch pipe. Several people have done this. A Smooth inner surface is important.

10 to 15hp REACTOR PIPE SIZES



File -10 to 15hp REACTOR PIPE SIZES

20_to_25hp GEET Reactor Plans



[File- 20t o 25hp GEET Reactor Plans]

Free internet Plans

The following introduction comes courtesy of [narivasant] from the Vortex heat exchanger group -These plans are not meant to demonstrate the capacity the GEET has or to be used to convert an auto engine over. Look at the GEET patent and compare it with the free internet plans. You need a valve (almost closed) in line with the exhaust gases to the bubbler. That is missing from the free internet plans. **Everything disclosed from Paul Pantone and GEET in their training videos sold through Tesla Tech is that you need 8 to 10 inches of vacuum in the bubbler and through the reactor**. It is better to be sucking on the reactor and bubbler than putting pressure into them. Exhaust gases sent to the bubbler sold be drawn in with vacuum rather than pushed in under pressure.

Most of [the videos] have lots of not very helpful talk but the talk is intermingled with %5 good technical data. Mike Hollar gives the best technical data. One video I have cuts him off near the end of a lecture when he started giving out too much good information.

Bob Colvin, a GEET distributor I think, designed the plans and they were made with lower efficiency on purpose it sounds like from Paul Pantone's description. It could well be that GEET the company was still trying to make a profit and rightly felt they had the right to keep the best design information for only those who paid a license fee or paid to attend a GEET training class.

The inner weld ridge is 1 thing that makes it a less efficient reactor and can only work with a 1/2 inch rod because a 9/16th inch rod would be against the inner ridge and forces the gases straight through rather than vortexing.

The free plans show the bubbler valve AFTER the bubbler but the Paul Pantone patent shows the bubbler valve BEFORE the bubbler. In the 1st case the bubbler will be under pressure and in the 2nd case it will be under vacuum. Some people have said they only got their reactors to work when the bubbler was under vacuum. But then the bubbler must have stiff sides that don't get sucked in.

The outside of the bubbler needs to be heated with exhaust gases flowing around it. The bubbler needs to handle the heat without melting. In other words, build it more like the Pantone patent than like the free plans and you will have better results. Still, as you may have seen on some of the videos on the internet, people can still get good results using a plastic bubbler.

Reactor Rod

The following is courtesy of Naresh -I don't remember seeing that but I've heard Paul Pantone and Mike Hollar in the full length versions of the GEET training videos and they say the rod needs to have iron in it. They say some materials work better than others but even that need to have iron in them. There might be a possibility that non-ferrous rods could work if the reactor is always aligned correct lt relative to the Earth's magnetic field.

There a French variations that are all stainless and as far as I'm concerned, until proven otherwise, I think they have diverged so far from Pantone's best designs that they are really making just fuel vaporizers with some of their designs. I think they still use a rod but some may be using stainless steel. I haven't studied the SPAD a lot, just some. I see it deviates from the best original GEET designs from Paul Pantone. I think water only GEET system works best with multiple reactors in parallel with the rod lengths tuned for water. All the information I've accumulated leads me to still think iron in the rod and pipes are better than aluminum, copper, or brass. I think some types of magnetic ceramic (ferrite) rods might work well.

The bottom line is it is easier to make a system less like Paul Pantones, like with no rod, but it is more likely to work only as a fuel vaporizer and not a reformer that also has the other unusual effects like magnetic fields during operation and transmutation of elements.

Even if just making a vaporizer, why not just go ahead and use a mild steel rod since it is still helpful just to get the vapors up against the inner pipe wall for better heat transfer if nothing else. But please not use Black pipe or galvanized pipe or it will quickly give a person the wrong impression that it doesn't work when in fact the pipe welds and roughness are stopping it from working. Mild steel DOM (Drawn Over Mandrel) or mild steel seamless tubing has a better change of working.

People who don't understand the electro magnetism's are more likely to ignore the parts of the design needed for making it work, like swirling gases. So then they prove to themselves it doesn't work simply by building it wrong when they ignore the parts they think are not important.

Reading the actual Pantone patent document on the US Patent Database (Patent No. 5,794,601). *DETAILED DESCRIPTION OF THE ILLUSTRATED EMBODIMENT. it possible for the fuel burning equipment to utilize as its fuel source fuels or other materials that are generally considered as not being suitable fuels for such fuel burning equipment. A steel reactor rod has been found satisfactory as have stainless steel, aluminum, brass, and ceramic reactor rods*.-End

So clearly, Pantone advises the use of any material for the rod, which is surprising, given that SS, brass etc are non-magnetic. In respect to the French SPAD which seems to use no magnetic rod, their claims in the Gtone document are listed in this document.

Also since people often put misleading information in patents to stop design theft and because the GEET training videos clearly state, and at length, that the rod material must contain iron, I think it best to use mild steel. Maybe the patent was filed before Paul started experimenting with turning the react6or to align with the Earth's magnetic field and it was at that point he realized and experimented with metals with higher magnetic permeability.

If someone has the time, it would be a great experiment to get a GEET going and put exactly the same shape rods of different materials and see the difference. GEET training material even says if the rod is hollow steel it won't work well, even if the outer shape is exactly the same.

Please check the "getting the reactor to work in the files section of the vortex heat exchanger group. Category Section 5 has a different link to a carburetor that has a float bowl, which is easier to use than an RC engine carburetor, also added Section 15 about smooth flow through the pipes.

Note- There is a place in the GEET training videos where they say stainless steel does not work for the rod material.At 3:00 minutes into this video, Mike Hollar , president of GEET , says use steel and not other materials.
http://www.youtube.com/watch?v=qMNCebzgCgg

At 6:18 minutes into this same video he says again that the rod material must have iron in it and at about 6:45 he says the electromagnetic field comes from the rod and the rod material must be able to support an electromagnetic field. Most stainless steels don't create much magnetism.

Note- Galvanized pipes are not a good idea because the coating of zinc makes a very poisonous gas when the pipes get too hot. But zinc on the reactor rod might make a good catalyst. If for some reason a reaction is not occurring and if the vapors flow through the reactor are not keeping zinc coated reactor rod from getting too hot then a small amount of the poisonous gas can come out of the engine exhaust. It has also been reported that an electroplating happens and crates deposits inside the chamber.

GEET Reactor Rod Material

GEET Reactor Construction Details

Paul's GEET patent is better. Many other variations are better. Some key improvements are: get an inner pipe with a smooth inner wall because typical steel pipe nipple has an inner weld ridge that blocks the swirl motion of the vapors. I've heard but not seen for myself that black steel pipe nipple made in the USA and sold through MSC Industrial Supply Company in the USA has a smooth inner wall. Else, DOM (drawn over mandrel) steel pipe is smooth inside.

Use a rod that provides 1/32 inch gap rather than 1/16th inch gap. Use a heated bubbler than can handle a partial vacuum without collapsing in. Even better are 2 carburetors for fuel and water. If using a bubbler, use a valve in line with exhaust gases to the bubbler. For Alternate small engine plans that includes the smooth wall pipes of dimensions that create a smaller rod to pipe gap. **Please view the vortex heat exchanger group files contained in this document. This file is listed below.**

The history behind the reactor magnetic In some of the GEET training videos sold through teslatech, Paul Pantone says that one day he was running his engine on a bench and connected to an exhaust analyzer. Mike Holler needed some space on the bench also so Paul turned his reactor to make more room. He noticed that the pollutions numbers on the analyzer went down as he turned the reactor. He then proceeded to turn the reactor all different ways for a long time while looking at the analyzer and discovered that it had very low pollution numbers when the rod nose end of the reactor was facing South.

I have heard of another person who accidentally put there already magnetized rod back into the reactor the wrong direction and their reactor still worked just as well. But Paul Pantone said when some of the GEET company distributors took demonstration reactor pipes apart and put them back together they stopped working if the pipes were inadvertently turned and not oriented like in the magnetic diagram in the vortex heat exchanger group.

There is a much about the magnetic still to be learned. When Paul Pantone uses a vertical reactor, he points the rod nose down and the vapors blow it up so it floats better in the wind, but in the Northern hemisphere where Paul lives down is North, not South like he originally did for a horizontally oriented reactor. But anyway he says he gets more power from a vertical reactor than a horizontal reactor.

Maybe the magnetic direction is not as important as point either North or South but I don't really know. I think the reason a vertical reactor gives a better reaction and more power is because it is easier for the rod to spin and that gives a better reaction. If not for any other reason because there is lower resistance to the flow of the vapors when there is drag only against the inner pipe wall rather than against a non-spinning rod also.

David Pantone was saying that smaller reactors work better with smaller engines. According to him, something like a reactor with a 9/16th rod might work better with a 20hp engine than a 10hp engine and a 5hp engine might work better with a 3/8 rod and everything else proportionately smaller.

If some of the parts are magnetized too strongly already from the factory then the reactor may never start working and so can never magnetize the parts the correct way. So it is still good to check and insure there are no strong magnetic fields on the parts from the factory or from builders cutting, welding, etc. Various types of working on the tubes can leave magnetic fields on them.

If there are any strong fields detected with a compass it may be best to heat up the parts to a very dull red in the dark as a way of erasing the magnetic fields. Heating a ferromagnetic metal above the Curie temperature will erase the magnetic fields.

Spinning the reactor Rod

It's not that hard. I [Naresh] had mine spinning by having the rear stop wire placed in the rear indention of the rod. I could hear it when it started spinning. It made a very slight noise of the centering bumps against the inner tube wall. If the centering bumps are rounded there is not much scrapping. Paul Pantone and others have left off their centering bumps and still got there rod spinning and held center side to side wise by the vortexing.

A rod spinning perpendicular to a magnetic field will induce current in the rod that will travel in a loop inside the rod and fight against its trying to spin. A rod spinning parallel with a magnetic field can induce current in the rod to travel out to its perimeter if it is electrically connected to something outside the magnetic field. That is how a homo polar generator works.

But the rod spinning is not like an AC generator unless there were a perpendicular magnetic field with a North and South pole that the rod was alternately spinning through. But the GEET does not have that type of magnetic field. If the rod had a magnetic field coming out its sides it could induce current external to it if it spins but again it does not have a magnetic field coming out its sides so it is not like an AC generator in that respect.

If it did then you are correct that the external coils would have to be wound different to pick up electrical energy. The diagram is for the purpose of showing you how Paul Pantone did it when he measured electrical power coming off the reactor and for the purpose of showing you how it worked, so I think it should not be redrawn some other way that Paul Pantone did not get current from.

Instead of changing the design of something that worked to some other design that is no longer the same, it is better to try and reproduce the same working design and try to reproduce the phenomena that made it work. I believe the generated current was from some more unusual phenomena. I've spent a long time investigating it and thinking about all the kind of things that you are now thinking about.

Reactor Rod from the GEET institute –Power point file

If you used the free plans (12" Rod) and you want to do an efficiency test is suggested that you want to do test of how long it runs with versus without the reactor, that you use E85 or E100 high ethanol fuel. Same for emissions testing with versus without the reactor. And in those cases you will need to put a shorter rod in it. Hopefully if you start with a 12 inch rod then after 30 minutes you will see a magnetic signature with the ethanol fuel that is shorter than 12 inches between the end points.

For the best fuel economy and longer run times and to use the bubbler for more oily fuels which I think was more Paul Pantone's intent, we recommend using diesel fuel or non-synthetic motor oil. That should work better initially with the 12 inch rod also.

You will still need to put a small amount of petrol in the fresh air valve to get the engine started. Also we suggest putting a small amount of electrolytes in the water/ fuel mix.

Something like a small amount of ammonia and a very small amount of battery acid. Salt is good also if you are brave. Paul Pantone did not have a problem with it but some people are afraid to have salt going into their engine.

If you find that the bubbler stays with more water or even starts filling up with water condensing out of the exhaust then you might want to try letting a smaller amount of exhaust into the bubbler and add a small amount of fresh air to bubble through with the exhaust. If you can close the exhaust back pressure valve completely and it engine keeps working then you know it is cracking well enough to release a lot of oxygen.

But also, in that closed loop configuration, insure the exhaust is not blowing out any of the other valves and is actually going all back into the engine.

If the fuel is petrol then you will have some left over in the bubbler. We have never seen a single video or heard of a single time that Paul Pantone ever used petrol and water in a bubbler. He uses a bubbler for oil and water with electrolytes and sometimes just water and electrolytes and no hydrocarbons at all.

Petrol is made up of a wide range of sizes of hydrocarbon molecules. The shorter chain molecule evaporate first and slowly more and more long chain more oily portions of the petrol are left behind. If your rod length is tuned for short or medium length hydrocarbons that evaporate first then the reactor will not work well with the long chain more oily hydrocarbons that tend to be left for last.

You will get better results with a bubbler if you use oily fuels and adjust the rod length for more oily fuels. Diesel fuel ought to work also in a bubbler. Petrol can be used just for starting the engine and getting the reactor warm enough to start working. The petrol can be put in the fresh air input valve to start the engine. If the engine will not keep running on cracked oil and water then something is not working, vacuum leaks, magnetic, rod length, etc.

If you want to run continuously on petrol then you need something that makes a fog of the fuel such that each droplet contains all fractions of evaporation points of the fuel. Paul Pantones has used small carburetors, Naresh however reports that when he tried that with a cheap small carburetor it made too large of droplets and that got the rod wet and the reactor stopped working. Now he tries to use other things to make very small droplets.

We think that this is the main problem people have had over the years, not the GEET itself, but controlling the vacuum & fuel. Firstly we need a constant minimum vacuum at all times & for idle. Secondly we need to increase that vacuum up to maximum gradually whilst introducing more fuel & air. The 2nd part being the pain in the backside to achieve. Naresh States: I tried using a smog pump as a vacuum pump and it worked great for giving me a few inches more vacuum but it had leaks around the bearings or bushings, I think for ventilation cooling of the bearings or bushings. I tried to plug them but still I couldn't get my reactor output gases to give me much energy for combustion in the engine but it might have been for other reasons and after I get all the other bugs worked out I might try it again.

Any vacuum pump for use with a GEET reactor needs to handle high flow volume and hot gases coming out of the reactor. The smog pump had carbon composite seals that worked well with hot gases.

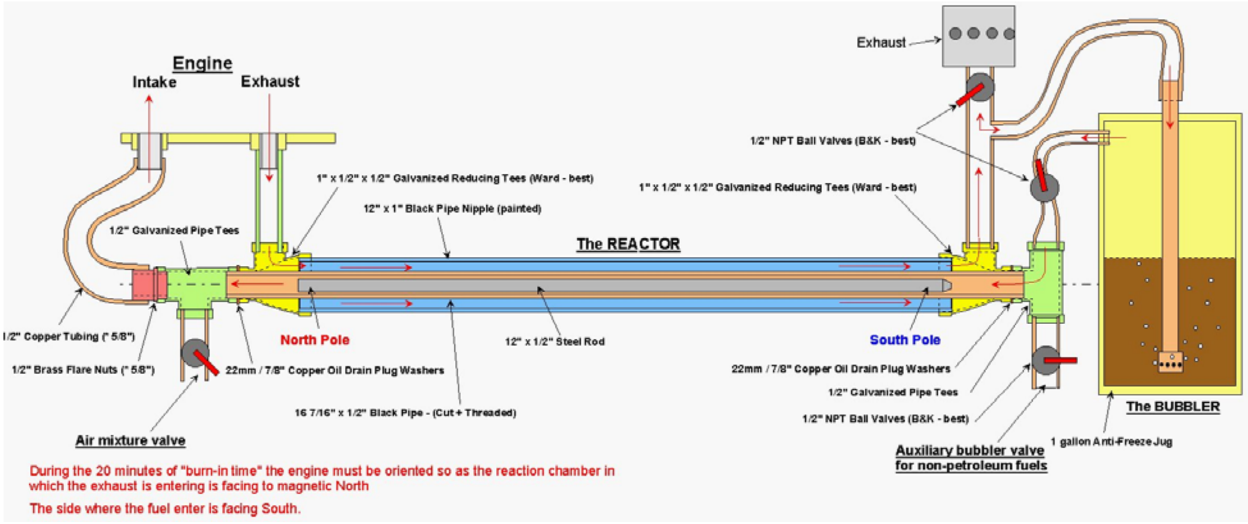
Water in the Bubbler

The exhaust is rich in water vapors. If the bubbler is not externally heated then it is common for that water to condense in the bubbler rather than having water in the bubbler vaporizing. When the lighter portions of the petrol vaporize it also has a cooling effect that can increase the amount of water that condenses in the bubbler. That may be one reason that on some of Paul Pantone's reactors he has only fresh air going to the bubbler.

On his bigger blue demo unit where he sends exhaust to the bubbler he also heats the outside of the bubbler to help keep the water content vaporizing rather than condensing.

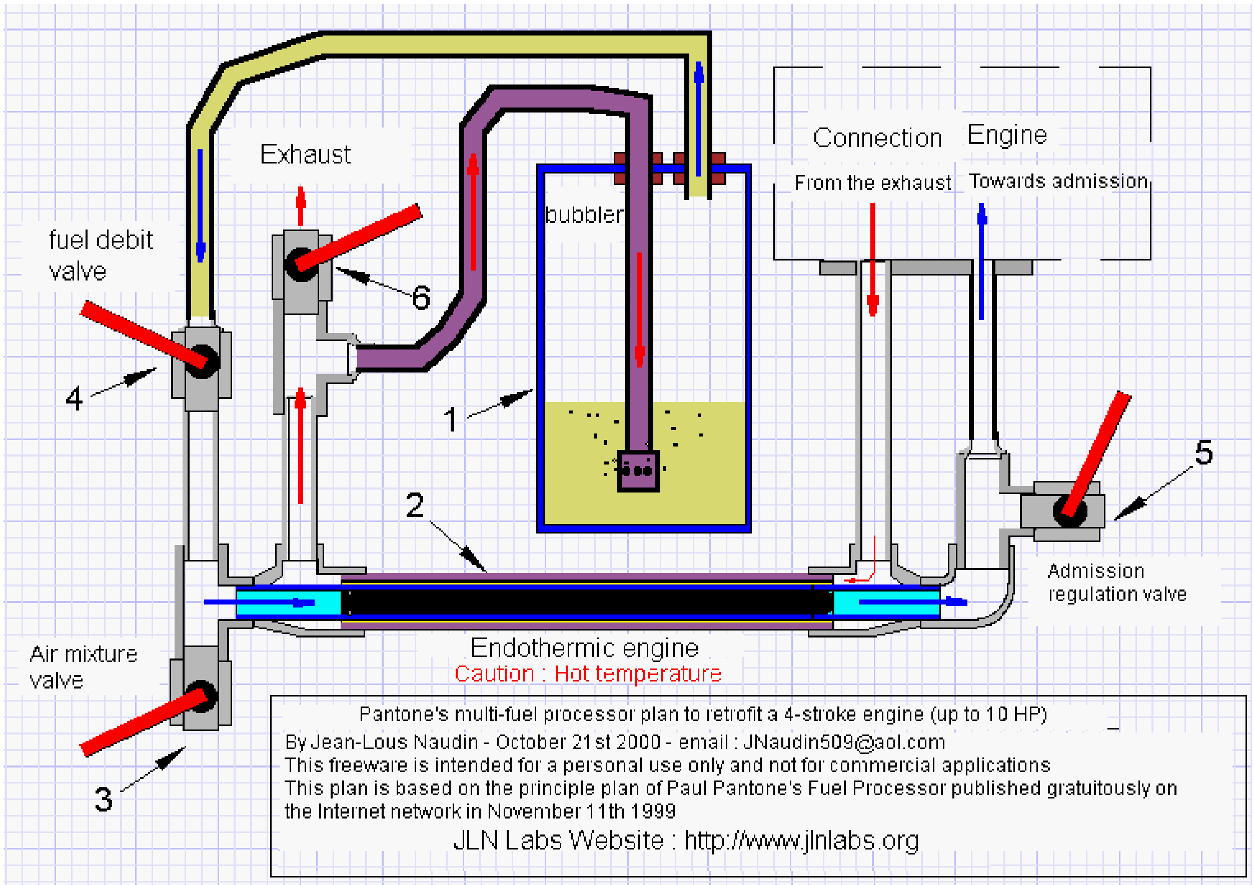
Free Plan's version

Do-It-Yourself
GEET Construction Plans
for a Small Engine (<20 HP)



Credits got to JLN labs and Rex Research. Please Copy and enlarge for a better view. An Alternative source for this diagram can be found at the following: http://www.rexresearch.com/pantone/geetbig1.jpg.

Here is another graphic kindly done by JLN labs, this was translated by Panacea France.



Step 1 ~ Tools needed - pipe wrench, crescent wrench, spring tube benders, pipe cutter, pipe flaring tool, allen wrench, soldering equipment, file, and screw driver. Obtain all your parts and tools needed for the conversion ahead of time. (Parts List at bottom)

Most professional plumbing supply stores stock higher quality parts compared to large home centers cheap plumbing parts. The savings aren't that much on a small project like this. The most crucial quality part is on the inner pipe, problems arise from inconsistent wall thickness, out of roundness, thick weld seams, etc on low quality pipe.

Step 1: Tools Needed ~

Step 2 ~ Strip down the engine removing the gas tank, muffler, and carburetor. Remove the mower blade and replace with a 12" diameter steel disk flywheel of the same thickness as the blade for safety.

Step 2: Stripped Engine ~

Step 3 ~ Take the 1"x1/2"x1/2" reducing tees and mount them on a 1" nipple (short pipe), and then using a lathe, machine the end smooth and fly cut (bore out) the hole in the end 27/32" (21mm) so that the 1/2" inner pipe will slide inside. This procedure can also be done by using a drill press to drill a 27/32" or 7/8" hole in the end of the tee and then use a file to smooth the roughness off.

The 1/2" pipe connector and 1/2" tee will each need to have one end smoothed off as well to receive the copper washers as a tight seal.

Step 3: Tee & Connector ~

Step 4 ~ Have a plumber or plumbing center cut your inner reactor 1/2" pipe to 16 + 7/16" and thread both ends. Use Black Pipe here because galvanized pipe gives off toxic fumes if heated too much. File the 12" x 1/2" multi-fuel steel rod to a bullet point on one end only. (7 + 3/8" x 1/2" for gasoline only) .

This will keep you out of trouble later if you can't remember which way the rod points. The engine will not run if the rod is put in backwards after it has a magnetic signature.

Assemble the parts in order as in the above picture using the 7/8" / 22mm copper. Washers used in oil drain plugs for cars. (2 - 1"x1/2"x1/2" machined reducing tees joined by the12" long 1" nipple, slide the 16 + 7/16" long 1/2" reactor pipe inside, add a copper washer on each end and then add the 1/2" tee, 1/2" NPT / 1/2" Brass Male Flare Fitting, and 1 1/2" nipple, and 1/2" Air Mixture Valve.)

Step 4: Plumbing Pipes & Rod ~

Step 5 ~ Assemble the other valve component subassemblies above. The 1/2" thick steel intake / exhaust adapter plate above is used only on some engines like *"Tecumseh"* and Overhead Valve Engines (picture 9). Add a 1/2" NPT / 1/2" Brass Male Flare Fitting to the Intake on the adapter plate.

Some *"Briggs and Stratton"* engines, etc usually already have the exhaust threaded for 1/2" pipe, but the intake is on the other side of the engine causing longer hose runs. Also a compression pipe connector or a piece of rubber hose with clamps will need to be connected from the engine intake to the Bubbler pipe.

(1/2" valve (Auxiliary Bubbler Valve), 1 1/2" x 1/2" nipple, 1/2" tee, 1 1/2" x 1/2" nipple, 1/2" valve (Throttle/Bubbler Valve), 1/2" to 1/4" pipe reducer bushing, half of 3" x 1/4" nipple.) and (Muffler, 1/2" ball valve (Optional - Back pressure valve), 3" x 1/2" nipple, 1/2" tee, 1/2" to 1/4" pipe reducer bushing, half of 3" x 1/4" nipple, 1 1/2" nipple.)

Step 5: Valve Components ~

Step 6 ~ Assemble the sub-assemblies onto the reaction chamber above making sure to install the 12" rod inside pointed away from the engine. Now it's time to start on the bubbler.

Step 6: Finished Reactor ~

Step 7 ~ Take 10 3/4" x 1/2" copper pipe and solder a copper 1/4" NPT - 1/2" pipe adaptor on one end and a 1/2" cap on the other. Drill a 1/16" hole through the cap, turn 90 degrees and drill through again, also one up through the bottom. Take the other 1/4" NPT - 1/2" adaptor and cut off the thin wall portion to make a pipe nut and file smooth for inside the Anti-Freeze jug.

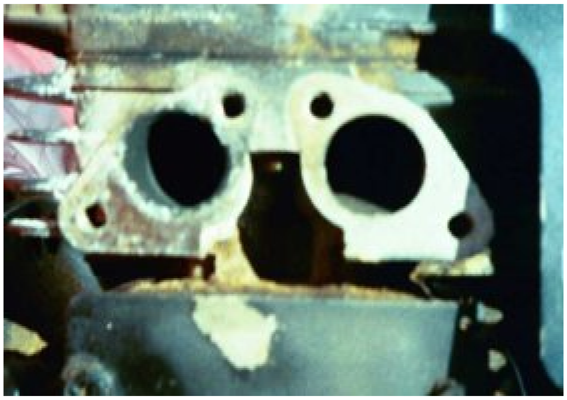
Step 7: Bubbler End & Pipe Nut ~

Step 8 ~ Take a 1 gallon anti-freeze jug and drill a 1/2" hole near the top of the jug and through the cap as illustrated. Assemble the parts together in the following order. (Hose, half of 3" x 1/4" nipple, 1/4" pipe connector, short 3/4" nipple, bushing, hole in jug, bushing, and pipe nut.) and ((Optional - Back Pressure Hose), half of 3" x 1/4" nipple, 1/4" pipe elbow, short 3/4" nipple, bushing, hole in jug cap, bushing, and soldered pipe.)

Step 8: Bubbler & Hoses ~

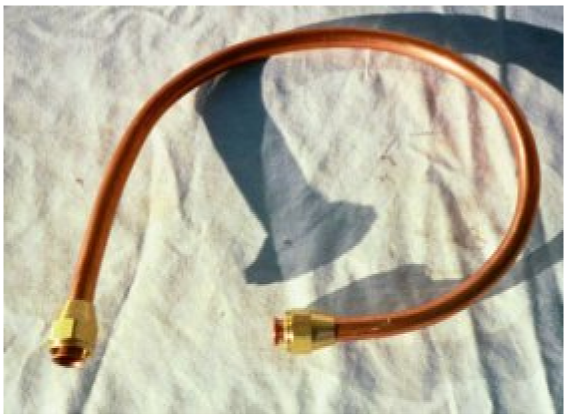
Step 9 ~ The port adapter was formed by cleaning the intake and exhaust ports off. Then dipping a finger in the exhaust port to get some soot to rub on wide masking tape taped over the ports. This then leaves a perfect template to then tape into a 1/2" thick piece of steel , then drill the mounting and the port holes and tap the ports with a 1/2" NPT thread tap.

Step 9: Exhaust & Intake ~



Step 10 ~ Use 1/2" tubing for 10 HP or less (* 5/8" tubing and flare fittings for 10-20 HP) with a 1/2" tubing spring bender and form a loop, then remove the spring. Slide the flare nuts on each end, and then slide the flaring tool on so that the pipe sticks out about 3/16", make the flared ends. (Air-Conditioning supply houses carry flare fittings if you have difficulty finding them.)

Step 10: Tubing Loop ~



Step 11 ~ Assemble all the parts onto the engine, and then add a 1" pipe support or 1 1/4" exhaust hangar. Fill the bubbler up no more than 1/4 full till you get used to using it (up to half full later), have someone steady the jug while starting the engine so it doesn't spill into the hoses. If wet fuel gets on the reaction rod it will stop running, you'll have to dry your rod and hoses out. You can hang it from the mower handle if you like later after it's started.

You must point the exhaust end of the rod due north while starting the engine the first time and let it run for 30 min to "burn in the rod". The rod will self center magnetically by itself after it's running or you can weld three bumps on each end to center the rod (file them to fit snugly).

Leave the optional back pressure valve full open, open the throttle about halfway and crack open the mixture valve, and start the engine by varying the air mixture valve. If it's very cold you will have to choke the engine by blocking off the air valve with your finger. Then slowly increase the throttle wide open while adjusting the air mixture valve.

The engine will turn over easily if you are near the right setting, if it's very hard to pull, readjust the throttle or air valves. Make sure to paint all external pipes and connectors with High Temperature Grill Paint or they will rust very quickly afterwards. (Except copper, brass or galvanized)

Step 11: Finished Conversion ~

Step 12 ~ for an installation on a generator, you can also use 90 degree elbows to keep the pipes within the cage. Mount the GEET Fuel Processor as far away as possible from the generator magnetic field so they do not interfere with each other. Also be very careful with credit cards in your pockets or video cameras, etc from getting too close to the engine while it's running so they won't be erased.

Step 12: 5 KW GEET Generator ~

Finally: Experiment with the optional "Back Pressure Valve" to run closed loop on alternative fuels, don't use hydrocarbon fuels, because they will be contaminated with water from the exhaust (Hydrogen and Oxygen combining). Different materials for the inner pipe and reaction rod. Different rod lengths and also threaded rods, engine side of the reactor locations for the air mixture and/or throttle valves, exhaust heated copper tubing from the bubbler to the throttle valve, 5 gallon bubblers, double bubblers for non-soluble fuels, vacuum gages, etc, etc, and also "alternative fuels". Additional note: It has been found that the 1/2" reaction rod inside the pyrolitic chamber gives a bit too much clearance. It is recommended to use a 9/16" steel rod for the reaction rod.

Parts List

Note: Some Engines only --- 1/2" Steel Adapter Plate with 4 flush 3/4" Allen Screws and 12" steel disk

Fuel Processor:

1 - 16 7/16" x 1/2" Black Pipe - (Cut + Threaded)
1 - 12" x 1" Black Pipe Nipple (painted)
1 - 12" x 1/2" Steel Rod
2 - 1" x 1/2" x 1/2" Galvanized Reducing Tees (Ward - best)
2 - 22mm / 7/8" Copper Oil Drain Plug Washers
1 - 2" x 1/2" Galvanized Pipe Nipple
1 - 1" Galvanized Pipe Hangar with Bolt & Nuts
4 - 1 1/2" x 1/2" Galvanized Pipe Nipples
1 - 3" x 1/2" Galvanized Pipe Nipple
4 - 1/2" NPT Ball Valves (B&K - best)
1 - 1/2" Galvanized NPT Muffler
3 - 1/2" Galvanized Pipe Tees
2 - 1/2" x 1/4" Galvanized Pipe Reducing Bushings
1 - Can Hi-Temp Grill Paint
27" - 1/2" Copper Tubing (* 5/8")
2 - 1/2" NPT / 1/2" Brass Male Flare Fitting (* 5/8")
2 - 1/2" Brass Flare Nuts (* 5/8")

Bubbler:

1 - 1 gallon Anti-Freeze Jug
4 - 1/2" Galvanized Hose Clamps
6' - 1/2" ID Clear Vinyl Hose - (cut in half)
2 - 3" x 1/4" Galvanized Pipe Nipples - (cut in half)
4 - 9/16" Galvanized Bushing Washers - (1/8" thick)
1 - 1/4" Galvanized Pipe Elbow

2 - 3/4" x 1/4" Galvanized Pipe Nipples
1 - 1/4" Galvanized Pipe Connector
1 - 10 3/4" x 1/2" Copper Water Pipe
1 - 1/2" Copper Pipe Cap
2- 1/2" x 1/4" NPT Copper Pipe Adapters

Walk through video theory of the FREE plans

http://au.youtube.com/user/neverchk

Modified Carby

Open source engineer Ady states that he has using a whipper snipper/chainsaw carby on his Peugeot & it has proved just about perfect for idling the 1.4Liter engine, , but needs to add another for power etc. So, that in mind, Ady state reasons that if one were use this carby on a small genset (5HP) it may work perfectly. He didn't modify it at all; he just used the choke plate to control the vacuum & fuel. Also, the mixture seems to be very, very adjustable via the screw on the side of the carby, very rich to very weak. The little "blister pump" is useful for starting from cold, even my Peugeot fires up on the GEET set up.

The GEET needs a 3 to one air fuel ratio, so the idea is to restrict the carby so you can achieve this. In the following, Panacea modified a 6.5HP Brigg's Genset carby and used washers to restrict the air in,

Panacea's Modified Carby

Made for Easy retrofitting onto the GEET reactor

Notes for a car engine. For those who do not know, since that GEET reactor's output gases go into the engine's intake before the car's original throttle plate, that little carburetor needs a control linkage to close it so it can maintain a good vacuum in the reactor at low RPM.

Large Auto Plans Courtesy of the GEET institute

GEET FUEL PROCESSOR PLAN SET SA-1

The following plans are designed to help you retrofit your internal combustion engine with the GEET Fuel Processor Technology. The diagram drawings represent working designs of basic individual parts and components needed for a manually operated retrofit application of an automotive type engine. Changes or variations of designs may be necessary to fit your specific needs (example: flange modifications, parts placement, space limitations etc.).

As long as the principles of the GEET Technology are adhered to and all system components are incorporated as specified, your retrofit application should function properly. WARNING!!—This information is classified as EXPERIMENTAL!! We cannot guarantee results. Please take time to educate yourself sufficiently before proceeding.

The following is for information purposes only. We do not control the materials used in construction, the methods of construction, nor the applications of construction or use of the finished product. Therefore, WE ARE NOT RESPONSIBLE FOR DANGEROUS OR UNDESIRABLE RESULTS, and we do not take responsibility for accidents due to negligence or ignorance!!WARNING!!—Gasoline is EXTREMELY FLAMMABLE and caution should always be used when working around it. Do not smoke around it, and keep it

away from open flames. Whenever possible you must use goggles, gloves and/or other protective gear.

WHAT IS A GEET FUEL PROCESSOR?

The GEET Fuel Processor (GFP) is a fuel refinery, plasma field generator, and fuel delivery system all rolled into one. The GFP is capable of dramatically increasing the efficiency and decreasing the pollution on an internal combustion engine. The GFP is also capable of using diverse fuels that the engine would normally not tolerate. For many applications, the GEET retrofit can pay for itself in a short period of time. The GEET Fuel Processor works on the principal of hot and cold air masses moving in opposite directions within a vacuum. By running the fuel vapor up through the center of the exhaust, in the opposite direction of the exhaust flow, we are simulating these air masses. The reaction rod in the GFP acts like a synthetic “Mother Earth”. As the masses shear against each other, friction causes the formation of electrical charges and the atmosphere (around these charges) becomes ionized. Plasma forms as “lightning” within the reaction chamber causing whatever fuel being used to “transmutate” into a new, cleaner, simpler based fuel we call “GEET Gas”. With the GFP, fossil fuels would no longer be as “polluting” or “harmful” to the environment as before, and the terminology “usable fuel” would take on a whole new meaning. To better understand how the GFP works, let’s look at the smaller sub-systems:

1)Liquid-to-Vapor Phase Changer—The GFP reactor utilizes a vaporized fuel, not a liquid fuel. Therefore we must incorporate a means to change the liquid fuel, by means of a phase change, into a vapor. The drier the vapor the better. A bubbler or even a modified venturi carburetor can sufficiently deliver an aerosol, which is small droplets of liquid fuel mixed with air, to the reaction chamber. The heavier the fuels the more difficult it is to vaporize them. The bubbler has delivered better results, in the use of heavier or diverse fuels and also seems to be more efficient in the vaporization of fuel. For ongoing operation, the bubbler will need to have a float system set up to regulate the level of fuel in it.

2)Reaction Chamber—This is a pipe in the center of the exhaust system which utilizes a heat transfer or plasma reaction to break the fuel down into a simpler, cleaner form before it is fed into the engine. By running fuel vapor through the center of the exhaust in the opposite direction of the exhaust flow we are simulating a natural phenomenon of hot and cold air masses colliding. This collision within a vacuum helps create the desired “reaction” needed in the GFP.

3) Air Management Valve (AMV)—The new “GEET Fuel” coming from the reaction chamber must be properly mixed with additional incoming fresh air as it is metered into the engine. This is the job of the Air Management Valve. This can be as simple as using

two ball valves or as complicated as machining either the unit shown in these plans or one of your own design.

We have diagramed a manually operated air management valve for small engine use, and a lever controlled unit for the larger engines. Understand that you have the freedom to mix and match styles and designs to best suit your application needs, space requirements, and your individual abilities.

For security and protection reasons, we are not including the specific clearances for each application. You will need to call GEET Management at (801) 913-5028 or write us at info@geet.com, to get these specifications. When built properly, the GFP should work as promised and in most instances may exceed your expectations.

The sub-systems of the GFP will now be explained in greater detail:

THE BUBBLER

One of the many methods of phase-changing the liquid fuel into a vapor is a bubbler. Included in this set of plans are three drawings of bubblers. Each has advantages and disadvantages, so will be up to you to study them and choose which is right for your application needs.

Air is drawn into the bubbler through the inlet at the top and goes through a tube to the bottom of the bubbler where it escapes and “bubbles” up to the surface of the liquid. As that air bubbles up through the liquid fuel, it creates vapors in the top portion of the bubbler which are pulled into the reaction chamber by the vacuum of the engine. A diffuser/bubbler plate and/or metal “scrub pads” placed in the bubbler will help break up the larger bubbles into smaller ones to produce better vapors, enriching the fuel mixture to the reaction chamber. Various other types of bubblers can be used also. The liquid level within the bubbler should never be high enough to allow the liquids to be sucked into the reaction chamber. Only the vapors are allowed in the reaction chamber for the process to work properly.

For continuous running applications, you will need to control the fluid level in the bubbler. One way of doing this is to attach a “Holley” float chamber to the side of the bubbler as shown in the drawings below. This is a mechanically operated system that can sufficiently do the job. If you use this method, the surface of the bubbler you are attaching the chamber to must be flat. A fuel bowl adapter (shown on the drawing) can be made to connect the chamber to a round bubbler. You will need to drill about 5 or 6 1 /4” holes across the bottom area of the bubbler, where the chamber attaches, so liquid can flow from the float chamber to the bubbler. Another 3 or 4 holes should be drilled in the bubbler, at the top area of the float chamber to equalize pressure between the two chambers.

Another way of controlling the fluid level would be an externally mounted electronic float switch. This could be adapted to either shaped bubbler. Parts needed would be an electric float switch, a horn relay, a propane shut-off solenoid, some 2” PVC pipe with solid end caps, 16-18 gauge wire, screw-in brass hose fittings, some 5/16” fuel line and hose clamps (See illustration).

The principle used here is that fluids will seek their own level. Therefore, the level of fluid in an externally mounted chamber will be consistent with the level in the bubbler. The advantage is that the level in the float switch will not be as affected by the constant bubbling action that takes place in the bubbler.

The electric float switches available are for very low amperage. The amount of power required to activate the propane shut-off solenoid would burn out most float switches. Therefore we use a relay, activated by the float switch. The relay then activates the shut-off solenoid. Adjust the float switch within the chamber to the desired height of the fluid in the bubbler. Raising the fluid level in the bubbler can give a richer mixture, but remember not to allow any liquids to be sucked into the reactor. Maintaining a fluid level approximately 1 to 2 inches over the bubbler/diffuser plate should give good results. You will need to experiment and determine what works best at your altitude. You may find that a fuel pump isn’t necessary. The vacuum within the bubbler may be sufficient to “pull” the fuel into the bubbler. A disadvantage to this is that too much fuel may be pulled into the bubbler if your shut-off valve doesn’t seal properly.

During the phase change from liquid to vapor, you will notice that when using gasoline your bubbler will become very cold (this should be the first fuel you use, because it is the easiest to use). The lighter elements in the gasoline will vaporize at room temperature, but the heavier elements will resist vaporization due to the cold atmosphere created. Eventually you will vaporize all the “lights” and end up with a bubbler full of “skunked” fuel. The heavier elements will require the bubbler to be heated to enhance their full vaporization. This can be accomplished by either surrounding the bubbler with hot exhaust from the engine, and/or directing some of the exhaust into the bubbler instead of fresh air.

The bubbler can be constructed of metal or some plastics such as PVC. When using plastics, make sure that the glue you use will not dissolve in the fuels you will be putting into the bubbler. We have found that JB Weld™ epoxy is good to use. If you choose to use exhaust gases to surround and heat the bubbler plastic should be avoided due to the hot temperature of the exhaust. If you choose to use exhaust gases to bubble the fuel with, there are several factors to consider:

First, when tapping into the exhaust system, come out in a “Y” fashion. If you “T” into the exhaust, there could be a siphoning effect that might suck the fuel into the exhaust. By coming out of the exhaust at an angle as “Y” under slight pressure, the siphoning is eliminating.

Second, the line carrying the exhaust to the bubbler must at some point rise above the top of the bubbler to eliminate drainage back into the line upon shut down.

Third, you must use the “manual valve” between the exhaust source and the bubbler inlet to control the amount of exhaust entering the bubbler to maintain a vacuum. Having the valve open to far could cause the vacuum to be lost and a “pressure” to be formed in the bubbler. We have noted in our testing that the efficiencies can be cut to 1/3 by pressurizing the bubbler (as opposed to having it under vacuum). The valve must also be closed when starting or stopping the unit to prevent fuel from being forced into the exhaust by back pressure from the engine.

Finally, all of the components exposed to the heat of the exhaust must be capable of handling this heat. Most ball valves use a Teflon™ seal inside that will melt with the heat of the exhaust. Furthermore, if using a PVC bubbler, the exhaust would melt the plastic at the point of inlet. We have used a metal screw-in lid in both the “top” and “bottom” applications of admitting a bubbling effect into the bubbler. The metal lid will dissipate the heat into the fuel before it comes into contact with the PVC. When using the “bottom” method for bubbling, make sure the inlet line or tube at some point rises higher than the bubbler to prevent the liquid fuel from running back out upon shutdown of the engine. The “valve” should be placed at the “higher” spot also.

In summary, the bubbler can be square, round, heated, not heated, tall or short (make sure it is tall enough to keep the liquids from being sucked into the reaction line!). Whatever the specific needs of your application are will determine if you should use a bubbler and how that bubbler should be built. For constant running applications, a means of controlling the level of fuel should be used. Be sure to use materials able to handle the fuels and temperatures to which they will be exposed. Either fresh or ambient air or exhaust gases can be bubbled up through the bubbler. The ambient air is easier to work with, but the exhaust gases may offer better results.

MODIFIED CARBURETOR

For certain application a bubbler may be impractical. Space limitations or other restrictions may require the use of a modified small engine carburetors. We have successfully used 8- and 10-HP carburetors of horizontal shaft Techumseh™ engines. The drawing shows that we restrict the venturi in the carburetor to pass more air over the pickup tube.

The stock carburetor is designed to deliver a 12:1 air/fuel ratio (AFR). For the GEET we need a 2 or 3:1 AFR. This is achieved by restricting the venturi diameter and forcing all of the air past the fuel feed tube. This can be accomplished by either drilling a small hole in the choke side of the venturi and mounting a baffle to direct all of the air over the discharge feed tube, or by using JB Weld™ to “secure” a washer into the venturi, restricting the inlet (leaving the choke on full may also help channel the air). Combinations of all of these tricks can be used to make the mixture richer to deliver the necessary fuel charge for a strong and powerful reaction.

The mixture adjustment screw can also be used to regulate the amount of fuel to be delivered into the reactor, helping to control engine power requirements. NOTE: If you adjust the air/fuel mixture going to the reaction chamber too much on the “rich” side, economy will decrease and pollution levels will increase.

THE REACTION CHAMBER

The heart of the GEET Fuel Processor is its reaction chamber. In a sense, it is the single most important part of the entire system. Ironically, it is also usually the easiest part of the system to build and tune! Most of the time if a system is not operating properly, it isn’t the fault of the reactor (unless there are leaks). Improper fuel mixtures, vacuum leaks, too much liquid being fed into the reactor, will cause problems before the reactor will.

The following three effects must be present in the reaction chamber in order to create the “reaction” (assuming the hardware is correct):

a.)Heat

b.) Vacuum on the inner chamber carrying the fuel.

c.)Cross flow of exhaust and fuel (hot in one direction, cold in the other.)

Upon initial start-up, you will instantly have vacuum and cross flow, but not heat. Therefore, you must start the engine on a suitable fuel such as gasoline until the engine and reaction chamber reach normal operating temperatures. This can be accomplished by either putting sufficient gasoline in the fuel you desire to run, or using a separate tank for start-up purposes. A valve can be placed in the fuel line to switch from the start up tank to the one you will be running off continuously. Our advice is to keep it simple.

The reaction chamber and rod (placed inside) need to be tuned to the size of the engine, the fuel being used, the RPM range of intended use, and whether you are using a single or dual reactor system.

Generally the exhaust pipe which surrounds the reactor chamber (pipe) should be sized large enough so as not to restrict the flow of exhaust from the engine. The diagram illustrates the inner pipe entering and exiting through elbows.

Steel exhaust pipe is recommended for the outer part of the reactor and “black” natural gas pipe is recommended for use as the inner pipe (reaction chamber) which the fuel passes through. The rod is placed in the inner pipe also. A means of preventing the rod from striking the ends of the reaction chamber on start up and shut down is necessary. We have found that springs that just fit inside the reaction chamber, will cushion the rod until the reactions start and the rod becomes naturally suspended by air flow and “electromagnetic” fields. In a pinch, a piece of twisted bailing wire will work also. Stainless steel can be used in all parts of the reactor, but is only required if the fuels being used are acidic in their natural state. As a general rule the parts should only be made of ferrous metals; or in other words, should be attracted to a magnet. This means no copper, brass, or aluminum.

Low grade steel dowel is what we use for the rod. The length and diameter of both inner pipe and reaction rod will only be given over the phone or the internet (info@geet.com) for the application you are building. This is done for our protection.

Remember that your exhaust gases will be exiting the engine and traveling in one direction, while the fuel will be traveling up the inner pipe in the opposite direction. Hot flowing south (down, in a vertical reactor), and cold flowing north (up, in a vertical reactor).

The GFP reaction chamber has been shown to generate its own electromagnetic field (EMF). If applying your GFP to a generator, route the reactor away from the generator. The EMF from the generator may interfere with the performance of your GFP combination that works for you. You will have to experiment on your own to get it just right.

FUEL INJECTION

We have developed a fuel vaporization unit that uses fuel injectors to vaporize the fuel which is then pulled into the reaction chamber. We built our own fuel injector pulsing electronics, and were able to tie it into the throttle positioning sensor, and thus were able to control the RPM of the engine quite effectively. Call us for more information, and to purchase the electronics. If your vehicle already has fuel injection, it may be possible to tap into the signals already being generated by your cars computer, and fire off separate fuel injectors in the GFP.

PUTTING IT ALL TOGETHER

On V-configurated engines you can run either a single reaction chamber or a dual. If you have the room, the twin set up may be the preferred method. If you don’t have the room, you can mount the reaction chamber either in one of the lines or downstream of the “Y” pipe.

The reaction rod is shaped like a bullet. The end being hit by the vapor stream is bullet shaped, and the other end has a dimple. This creates “bow shock”, and “stern shock”, which we have found creates a more efficient reaction. The rounded end should be 1/2” long and the length of the rod is measured from the base of the bulleted end.

AIR MANAGEMENT VALVE

Of all the aspects of building the GFP, the air management valve (AMV) is the part that may give you the most trouble. The function of the AMV is two-fold; first to control the amount of air and fuel, and secondly to control the engine speed and power output.

For the automotive or stationary water cooled engine, it is recommended that you use the multi-butterfly valve assembly shown in the drawings. This was designed to coordinate and regulate changes in the flow of air and fuel to the bubbler (fuel vaporization module), reaction chamber, and to the engine. It is necessary for use where the RPM of the engine constantly changes (such as automotive). The AMV mounts on the intake manifold of the engine and the valve levers are connected to each other and then to the throttle linkage of your application. The difficulty comes in adjusting the proper movement of each valve together.

Example: The AIR valve needs to open wide enough to allow sufficient air to the intake manifold. The FUEL valve from the reaction chamber should open just enough to allow “GEET Gas” to be mixed with that air. The BUBBLER valve (Fuel Vaporization Module, FVM) must ONLY open far enough to allow the necessary fuel vapor into the reaction chamber, but remain closed enough to maintain the needed vacuum within the bubbler (FVM). The drawing shows levers with holes drilled and numbered. This can help you make small adjustments at a time until you get the right

All of the necessary elements must come together into a usable package. To review, there are three sub-assemblies:

1)The phase change mechanism, which is a bubbler, Modified carburetor, fuel injectors, or other device.

2)The reaction chamber, where the “magic” happens.

3)The air-management valve which controls the Air/Fuel ratio and engine speed.

Certain things cannot be related on paper. You will learn more by running the engine for the first time than could be taught on a hundred pages of print.

SUMMARY

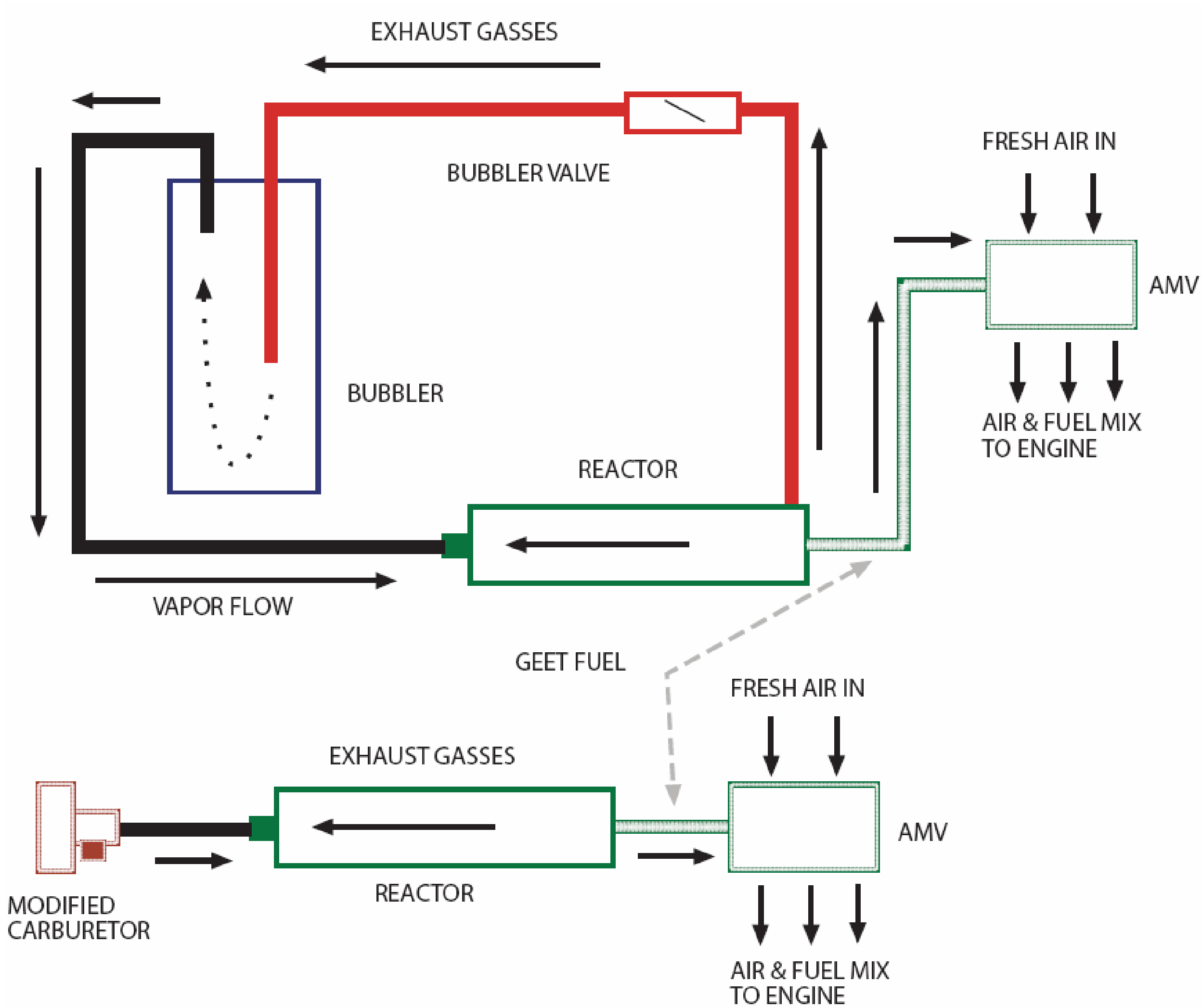
Believe it or not, the basic GEET system isn’t too overly critical of sizes and dimensions. It is however, extremely critical of vacuum leaks. If things aren’t working well for you, check the vacuum first. We recommend you start on something small with manual controls to learn how the GEET works.

Credits to

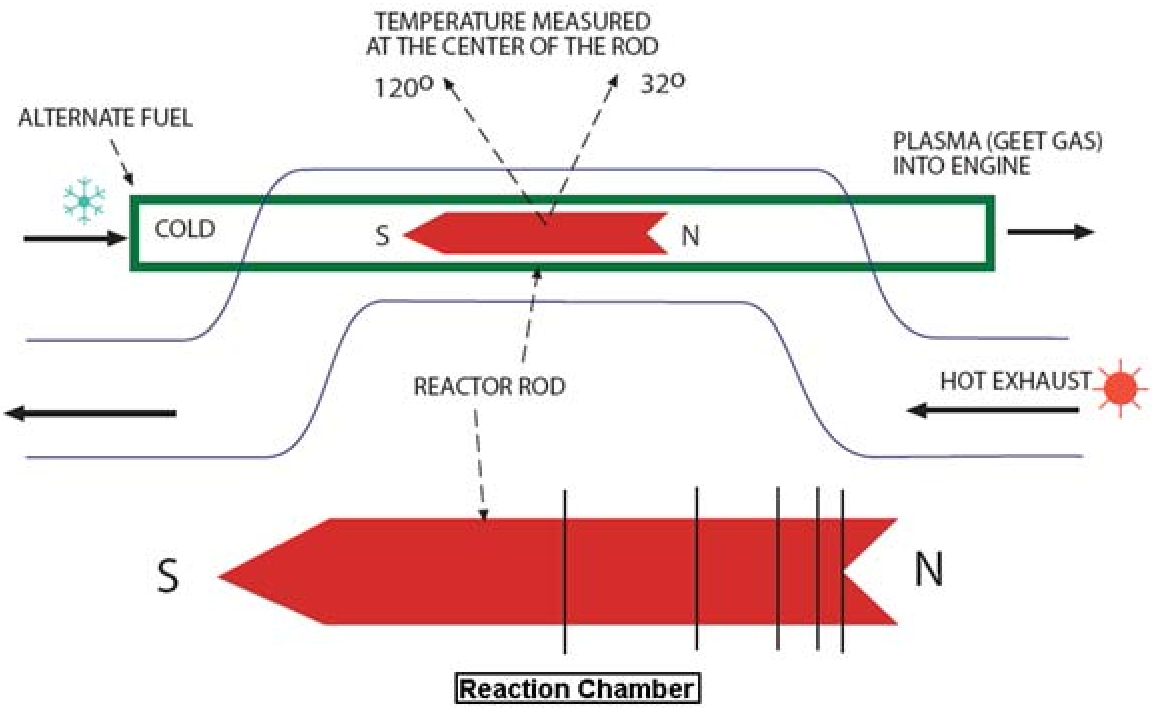
Paul Pantone

Inventor of the GEET Technology

Drawings

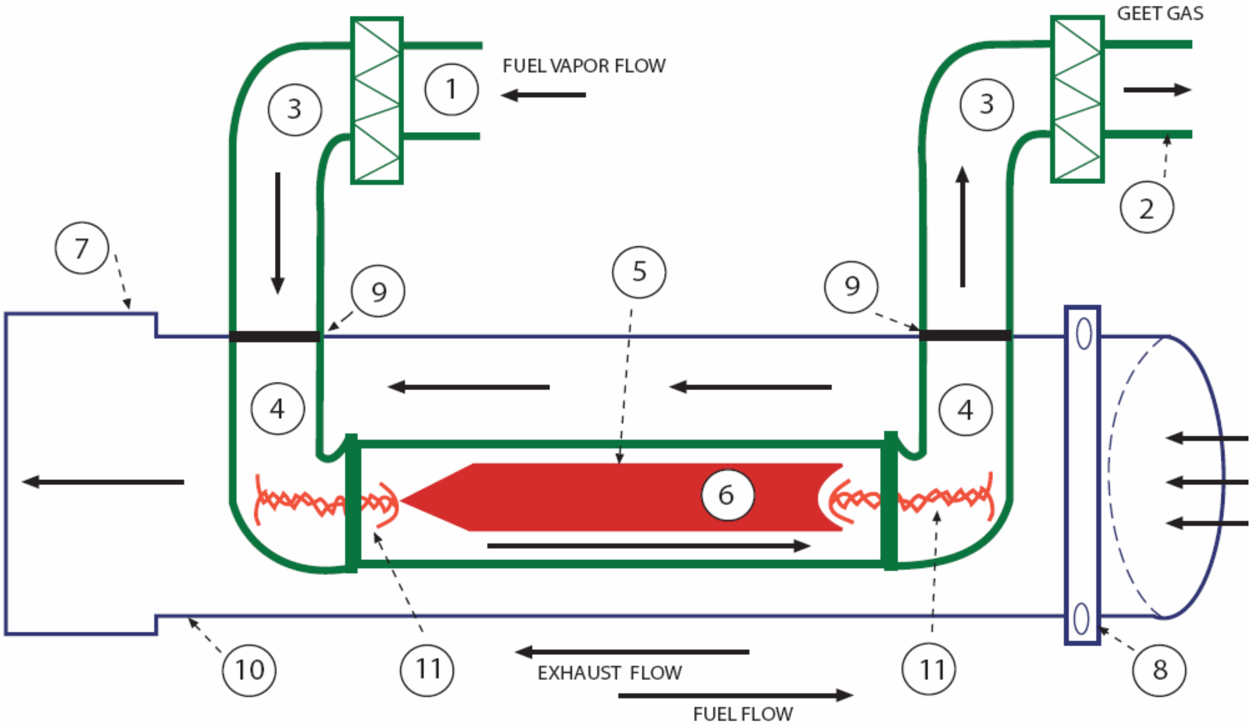


General Overview



A new Rod must be initialized by running the reactor for 35 minutes when horizontal, 17 minutes if vertical. If horizontal, the reactor needs to be placed so that the vapor is being pulled to the north during the initialization process. If the Reactor is vertical the Reactor should be set up so that the vapor is pulled upwards. In a horizontal reactor, the longest reactor rod you will ever use is 12 inches, in a vertical reactor it will be 6 inches. Install the longest rod, and initialize. Then use a small compass (where you can get within 1/4" of the needle) and read the signature on the rod, this is the length of Rod you need for your fuel.

3.14 (Pi) Spin over the length of the Reactor Rod The speed of the plasma doubles by the time it has gone half the distance of the Reactor Rod, then doubles again in the next 1/4, and in next 1/8 etc.



Large Engine Reaction Chamber

1. Copper or Stainless Steel(s.s.) tubing from bubbler

2. Copper or s.s. tubing to Air Management Valve

3. Brass compression or flare fitting for tubing

4. Black pipe elbow

5. Inner pipe of Reaction Chamber

6. Reaction Rod

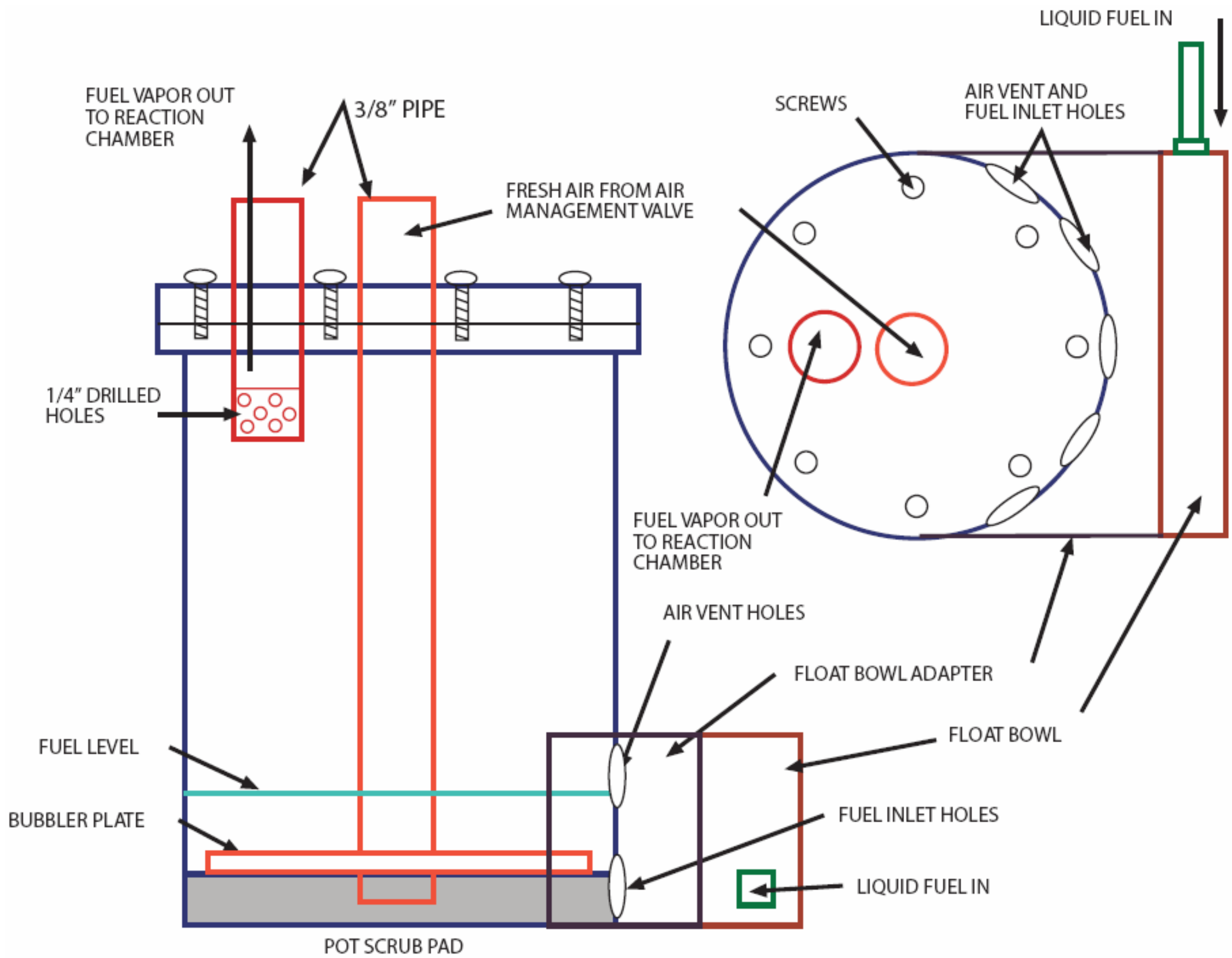
7. Flared end or flange to fit exhaust line

8. Flange to fit manifold side of exhaust line

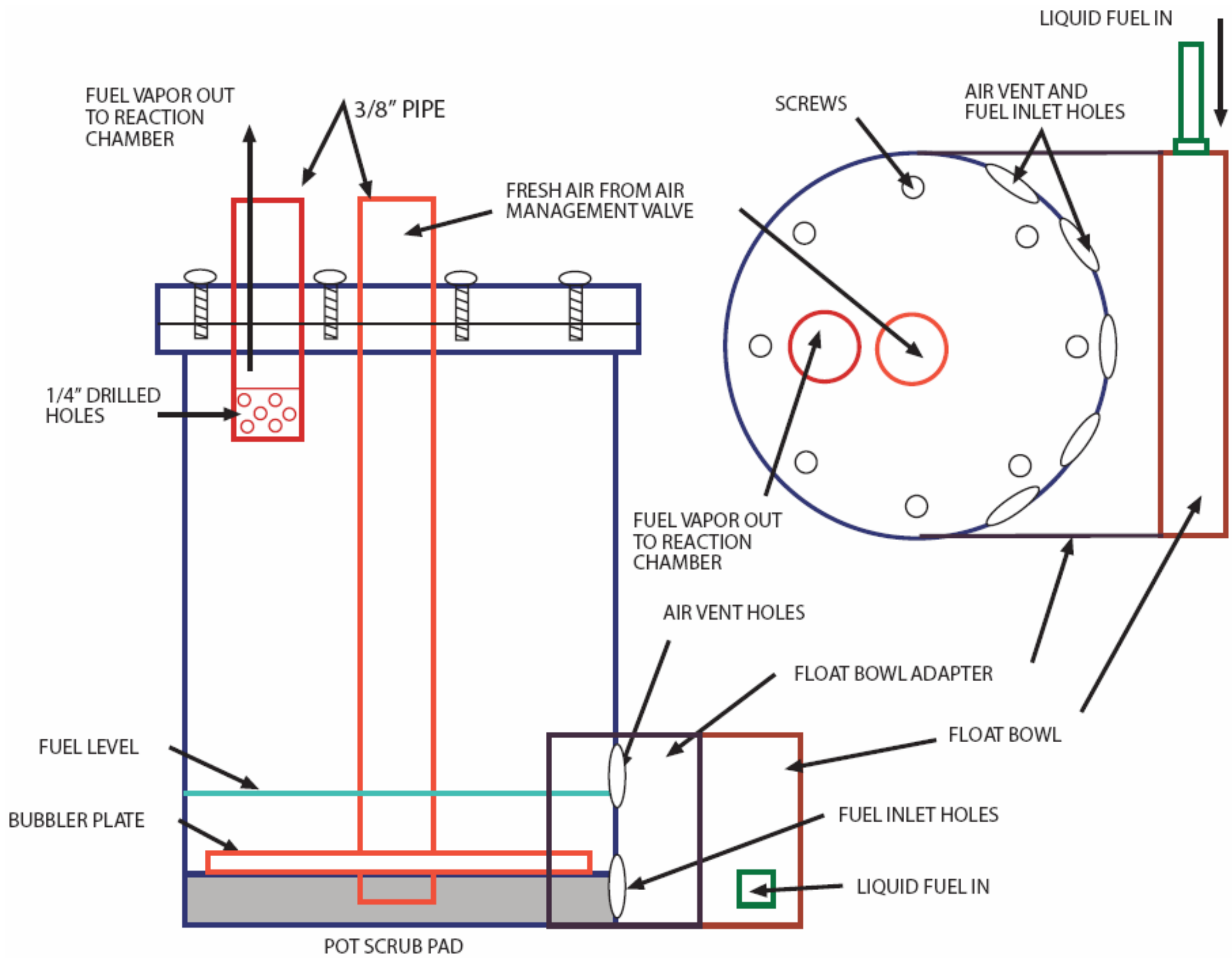
9. Weld to seal

10. Exhaust Pipe (1/2”-3/4” larger than inner reaction pipe)

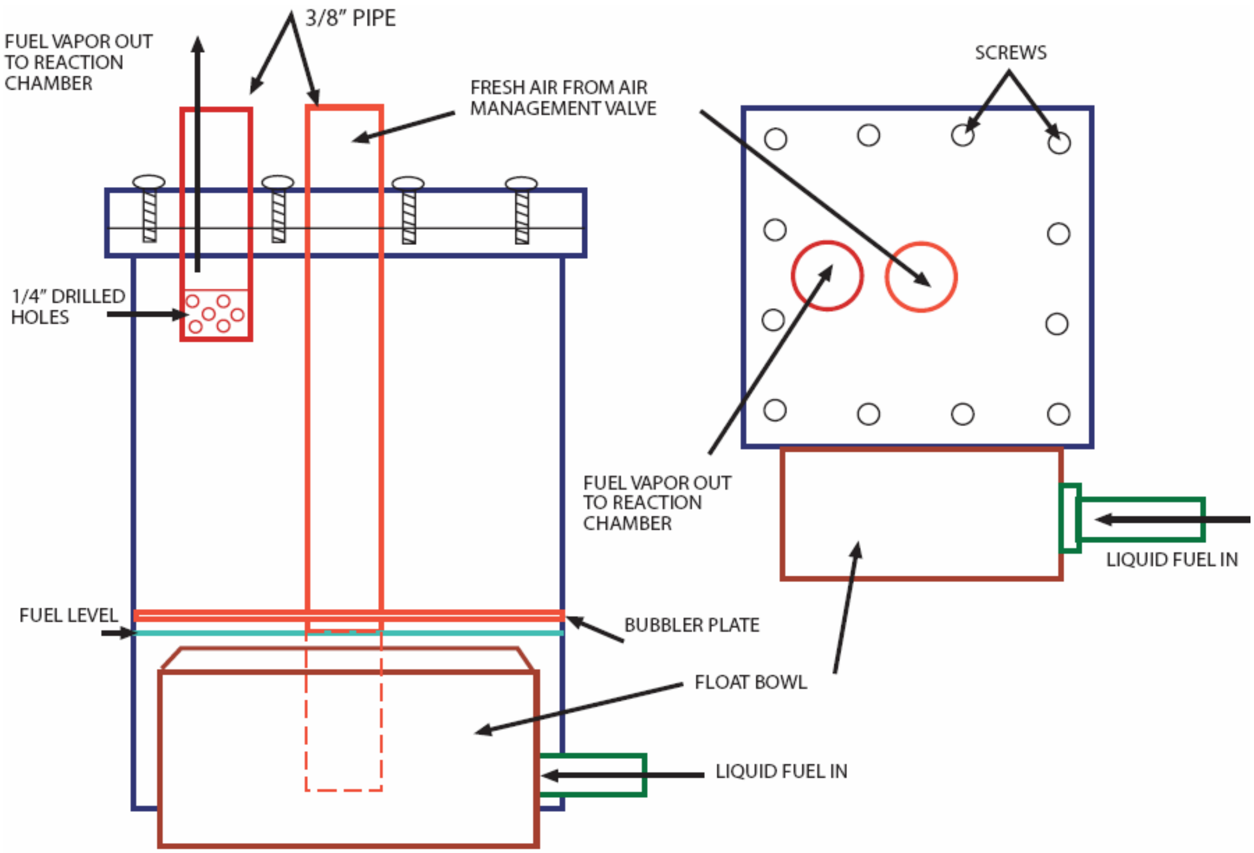
11. Twisted bailing wire or spring to keep rod out of elbows



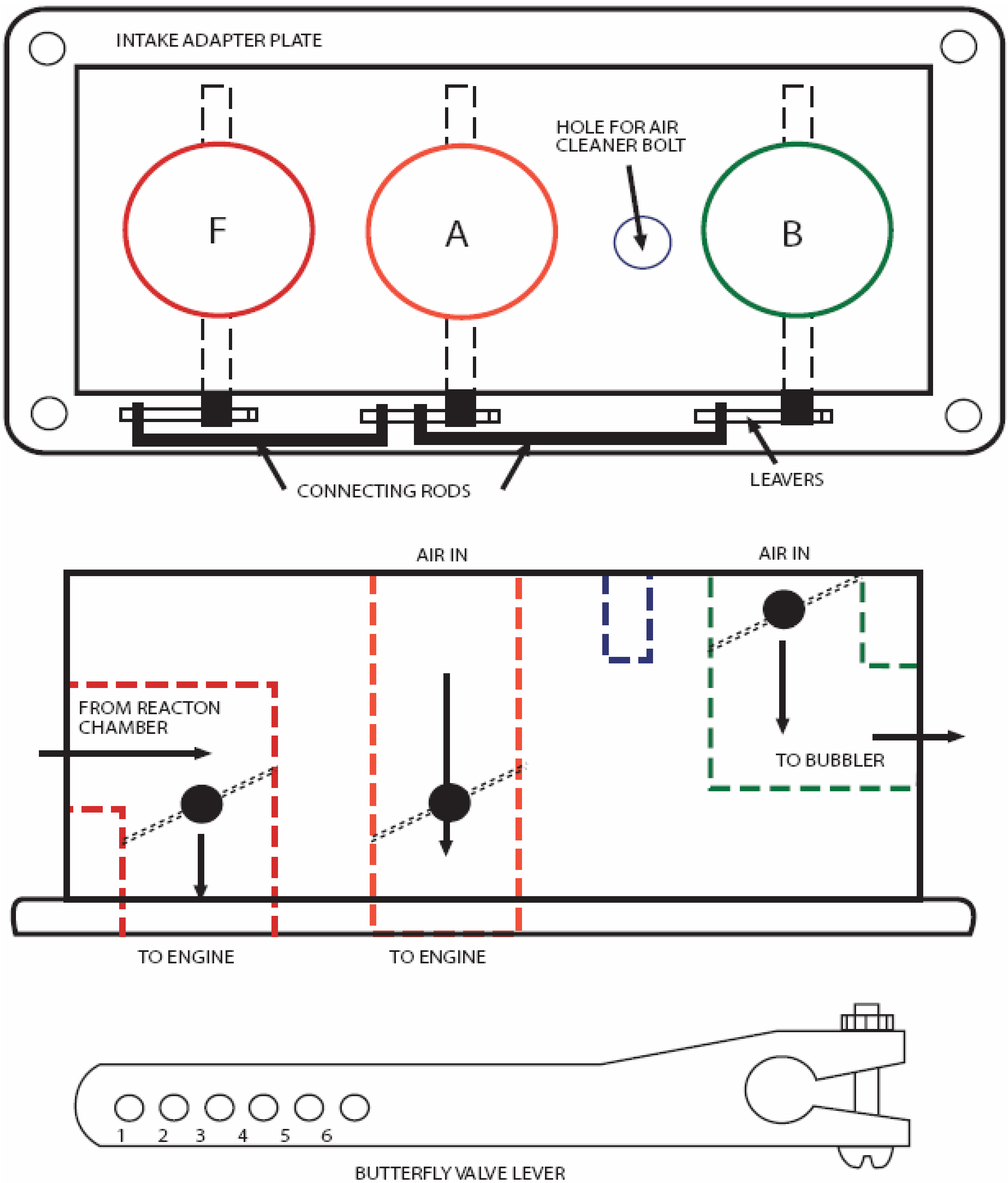
Round Bubbler



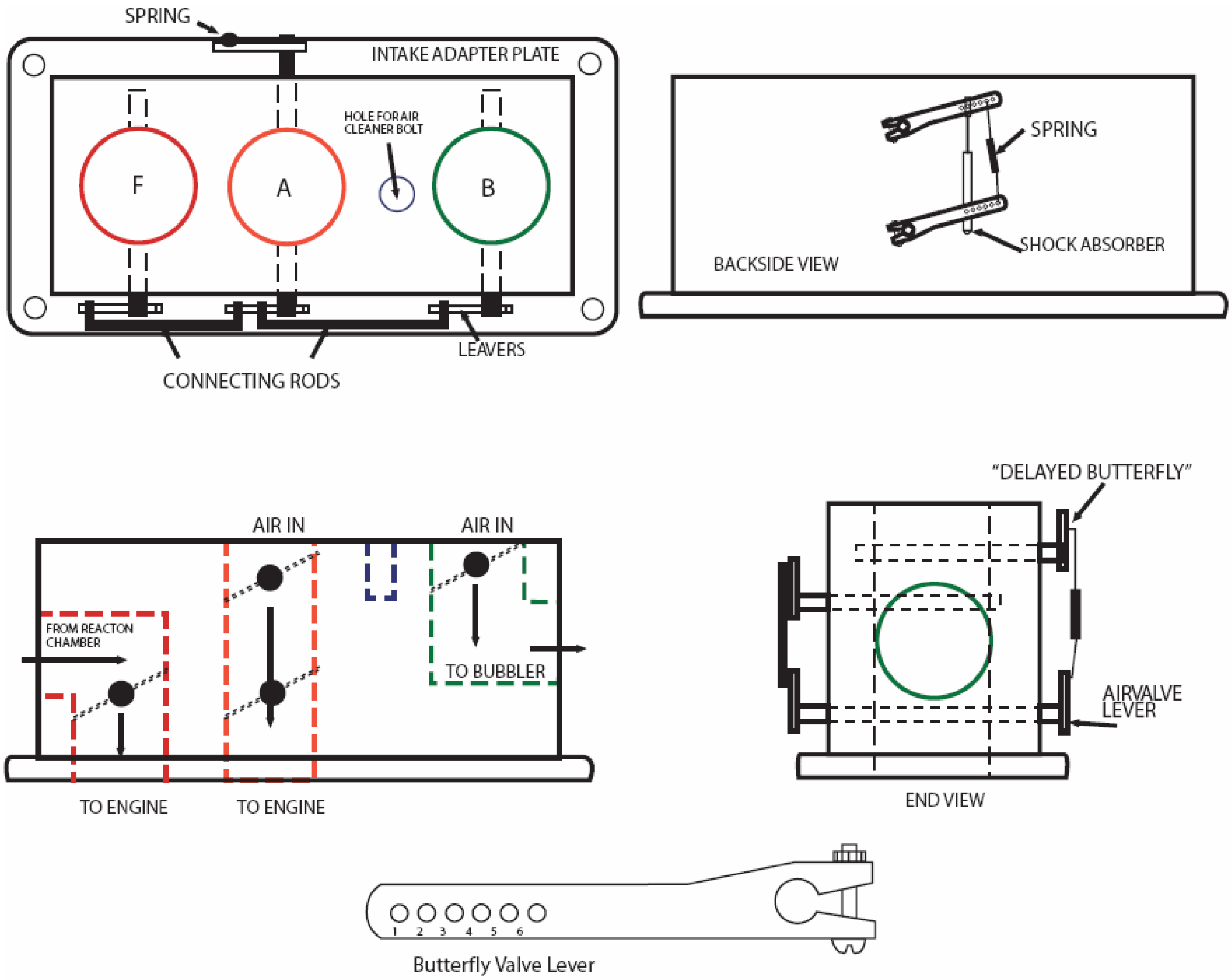
Round Bubbler



Square Bubbler



Air Management Valve



Large Engine Air Management Valve (delayed valve supplement)

Hilux GEET conversion construction by the Vortex heat Exchanger group.

[Hilux GEET conversion construction]

GEET Car by "geetuser" from the Vortex heat Exchanger group

It would be better for us to implement GEET on vehicles if we want to save on fuel and minimize pollution. Implementing GEET fully requires us to make that fuel mixing valves which majority of us don't have the resources to do it for now.

My proposal is to implement GEET as fuel supplement for the time being while we still don't have that fuel mixing valve at hand. 40% savings on diesel engines at City driving is very easy to achieve. 100% to 400% fuel efficiency on highway driving on gasoline engines is very possible and easy to achieve. You cannot find any gadget or fuel saving devices on the market that can do the same savings as geet. Construction is very simple, you can even follow the free internet plans for small engines and implement it in cars and you'll still get great results.

For those that don't have time to do experimentation, better start copying the internet plans, implement it on your cars and don't worry that it will not work for it will surely work when welded correctly with no leaks. We've implemented it on several engines, same 12" steel pipe (304 stainless steel for longer life, does not rust) with 12" steel rod inside held by springs for it not to rattle or cause noise, all placed inside the exhaust pipe. Rain water is used in a bubbler of different containers like 1 liter dextrose glass bottles, mason jar, plastic containers and even a vodka bottle. What is important here is you bubble the air through the water, let it pass through the reactor and connect it to the intake manifold below the throttle plate for greater vacuum during idle and cruising and that's it you have geet helping you save some fuel and engine oil.

I said engine oil since it will not get dirty easily when using GEET, thus, extend your engine oil life. With cleaner oil, your engine will also be clean which means you also extend its life. Things that you should do to make things work well:

Gas weld (not arc weld) the exhaust pipe joints and reactor joints. Once installed, initialized the reactor by idling your engine for 20 to 30 minutes while exhaust direction around the reactor is going south and water vapor from the bubbler is going north inside the reactor. North here is same direction where your compass will point to. Once initialized, you’re ready to go wherever direction you want.

Just fill your bubbler with water around 1/3 to 1/2 its total height with enough room above for water splashing. For stationary engines like generators regardless of size, GEET is very easy to implement. All you need are ball valves to control the air input to make it work. You'll be amazed with cool exhaust gases coming out of the exhaust pipe while no bad smell emissions. I've done it with a car engine configured as stationary engine, it's easy and it works.

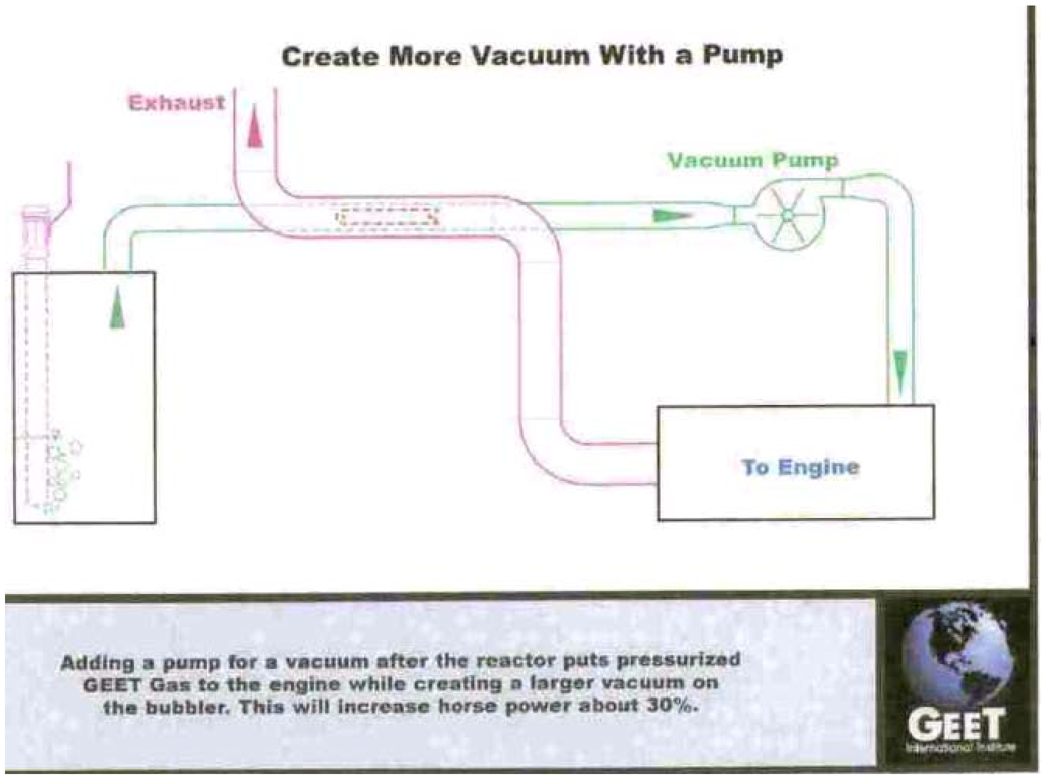
If you don't have time and space to plant trees, GEET is your way of cleaning the planet from air pollution. Let's start implementing GEET at home and with your friends and start implementing now, it will work and it will always work so stop asking the wrong question "does it work?", instead start asking yourself where to get those pipes and GEET rods that you can use to implement GEET. Find someone who can do welding for you if you can't weld, share GEET with him and enjoy fuel savings while saving your engine and the planet. Don't forget to donate some of your savings ($5, $10, or whatever you can) to help Paul Pantone, he still have a lot to do for humanity.

It is only from working together that we can save ourselves and our children's future, let’s start doing it now. More GEET users for 2009 and beyond. :)- geetuser -End

3 Additional Ways to Get Greater Efficiency

The following is courtesy of Naresh. If everything is built correctly so your reactor is really cracking the fuel and water then there are still at least 3 additional things I've seen to increase overall efficiency even further.

1)Put a suction pump in line with the reactor output to the engine intake manifold. Like here:



It says Paul Pantone got an additional 30% increase in horse power by using a suction pump. I was initially thinking a smog pump's plastic vanes would melt from the heat but more recently I'm starting to think that the additional vacuum will make the fuel and water vaporize more and that makes them get a colder faster before going into the reactor. So maybe even after heating up through the reformer, the cracked vapors might not be hot enough to melt a smog pump's vanes so I'm going to try this.

2. Restrict the exhaust flow just in the area where the rod is, by using an aerodynamic taper to a smaller size like is shown in the 3rd diagram on sheet 5 of this file:www.geetfriends.net/netdocs/geetro_rev.pdf.

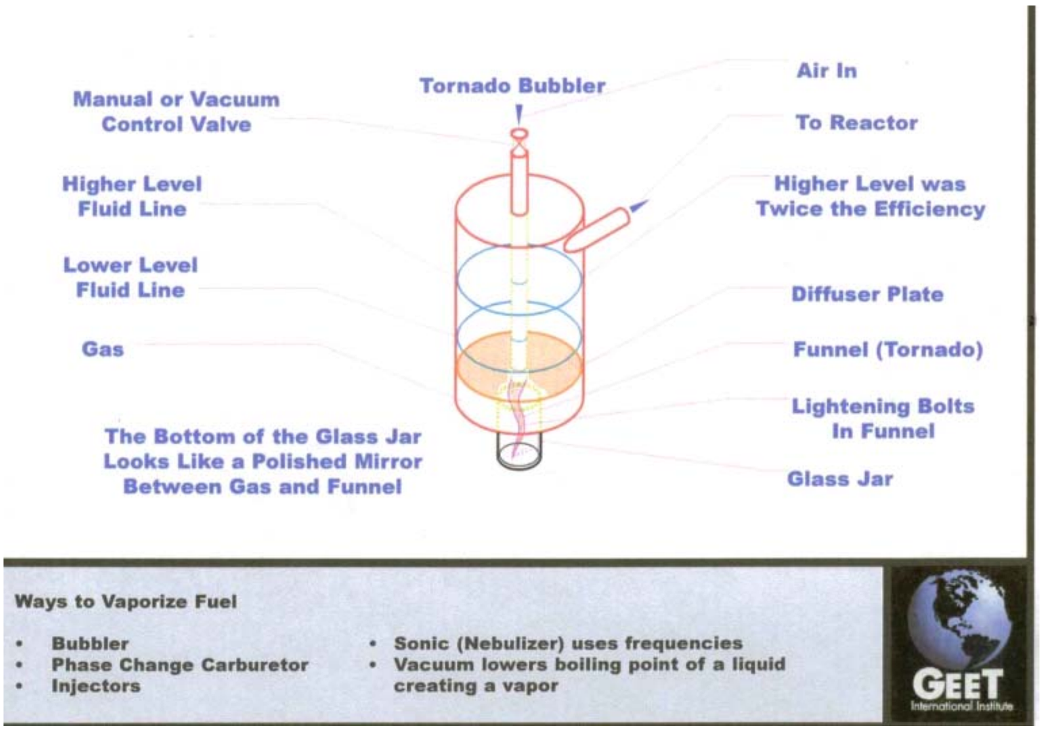
Paul Pantone got an additional 20-30% increase in horsepower from doing this.

3. Use multiple reactors in parallel, or at least with the inner pipe portion in parallel. Jean Chambrin's design is similar a reactor with multiple pipes in parallel and his patent says

he got it to run on only water even though he preferred to have some alcohol in with the water. Similarly, Bryishire at youtube.com says he got his truck working with 10% oil and 90% water by using 2 reactors with the exhaust in series and the inner pipes in parallel.

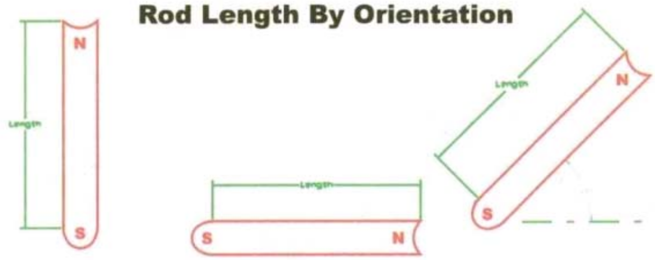
GEET Pantone group files courtesy of OregonHerbal

Tornado Bubbler



Rod Details

Rod Length By Orientation



FUEL	#1	#2	#3
Water	1.25" [32mm]	2.5" [64mm]	1.875" [48mm]
Gas	3.625" [92mm]	7.25" [184mm]	5.4375" [138mm]
Diesel	4.5" [115mm]	9" [229mm]	6.75" [171mm]
Crude Oil	6" [156mm]	12" [305mm]	9" [229mm]

Rod Clearance by Fuel Type:

Small chain fuels require smaller clearances.

Larger chain fuels require larger clearances.

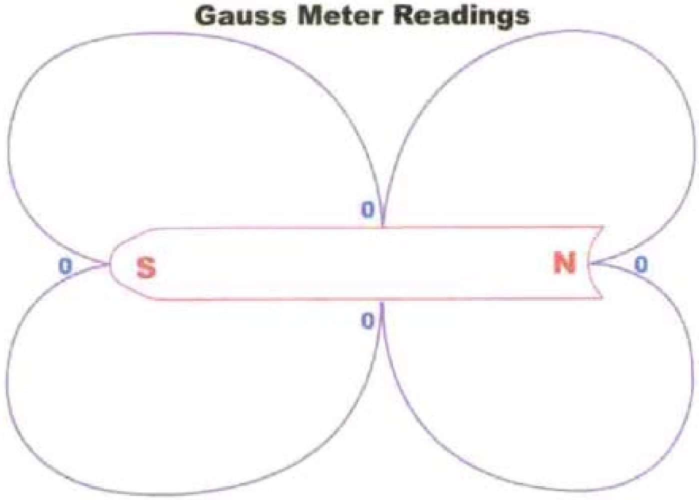
0.03125" [0.8mm]
0.0625" [1.6mm]
0.09375" [2.4mm]
0.125" [3.2mm]

1. Orientation up requires fixed length by fuel type.
2. Orientation horizontal in a fixed North-South direction requires fixed length by fuel type.
3. Orientation at angle off of vertical requires adjustment to length by fuel type.

GEET International Institute

Gauss Field

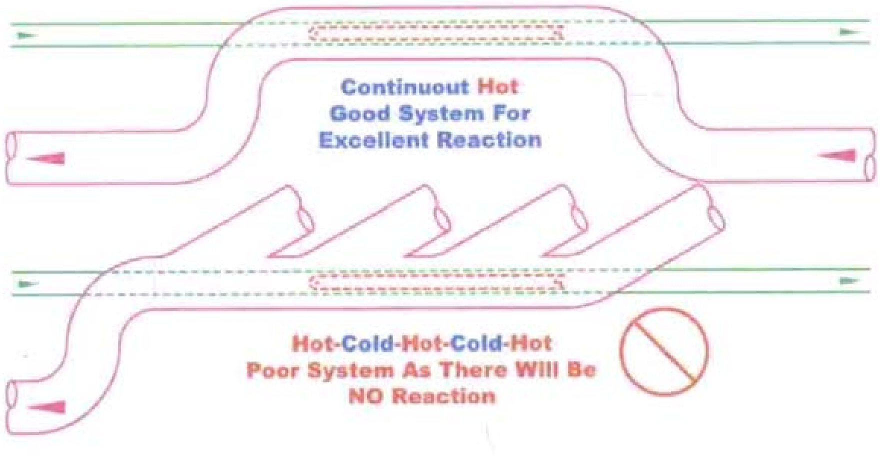
Gauss Meter Readings



Gauss readings are 0 at the midpoints of the sides and ends. When any GEET generator unit is running the Gauss reading is about 20 times stronger closer to the unit than when it is not running. This effect gradually deteriorates as you move away from the unit until at about 4 1/2 feet the Gauss reading is 0.

23 Reactor Details

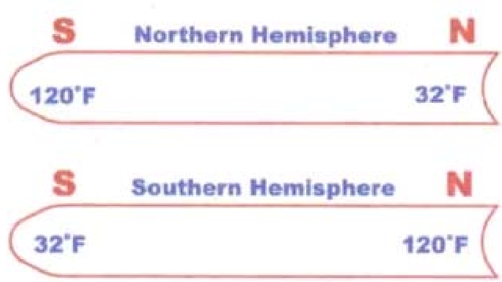
Heat In Exhaust Required For Reaction



Continuous heat from one cylinder of a V8 will have a good reaction and make enough fuel in the GEET reaction chamber to run the entire engine.

11 Reverse Hemisphere Spins

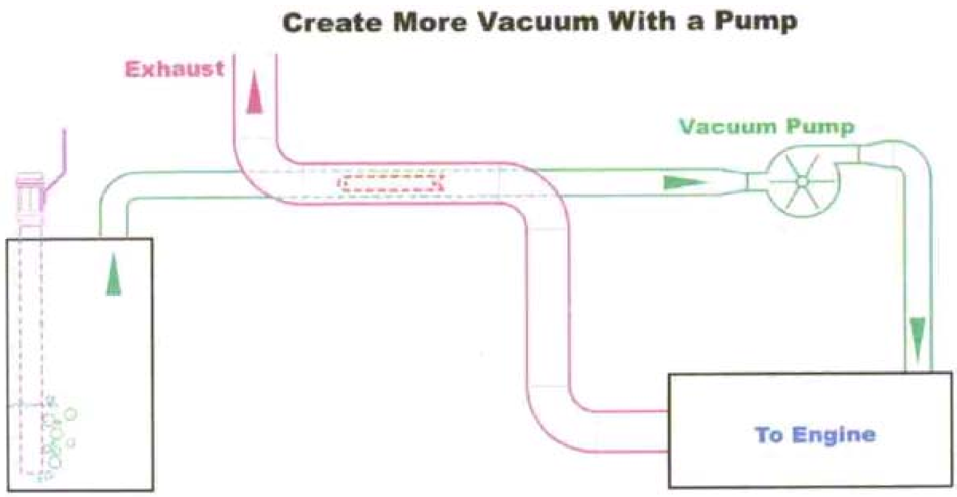
The University of Berlin discovered 32°F at the center of the North end of the rod and 120°F at the South.



Additional proof of Double Axis and electron movement & electron absorbtion.

This phenomenon has only been tested in horizontal positions in both hemispheres and they will reverse the temperature field. Will this be the case when tested in a vertical position?

19 More Horsepower



Adding a pump for a vacuum after the reactor puts pressurized GEET Gas to the engine while creating a larger vacuum on the bubbler. This will increase horse power about 30%.

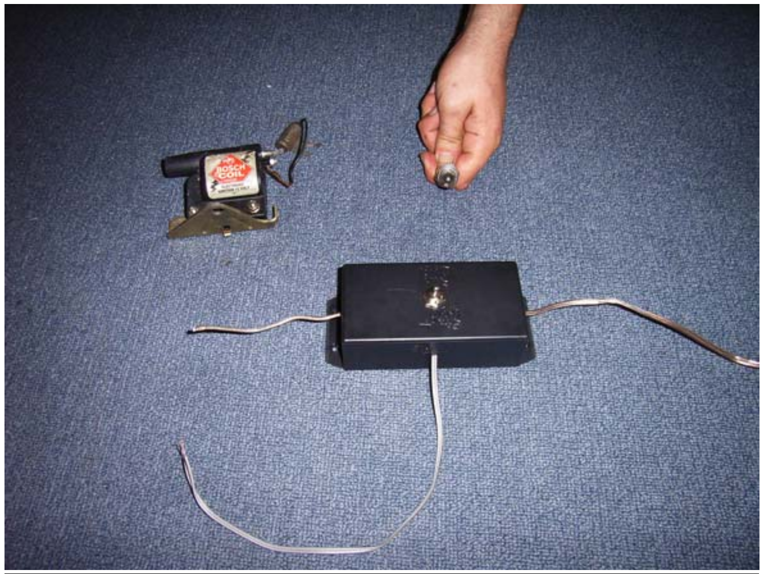
GEET institute files- File-down load –coming soon

Vortex Heat Exchanger Group Files

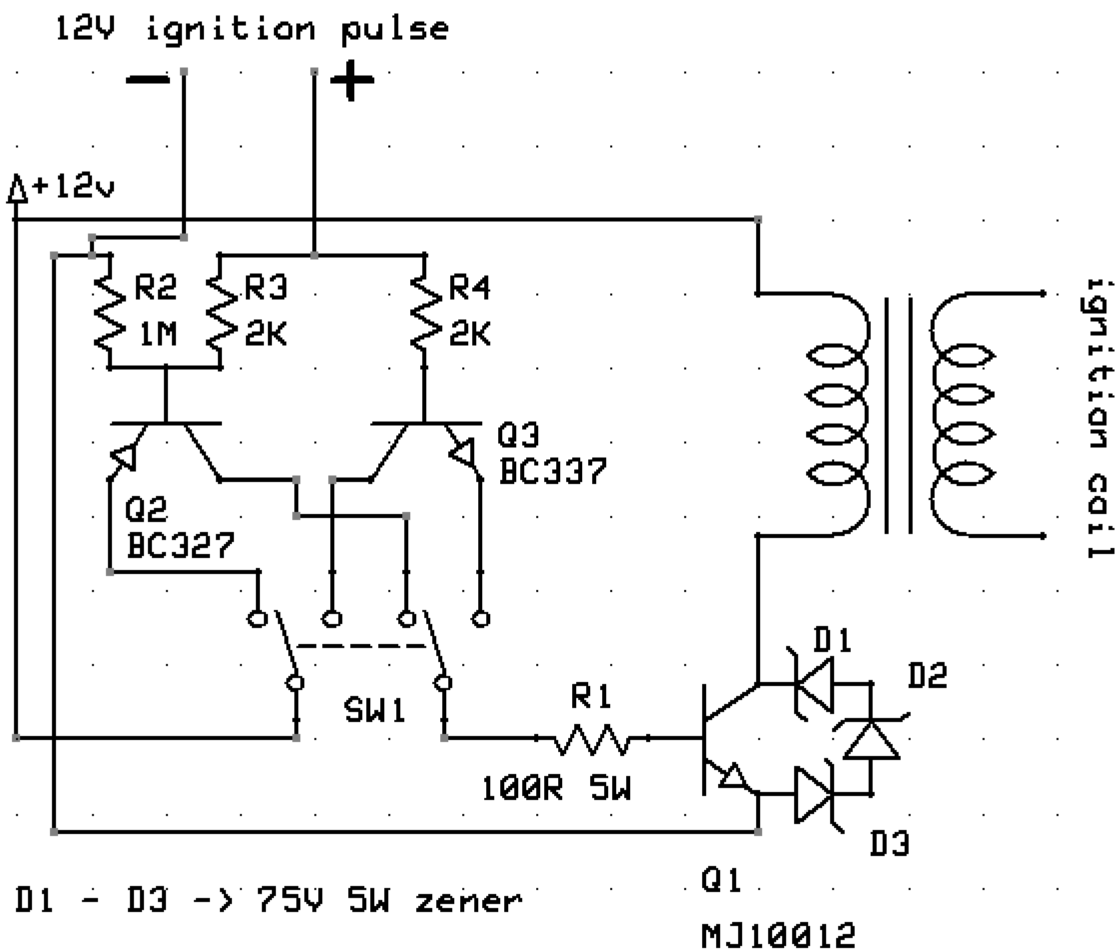
The following file is courtesy of the vortex heat exchanger group**. Naresh has provided the complete reference of beginner instructions to research material and more. This is a 170mb file which has everything need to get started and to understand the complete fuel reforming process. PLEAE READ THROUGH THESE FILES BEFORE STARTING A GEET. The file is zipped and contains an off line viewer to view the files.**

Vortex heat exchanger group's Messages down load

GEET ignition boost system



Panacea's GEET boost ignition system for a 6.5HP genset



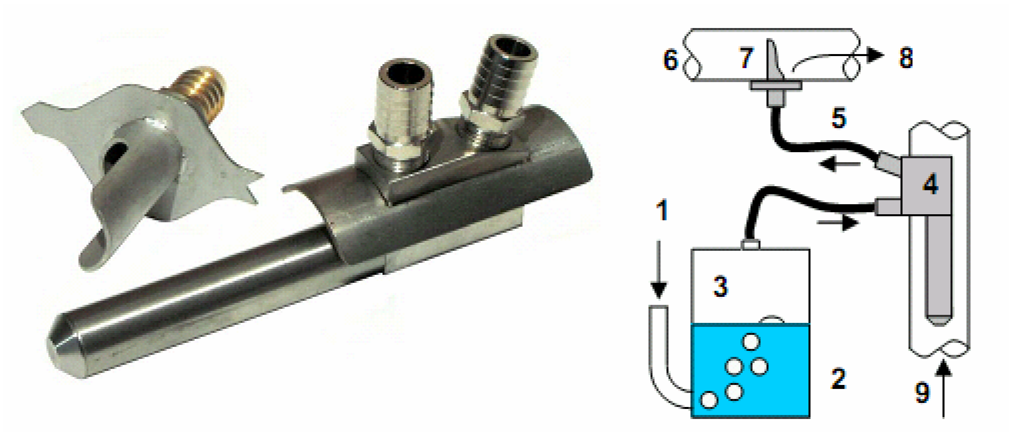
Pantone engine in "SYSTEM G"

Please note –Panacea prefers to present only Paul Pantone's original designs. And not confuse new comers with other designs that do not necessarily create the same electromagnetic phenomena inside as Paul's designs did. If you present new comers with all the other variations by other people then if they don't actually fuel reform then people think GEET does not work when in fact it is only everybody else's variations of GEET that are not always working. **The SPAD and G pantone system are not intended to be a substitute for the GEET!.**

It is our understanding is that an organization called the Association APTE has an agreement with Paul Pantone to use the technology. The SPAD is a hybrid of the Pantone technology and strictly speaking is more of a bubbler than the original GEET tech. When Panacea talks with the SPAD company, we deal with David Dieulle and Christophe Tardy who are executives of the SPAD company Hipnow and are both members of APTE an engineering group.

Retrokit

This SPAD is a compact performance optimizer and works with water .For naturally aspirated or turbo diesel engines – all horse powers. **For use in Agriculture, Industry, Public works, Marine (various), Irrigation, etc.**



Performance is said to be a 10 to 50 % reduction in fuel consumption, depending on operating conditions. This device reduces pollution, Increases engine life and reduces noise levels. NB : non for use with on-road vehicles.

How it works-Ambient air (1) is drawn into the bubbler and mixes with water (2) to make humid air (3). This aerosol is transformed by the reactor(s) (4) into a synthetic gas (5) that mixes with air coming from the air filter (6) via the diffuser (7) toward the engine air intake (8) or turbo. Fuel combustion is improved and fuel consumption drops. Cleaner (less polluting) exhaust gas (9) provides the energy necessary for the transformation that occurs in the reactor (4).

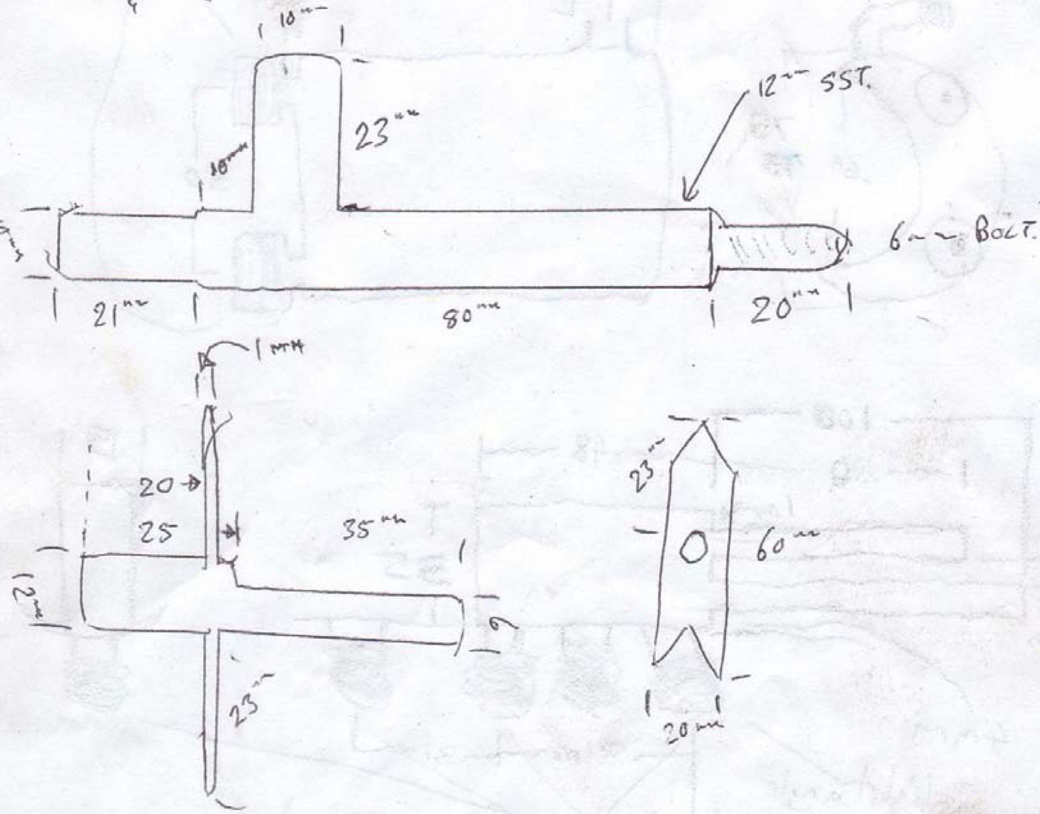
	E1 - 45	E2 - 70	E3 - 90	E4 - 100
Weight (g)	450	750	1400	1500
Ext. dimension. (mm)	174x75x45	193x55x50	195x90x55	197x89x50
Number of reactors	1	2	3	4
Diffuser	DIF1	DIF2	DIF2 ou DIF3	DIF3
Supply and exit	3/8" barb M	1/2" NPT F	1/2" NPT F	3/4 " NPT F
Attachment	2 hose clamps	1 M10 bolt	1 M10 bolt	1 M10 bolt
Holes in exhaust pipe	rectangular 20x70	2 x Ø 26 + Ø 11	2 x Ø 26 + Ø 11	2 x Ø 32 + Ø 11
Center distance (mm)	-	66	66	64
Exterior material	Stainless steel	Stainless steel	Stainless steel	Stainless steel
Minimum exhaust pipe diameter (mm)	45	70	93	97

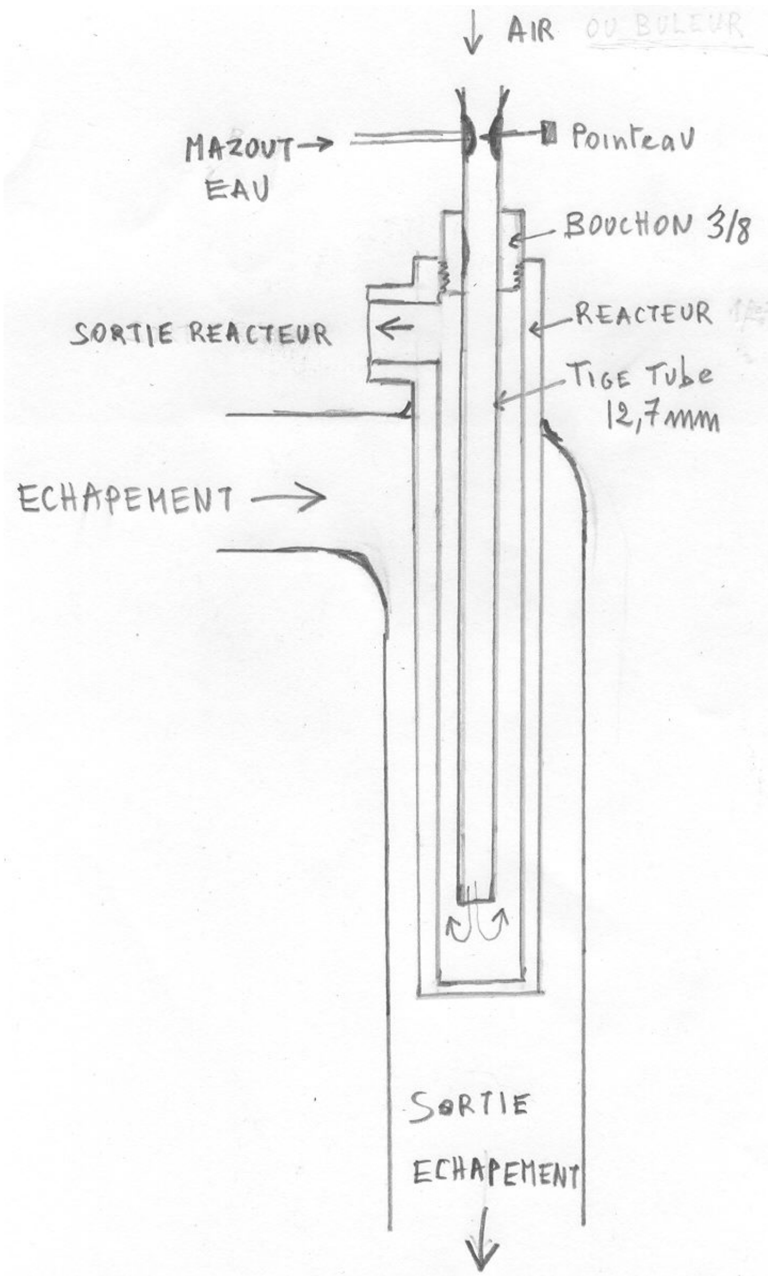
[Retrokit installation - English]

Nano Kit

NANO RETROKIT
F08A000457

10mm STAINLESS STEEL TUBE (SST)
12mm " " " "





A cheap nano size GEET Reactor could be made from drilling holes in the top and bottom of a horizontal exhaust pipe and installing vertically a 1/8 inch NPT pipe threaded on the ends or maybe even threaded the whole length. Use 3/8 stainless washers and 1/8NPT low profile nuts at top and bottom to hold tight in the exhaust pipe and then fittings go on top and bottom after the nuts.

Then install a stainless rod. A stainless 1/8 pipe might be best also rather than typical 1/8 inch NPT lamp pipe. Then a fitting at the bottom that goes to tubing to a water bubbler with electrolyte. Let air into the bubbler through a small vacuum valve set to get some vacuum maintained at the reactor pipe input. Then send the top of the reactor tube to the intake manifold vacuum.

Ideas how the the rod be placed inside the pipe as it will it float/spin/rest.

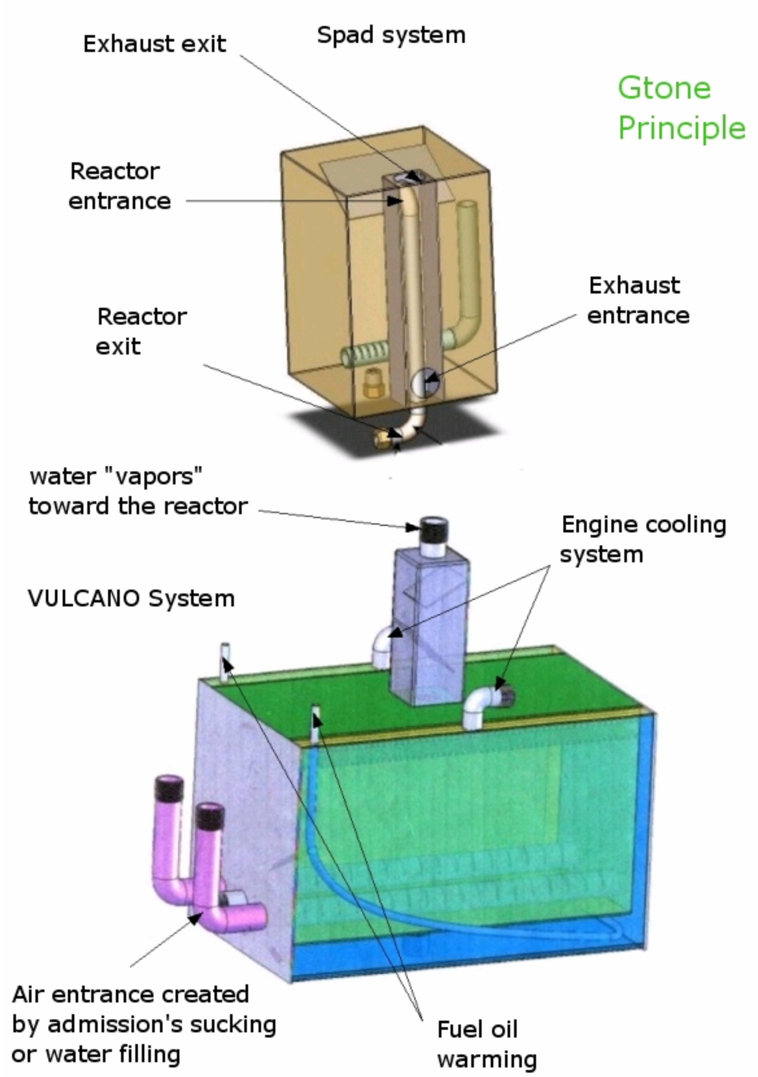
This is difficult at a small scale but the rear stop should be rigid dead center with a point for the rod rear dimple to pivot and spin on. There needs to be something to hold the rod higher than where the vapors come in at the bottom. The front stop needs to be centered well enough that the rod doesn't fall to the side and jam to the side of the front stop. That is a problem I have been having with a small GEET I've been trying to get working. I might try threading a hole through the end plugs and put rigid threaded rod with a jam nut to hold it tight. But according to Paul Pantone the end stops need to be non-magnetic.

The front stop can be big diameter if it doesn’t go up past the side port where the vapors first enter. The main thing is the vapors need to hit the nose not the side of the rod as they come in to help get the rod up floating in the wind or spinning against the rear stop.

There is no spinning rod inside the Hypnow kits . The rod might be rigidly held with no chance to spin or move forward or aft. If both the inner pipe and rod surface are mirror smooth then maybe it can still work almost as well even if it doesn't spin. It is a matter of how much drag there is on the vortexing vapors. A spinning rod lowers the drag on the vapors so they can vortex faster.

S.P.A.D System

Gtone



The principle in a few words: The most simple and popular version known today differs from the Panton's assembling; it was discovered by a farmer. It is about boosting/doping admission gas with 'water gas' produced continuously and without any storage so there's nothing to fear from the engine.

In order to simplify, we talk about 'water doping' This version is easier because it does not bubble fuel and water together Only one reactor is indispensable. The gas produced - either by the one-way exhaust heat (bubbling + reaction), or by the heat from the engine's cooling system (bubbler) - and the one produced by the exhaust (reactor), is sucked up to a place in the admission circuit as an air complement.

What is the result? After a few minutes' warm up to start up the reactor Gtone :

1- combustion is far better

2- the engine is quieter
3- The consumption is reduced by 30% to 80% in most cases and divided by two or three in the case of some tractors. To get a similar power, one have to press less on the accelerator pedal, and shift the gear less
4- Pollution is reduced in remarkable proportions : 90%-95%, there is no dark smoke mark on the white tissue placed at the end of the exhaust pipe
5- The engine lubrication oil stays clean longer

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SPAD how to PDF

SPAD Promo videos

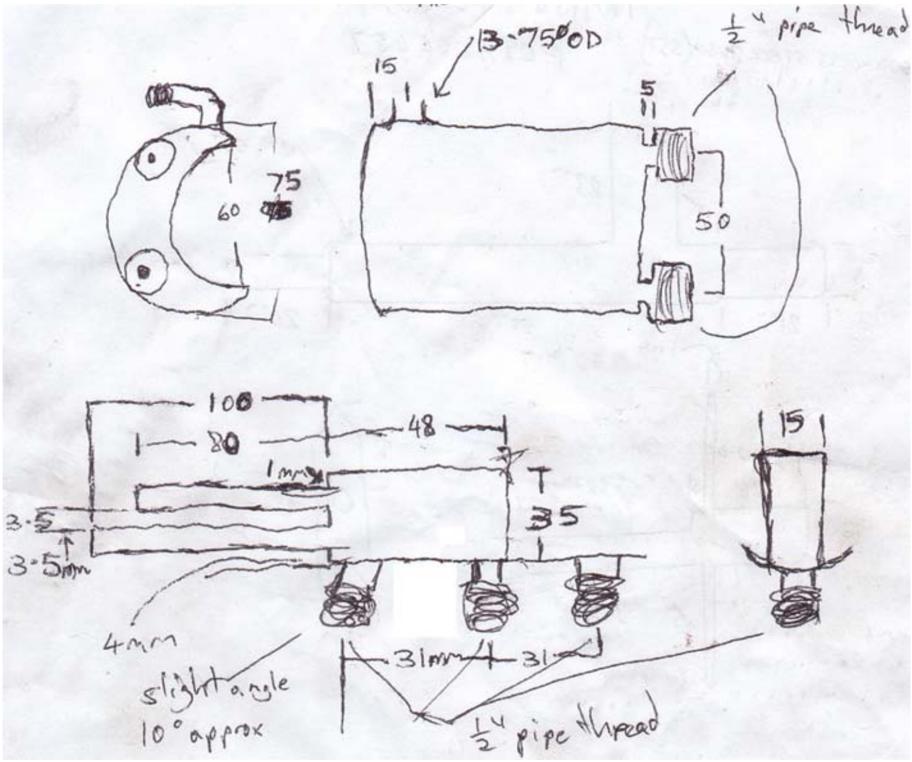
http://fr.youtube.com/hypnow

More information can be found at http://www.hypnow.fr/

The following is a homemade Pantone system "G" in combination with chainsaw carburetor to maintain the vacuum in the intake manifold on ford 2,0 i DOHC petrol engine.

System "G" on Petrol engine

Ecopra Kit



More information can be found at ECOPRA

Ecopra Videos

Youtube channel : http://www.youtube.com/user/davidbfilm

Video has english subtitles : http://www.youtube.com/watch?v=yBOVExJuC7w

Easy installation :http://www.youtube.com/watch?v=aZCmTPn4fpg&NR=1

Related Patents and Information Patents

Original US Patent # 5,794,601 US Cl. 123/538 ~ August 18, 1998

Fuel Pre-treater Apparatus and Method -Paul Pantone

Abstract ~ A novel fuel pretreated apparatus and method for pre treating an alternate fuel to render it usable as the fuel source for fuel burning equipment such as internal combustion engines, furnaces, boilers, and turbines, includes a volatilization chamber into which the alternate fuel is received. An exhaust plenum may enclose the volatilization chamber so that thermal energy supplied by exhaust from the fuel burning equipment can be used to help volatilize the alternate fuel. A bypass stream of exhaust may be diverted through the alternate fuel in the volatilization chamber to help in volatilizing the alternate fuel and help carry the volatilized fuel through a heated reactor prior to its being introduced into the fuel burning equipment. The reactor is preferably interposed in the exhaust conduit and is formed by a reactor tube having a reactor rod mounted coaxially therein in spaced relationship. The exhaust passing through the exhaust conduit provides thermal energy to the reactor to pre treat the alternate fuel.

Current U.S. Class: 123/538; 123/557; 123/575
Intern'l Class: F02M 031/18
Field of Search: 123/538,557,575,1 A,568

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5,443,052 ~ Aug., 1995 ~ Aslin ~ 123/575.

Other References ~

PCT Publication No. WO 96/14501--May 17, 1996.
Marin Independent Journal, Nov. 1992 "In quest of perfect engine".
Exotic Research Report, vol. 1 #2 Apr.-Jun. 1996, pp. 23-26.

Description ~

BACKGROUND OF THE INVENTION

1. *Field -*

This invention relates to fuel burning equipment and, more particularly, to a novel fuel pretreater apparatus and method for making it possible for such fuel burning equipment to utilize as a fuel a material not otherwise considered suitable as a fuel for such equipment.

2. State of the Art

Most fuel burning equipment in use today is designed to burn a particular fuel. For example, internal combustion engines are designed to burn gasoline or diesel fuel, furnaces and boilers to burn natural gas, oil, or coal, and turbines to burn kerosene or jet fuel. Fuels or other materials other than the fuels for which the equipment is designed to burn cannot generally be used in such equipment.

For example, in internal combustion engines, particularly in light of the extreme sophistication of many current engines, not only for fuel economy but also for reduction in the emitted pollutants, great care is taken in the selection of the fuel grade particularly as to its quality prior to its introduction into the internal combustion engine. One does not consider crude oil or recycled materials such as used motor oil, cleaning solvents, paint thinner, alcohol, and the like, as a suitable fuel source for an internal combustion engine. Further such materials would not be considered suitable fuels for furnaces, boilers, turbines, or most other fuel burning equipment. In addition, one would not consider using such fuels if contaminated by water, nor would one consider using non fuels such as used battery acid or other waste products as fuels for fuel burning equipment.

SUMMARY OF THE INVENTION

The present invention is a novel fuel pretreater apparatus and method for fuel burning equipment. This novel fuel pretreater enables the fuel burning equipment to utilize as fuels combustible products selected from material such as crude oil or recycled materials such as motor oils, paint thinners, solvents, alcohols, and the like and noncombustible products such as battery acid. Any substance that can be preheated and then burned in the fuel burning equipment will be referred to as alternate fuel. This alternate fuel is introduced as a liquid into a volatilization chamber. The volatilization chamber may be heated to aid in volatilization and in most cases may be advantageously heated by thermal energy from the exhaust in the exhaust conduit of the fuel burning equipment. A portion of the exhaust may even be bubbled through the alternate fuel to assist in the volatilization of the alternate fuel. The fuel vapor produced in the volatilization chamber is drawn through a heated thermal pretreater. The thermal pretreater may be mounted, preferably concentrically, inside the exhaust conduit to be heated by the exhaust gases. The thermal pretreater serves as a reactor and is configured as a reactor tube having a reactor rod mounted, preferably concentrically, therein with a reduced annular space surrounding the rod. The volatilized alternate fuel passes through this annular space where it is subjected to thermal pretreatment prior to being introduced into the intake system of the fuel burning equipment.

THE DRAWINGS

The best mode presently contemplated for carrying out the invention is illustrated in the accompanying drawings, in which:

FIG. 1 is a block diagram of a basic fuel pretreating apparatus of this invention;

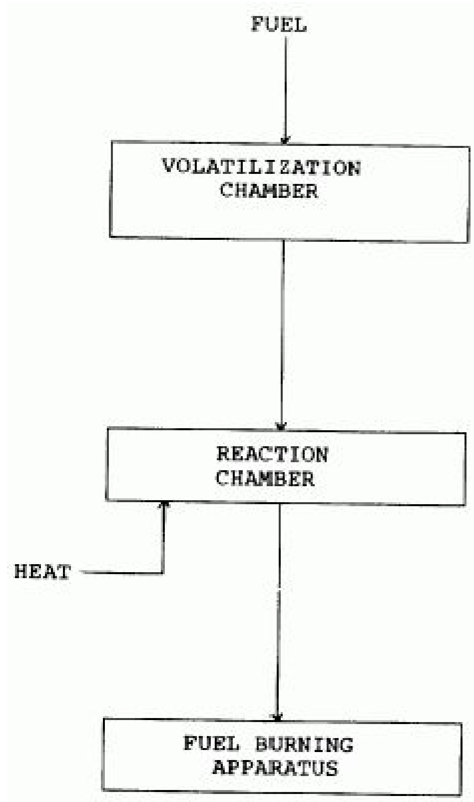


FIG. 2, a schematic flow diagram of the novel fuel pretreater apparatus of this invention shown in the environment of an internal combustion engine; and

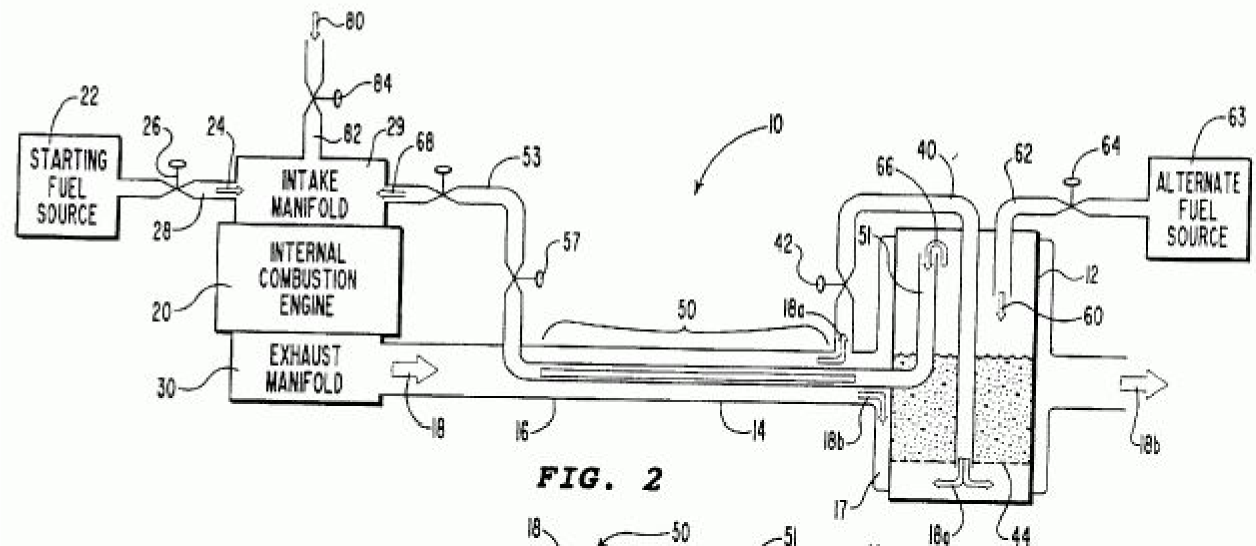


FIG. 2

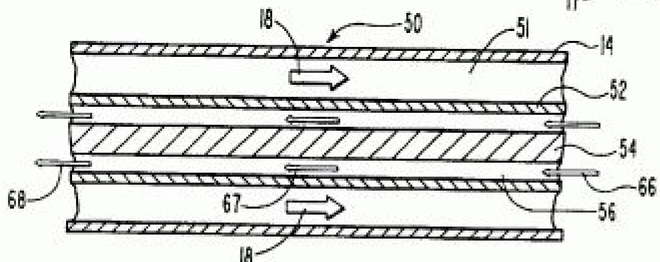


FIG. 3

FIG. 3, an enlarged cross-sectional view of a schematic of the reactor portion of the fuel pretreater of FIG. 2.

DETAILED DESCRIPTION OF THE ILLUSTRATED EMBODIMENT

The invention is best understood from the following description and the appended claims taken in conjunction with the accompanying drawings wherein like parts are designated by like numerals throughout.

The present invention is a unique apparatus and method for pretreating materials to be used as fuel for fuel burning equipment such as internal combustion engines, furnaces, boilers, turbines, etc. The pretreatment makes it possible for the fuel burning equipment to utilize as its fuel source fuels or other materials that are generally considered as not being suitable fuels for such fuel burning equipment. These alternate fuels include almost any liquid hydrocarbon such as crude oil or recycled material such as motor oil, solvents, paint thinners, and various alcohols, to name several. These alternate fuels may even be contaminated with water or may be material such as used battery acid which is not considered combustible or a fuel. Importantly, as shown in FIG. 1, the alternate fuel is volatilized in a volatilization chamber and is then subjected to a high temperature environment in a heated reaction chamber prior to its being introduced into the intake system of the fuel burning equipment. The reaction chamber provides a heated reaction zone with a reaction rod therein about which the fuel flows. It is this flow through the heated reaction zone about the reaction rod which makes the fuel suitable for burning in the fuel burning equipment. In most cases, since the fuel burning equipment involved will produce high temperature exhaust gases, in order to save energy, the heating for the reaction chamber will be provided by the exhaust gases from the fuel burning equipment. The reaction chamber will thus usually be positioned in the exhaust conduit, whether an exhaust pipe, flue, chimney, etc., leading from the fuel burning equipment. It is believed important that the fuel flow through the reaction chamber be opposite the flow of exhaust gas in the exhaust conduit so that the most intense heating of the reaction chamber is at the end thereof where the fuel exits the reaction chamber. Currently, it is not known precisely what happens to the volatilized alternate fuel in this high temperature environment although one speculation is that the larger molecules are broken down into smaller molecular subunits of the heavy molecules.

In any event, I have found, for example, that I am able to satisfactorily operate an internal combustion engine using as my fuel source materials generally considered to be totally unsuitable as fuels for an internal combustion engine. For example, in one experimental run I was able to successfully operate an internal combustion engine using recycled motor oil. In another experimental run I was able to operate the internal combustion engine using crude oil as my sole fuel source. In yet another run I was able to use waste battery acid as my sole fuel source.

However, I should state at this point that when the reaction chamber is heated by exhaust gases from the engine, in order to generate sufficient thermal energy necessary to volatilize the alternate fuel in the volatilization chamber, it is necessary to operate the internal combustion engine initially using ordinary gasoline. This step is necessary since, absent my unique pretreatment process, it is impossible to operate an internal combustion engine with the alternate fuels that I am using. Accordingly, the internal combustion engine is started and operated for an initial period until sufficient thermal energy has been generated in order to initiate the volatilization and the pretreatment processes. Once these processes are self sustaining, the fuel system is switched over from the gasoline system to the alternate fuel system. The internal combustion engine continues to operate for as long as the alternate fuel is supplied or until the internal combustion engine is switched off.

Similarly, with other fuel burning equipment, when the reaction chamber is positioned in the exhaust conduit, conventional fuels are supplied to the equipment upon start up and until sufficient thermal energy is supplied to the reaction chamber to produce fuel usable in the equipment from the alternate fuel.

The invention will be illustrated and described in detail with respect to an embodiment thereof for use with an internal combustion engine. Referring now to FIG. 2, the novel fuel pretreater apparatus of this invention is shown generally at 10 and includes a volatilization chamber 12 and a fuel pretreater section 14 incorporated into an exhaust conduit 16. Volatilization chamber 12 is enclosed in an exhaust plenum 17 through which a stream of exhaust 18 passes. Exhaust 18 is produced by an internal combustion engine 20 which can be any suitable internal combustion engine ranging in size from a small, one-cylinder internal combustion engine to a large, multi cylinder internal combustion engine. Internal combustion engine 20 is shown herein schematically particularly since no claim is made to an internal combustion engine, per se, only to the novel fuel pretreater apparatus 10 shown and claimed herein.

Internal combustion engine 20 includes a fuel tank 22 which supplies a starting fuel 24 and has a valve 26 for controlling the flow of fuel 24 through a fuel line 28 into an intake manifold 29. Fuel 24 enters internal combustion engine 20 through an intake manifold 29 either through carburetion or fuel injection (not shown), both of which are conventional systems for introducing fuel 24 into internal combustion engine 20 and are, therefore, not shown herein but only indicated schematically through the depiction of intake manifold 29. Fuel 24 is ordinary gasoline and provides the necessary starting fuel for internal combustion engine 20 until sufficient thermal energy has been produced in order to sustain the operation of volatilization chamber 12 and pretreater section 14. Thereafter, valve 26 is closed and internal combustion engine 20 is operated as will be discussed more fully hereinafter. Internal combustion engine 20 produces exhaust 18 which is collected from internal combustion engine 20 by an exhaust manifold 30. Exhaust 18 is then directed through exhaust conduit 16 into fuel pretreater 10 where it provides the necessary thermal energy for the operation of fuel pretreater 10.

Exhaust 18b represents a portion of exhaust 18 and passes through plenum chamber 17 surrounding volatilization chamber 12 prior to exiting exhaust conduit 16. Exhaust 18b represents the residual portion of exhaust 18 since a bypass 40 diverts a portion of exhaust 18 (shown as exhaust 18a) into volatilization chamber 12. Plenum chamber 17 acts as a heat exchanger for transferring thermal energy from exhaust 18b to volatilization chamber 12. A valve 42 controls the amount of exhaust 18a diverted into volatilization chamber 12.

Volatilization chamber 12 receives a quantity of alternate fuel 60 through a fuel line 62 from an alternate fuel source 63 with the flow thereof being controlled by a valve 64. Alternate fuel 60 accumulates as a pool of alternate fuel 60 in the bottom of volatilization chamber 12. Bypass 40 directs exhaust 18a into the bottom of the pool of alternate fuel 60 where a bubble plate 44 disperses exhaust 18a upwardly into the pool of alternate fuel 60 in order to assist in the volatilization of alternate fuel 60. However, the primary source of thermal energy for the volatilization of alternate fuel 60 is supplied by exhaust 18b as it passes through plenum chamber 17. The volatilized alternate fuel

60 is shown as volatilized fuel 66 which passes into an inlet 51 which is the end of reactor tube 52 extending upwardly into volatilization chamber 12.

Referring also to FIG. 3, an enlarged segment of pretreater section 14 is shown generally as a reactor 50 which includes a reactor tube 52 located concentrically inside exhaust conduit 16. A reactor rod 54 is mounted concentrically in spaced relationship inside reactor tube 52 to provide an annular space or reaction chamber 56. As shown, exhaust 18 passes through an annular space 51 surrounding reactor tube 52 where it transfers a portion of its thermal energy to reactor tube 52. Volatilized fuel 66 passes counter currently through the annular space of reaction chamber 56. The turbulent mixing of volatilized fuel 66 as it passes through reactor 50 in combination with the thermal energy imparted to it from exhaust 18 along with what is believed to be a catalytic reaction therein initiated by reactor rod 54 produces a pretreated fuel 68. Pretreated fuel 68 is then directed through an intake line 53 (which is an extension of reactor tube 52) into intake manifold 29. A valve 57 in intake line 53 controls the flow of pretreated fuel 68 into intake manifold 29. Supplemental air 80 is introduced into pretreated fuel 68 through an air intake 82 with the flow of supplemental air 80 being controlled by a valve 84.

The presence of the reactor rod has been found important to operation of the invention. The make up of the reactor rod does not appear to be important. A steel reactor rod has been found satisfactory as have stainless steel, aluminum, brass, and ceramic reactor rods.

Steady state operation of internal combustion engine 20 involves exhaust 18 contributing thermal energy to reactor 50. A portion of exhaust 18 is diverted as exhaust 18a and bubbled through the pool of alternate fuel 60 in the bottom of volatilization chamber 12. Exhaust 18a combines with the volatilized fuel from alternate fuel 60 to provide volatilized fuel 66. Volatilized fuel 66 is drawn into inlet 51 thence through reaction chamber 56 of reactor tube 52. The balance of exhaust 18b passes through plenum chamber 17 where a substantial portion of the balance of the thermal energy in exhaust 18b is transferred into alternate fuel 60 to assist in the volatilization of the same.

The method of this invention is practiced by starting internal combustion engine 20 using starting fuel 24 obtained from starting fuel tank 22. The flow of starting fuel 24 through fuel inlet line 28 is controlled by valve 26. Valve 84 is opened initially to allow the free flow of air 80 through air intake 82 during this starting phase of internal combustion engine 20. Internal combustion engine 20 generates exhaust 18 which is collected in exhaust manifold 30 where it is then directed into exhaust conduit 16. Exhaust 18 contains a significant amount of thermal energy resulting from the combustion of starting fuel 24 in internal combustion engine 20. A portion of the thermal energy in exhaust 18 is used to heat reactor 50 and then to volatilize alternate fuel 60.

Specifically, exhaust 18a is diverted through exhaust bypass line 40 into volatilization chamber 12 where exhaust 18a is dispersed by bubble plate 44 into alternate fuel 60. Exhaust 18a transfers its thermal energy to alternate fuel 60 and also provides a carrier stream for the volatilized products of alternate fuel 60 so that this combination becomes volatilized alternate fuel 66 which is then drawn into intake 51. At this point it should be noted also that valve 84 is partially closed in order to create a partial vacuum in pretreated fuel line 53, which means that a partial vacuum will also be created in intake 51. Simultaneously, valves 42 and 57 are selectively controlled in order to suitably recirculate the flow of exhaust 18a and volatilized alternate fuel 66, respectively. In the meantime, the balance of exhaust 18 becomes exhaust 18b which passes through plenum chamber 17 where it transfers its thermal energy into volatilization chamber 12 and alternate fuel 60 therein. Accordingly, a major portion of the balance of thermal energy in exhaust 18 after exhaust 18 has passed through reactor 50 is transferred into alternate fuel 60 for the volatilization of the same.

Volatilized alternate fuel 66 is directed into reaction chamber 56 where it is subjected to the pretreatment process of this invention by becoming reaction fuel 67 and then pretreated fuel 68. At the present time I am unable to state with any degree of certainty precisely what happens to reaction fuel 67 in reaction chamber 56. However, I have found that the larger molecules in volatilized fuel 66 appear to be broken into fragments with some type of reaction taking place. Specifically, I have found that a portion of the length of reactor 50 becomes quite hot, substantially hotter than could otherwise be accounted for from the thermal energy from exhaust 18 alone. This surplus thermal energy implies that some form of reaction is occurring in reaction fuel 67 as it is transformed into pretreated fuel 68. For example, in one prototype of the invention, the end of the exhaust conduit 16 positioned adjacent the end of reactor 50 closest the exhaust manifold 30 maintained a temperature of between about 500.degree.-700.degree. F. The portion of exhaust conduit 16 positioned along the central portion of the reactor 50 had a temperature between about 600.degree.-900.degree. F., while the position of the exhaust conduit 16 positioned adjacent the end of the reaction chamber where the volatilized alternate fuel entered was at a temperature between about 200.degree.-300.degree. F. Thus, the position of the exhaust conduit along the central portion of the reactor 50 reached temperatures higher than would be expected from the temperature of the other position of the pipe. Pretreated fuel 68 is directed into intake manifold 29 where it becomes the fuel source for internal combustion engine 20.

The change over from starting fuel 24 to pretreated fuel 68 is accomplished by the careful adjustment of valves 26, 84, 57, and 42. In this manner, the operation of internal combustion engine 20 is smoothly transferred from sole reliance on starting fuel 24 to reliance entirely on pretreated fuel 68. Using the novel teachings of this invention, I have run internal combustion engine 20 on alternate fuel 60 selected from materials generally considered to be totally unsuitable as a fuel for internal combustion engine 20. These alternate fuels have included crude oil and recycled materials such as motor oil, paint thinners, alcohols, and the like. Also, such fuels having some water content have also been used. Many of these alternate fuels are waste products for which disposal is a significant problem. By being able to use such waste products as fuel, a major source of pollution is eliminated. Tests on the exhaust generated by the engine 20 burning the alternate fuels have indicated that such exhaust is much cleaner than exhaust normally generated by such engines when burning gasoline in normal manner (gasoline can be used in the system as the alternate fuel of the invention to operate the engine more efficiently and without significant pollutants in the exhaust).

The dimensions of the reaction chamber and the reaction rod are such that the rod forces the volatilized fuel to flow substantially along the wall of the reaction chamber. For a 350 cubic inch V-8 Chevrolet engine, a reaction tube of about one-half inch inside diameter is placed substantially concentrically in an exhaust pipe from the engine. The reaction rod has a diameter to leave a concentric space between the reaction rod and inside wall of the reaction tube of about 0.035-0.04 inches and the reaction rod is between about ten inches and twelve inches in length. Lighter fuels, such as gasoline, work with the larger spacing between the reaction rod and reaction tube wall and the shorter rod while the smaller spacing and longer length may be required for heavier fuels such as crude oil since the heavier fuels generally require more heating and velocity through the reaction zone. Similar dimensions have been found satisfactory for use with single cylinder engines such as those having up to about fifteen horsepower. The smaller engines seldom require a reaction rod greater in length than about four inches. Similar dimensions will be used with other internal combustion engines.

The various dimensions indicated are examples only and can vary, usually depending upon the type and size of engine, lfuel volume required, and the type of alternate fuels to be used. The important thing is that the passage for the volatilized alternate fuel through the reaction chamber be such as to cause the reaction to take place to convert the volatilized alternate fuel to the reaction fuel which is satisfactory for operating the engine.

While the invention has been described in detail in connection with an internal combustion engine, the invention can be used equally as well and in similar manner with any fuel burning equipment. Thus, it can be used to treat material so it can be used in fuel for furnaces and boilers in place of the normal natural gas, fuel oil, or coal, or to power turbines in place of the normal kerosene or jet fuel. The reaction chamber can be positioned in the exhaust conduit, such as a flue or chimney, similarly as it is placed in the exhaust conduit from the internal combustion engine shown.

Rather than heating the reaction chamber with exhaust gases from the fuel burning equipment being powered, and such heating is presently preferred because such heating is integrally a part of the equipment used which appears to provide optimum results, the reaction chamber could be heated by other means. Such other means, however, should be arranged to provide similar heating and heat gradients as are provided by the exhaust gas.

Whereas the volatilization chamber is shown as heated by the exhaust gas, the volatilization chamber could be heated by other means or, depending upon the material used as fuel, the volatilization chamber might not be heated at all. The important thing is that the material to be used as fuel is volatilized in the volatilization chamber so the volatilized material is drawn into the reaction chamber. As used herein, the volatilization chamber does not have to be a chamber as such, but may be any means which volatilizes the alternate fuel. It could be a carburetor or an injection nozzle or other volatilizing or spray means. Further, it is not necessary that exhaust gas be combined with the volatilized fuel as it is in the embodiment described. It has been found that in most cases the invention works satisfactorily without exhaust gas in the volatilized fuel. In most instances the volatilization fuel will be drawn through the reaction chamber by a low pressure or a pump at the fuel inlet of the fuel burning equipment.

The fuel pretreater of the invention is a novel discovery in that it allows me to successfully operate fuel burning equipment using alternate fuels. As such, I am able to achieve several highly desirable goals, namely, the extraction of valuable energy from alternate fuel while at the same time removing alternate fuel from the waste stream; or, in the case of crude oil, using this material directly thereby eliminating the need to subject the same to the expensive and capital intensive refining processes.

The present invention may be embodied in other specific forms without departing from its spirit or essential characteristics. The described embodiments are to be considered in all respects only as illustrative and not restrictive. The scope of the invention is, therefore, indicated by the appended claims rather than by the foregoing description. All changes which come within the meaning and range of equivalency of the claims are to be embraced within their scope.

The above information is courtesy of REX research. Please consider helping them stay alive and supporting them, they rely on the purchasing of their CD's to live.

Related patents and information

Korean GEETs:

http://v3.espacenet.com/textdoc?DB=EPODOC&IDX=WO02053901

http://v3.espacenet.com/textdoc?DB=EPODOC&IDX=WO02053902

http://v3.espacenet.com/textdoc?DB=EPODOC&IDX=KR20010078436&F=0

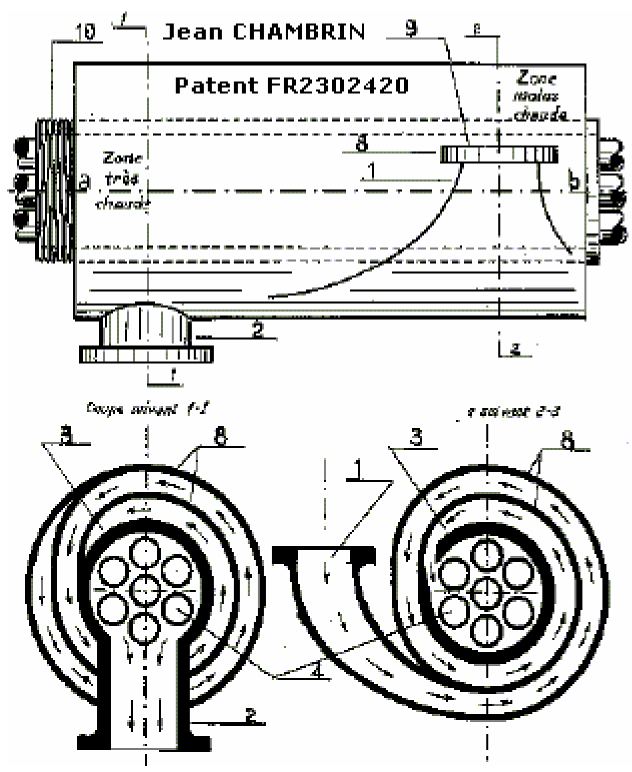
http://v3.espacenet.com/textdoc?DB=EPODOC&IDX=WO02055867

PICC GEET:

http://v3.espacenet.com/publicationDetails/mosaics?CC=US&NR=2009038591A1&KC=A1&F\T=D&date=20090212&DB=&locale=

1974, Jean CHAMBRIN & Jack JOJON, France, car running with 60% of WATER. Patents : WO8203249A1: "A reactor for transforming water and carburant for use as a fuel mixture", and WO8204096A1: "A reactor to transmute the matter which using any fuel in its solid, liquid or gaseous state".

And French patent 2,302,420 :



WO8204096, A REACTOR FOR TRANSFORMING AND CARBURANTS FOR USE AS A FUEL MIXTURE, from http://v3.espacenet.com , Data supplied from the esp@cenet database.

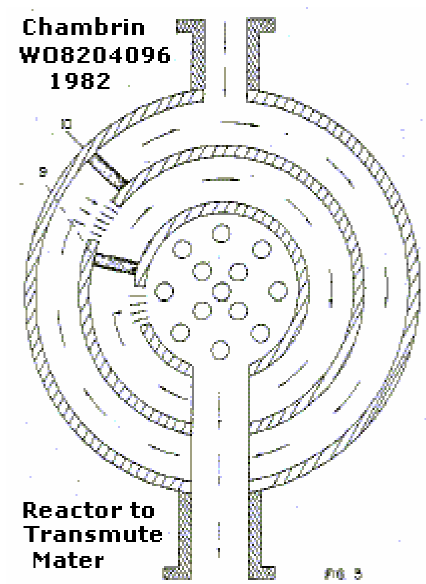
Publication date: 1982-11-25, Inventor: CHAMBRIN JEAN PIERRE MARIE (BR), - international: F02B43/04; F02B43/00; (IPC1-7): F02B43/08, Also published as: EP0078799

(A1), FI821543 (A), EP0078799 (A0), BE893151 (A), PT74890 (B).

Description: An apparatus that enables the **running of any engine, turbine, boiler, heater, etc., regardless of the fuel used, due to its capacity of transmuting such carburant**, once they contain dihydrogen oxide or are associated to it, into a new fuel.

To **start the transmutation process** it is only necessary to **reach the adequate temperature** for the process, irrespective of the fuel used - **gasoline, ammonia, kerosene, ethylic or methylic alcohol, or any carburant available (either in solid, liquid or gassy state)** - combined with a **hydric element** Contrary to what one can imagine, **this temperature does not reach extraordinary levels** since, in this case, it is only one of the necessary elements to the accomplishment of the phenomenon.

<u>The assembly of the REACTOR itself is the main condition to its functioning</u>.



Once we had the necessary conditions to set in the process, **the REACTOR can even be fed only with dihydrogen oxide** (H2O=WATER). Although the phenomenon proved satisfactory, also, in this case, the use of other carburant, mainly the alcohols, even though in **minimal proportions (5 <u>to 95%</u> of dihydrogen oxide**), is also important. It was verified that the carburant which are firstly used to set in the process can also stabilize the transmutation, as the proportion of dihydrogen increases, **keeping it within the limits of the necessary safety**.

A formal explanation to the said process, considering the use of the REACTOR TO TRANSMUTE THE MATTER, may be given by **its capacity of producing hydrogen at relatively low temperatures** with the **support of the exhaust gases of the engine** to which it is attached, and the **hydrogen transmutation into other gases**, with occasional and consecutive changes of the elements, **causing an electromagnetic reaction of the physical field**, by an **elastic compression of these gases**. Since a starting mechanism of the process is determined, the calories wasted to set the engine into motion, which can be either conventional, gasoline or diesel consuming, or boilers, turbines, etc., are also **used to produce a fuel which will be re-used**.

Hence, one can say the REACTOR TO TRANSMUTE THE MATTER is **an apparatus to produce calories**. For example, if 2,000 Kcal (two thousand kilo/calories) is introduced in the REACTOR it will be **possible to multiply these calories by 100 (a hundred), 1,000 (a thousand) and even 100,000 (a hundred thousand)** according to what it is chosen to be used. The only **condition to have a progressive multiplication of the calories without problems is to provide a cooling apparatus** like the one used in combustion engines during operation.

Another important aspect of the process accomplished with the REACTOR TO TRANSMUTE THE MATTER is the necessary obtention of the **molecules strike, as intensive as possible**. The bigger in intensity and molecules the strike is, **more calories will be produced** and consequently more potentiality it will have.

The REACTOR TO TRANSMUTE THE MATTER (Fig. 1-2), which is installed, in case of engines, between the carburetor (Fig. 1-1), already modified, and the engine block, **processes the fuels, or the hydrogen oxide**, before their admission in the engine (Fig. 1-3).

The outer side of the REACTOR must be conceived **to receive the gases inlet to the engine** (Fig. 1-3), the **exhaust gases outlet** of the engine (Fig. I-5), which has **a ball to decompress the gases** (Fig. 1-5), and the feedback pipe (Fig. 1-6).

After innumerable experiments and **considering the velocity of the molecules**, the REACTOR TO TRANSMUTE THE MATTER has a **cylindrical shape** (Fig. 1-2 and Fig 2 - longitudinal section) with two or more tubes inside (Fig. 2-7) according to its use. These tubes are placed **leaving 5 to 10 mm between each other**, depending such variation on the dimensions of the engine or apparatus to which the REACTOR is attached. The width of the REACTOR will also be determined according to the type of engine or apparatus employed.

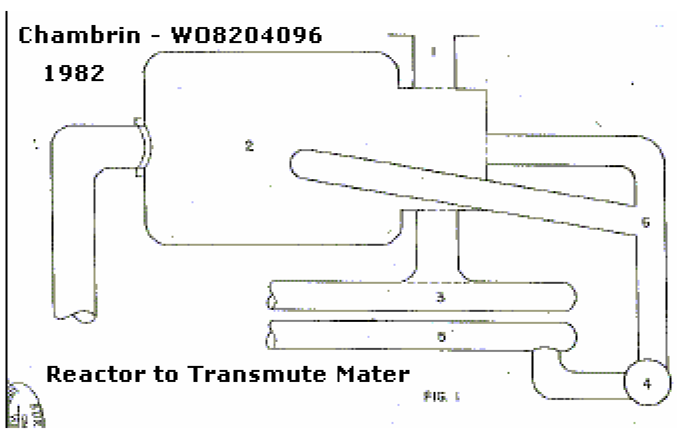
The builder of the REACTOR TO TRANSMUTE THE MATTER must consider in his calculations mainly the **production of hydrogen and of the several other gases** that feed the engine, turbine, boiler, etc.

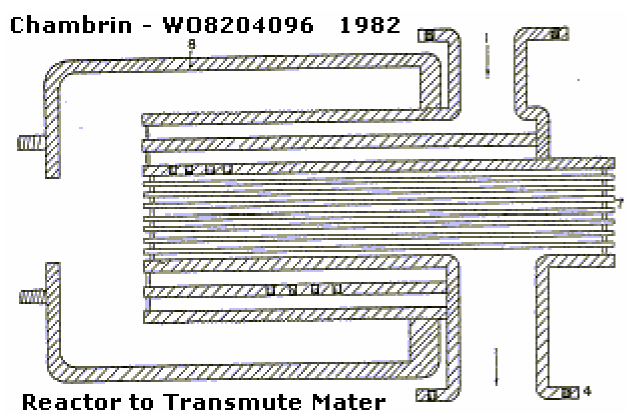
The invention of the REACTOR TO TRANSMUTE THE MATTER has a cylindrical shape because it helps to **speed up the velocity of the molecules**. A shock barrier is placed

Longitudinally (Fig. 3 - cross section - 9 and 10) to multiply the fractioning of molecules, intensifying, there fore, the calories producing process. On the other hand, it is also necessary a **constant pressure of the exhaust gases** next to the REACTOR (Fig.1-6) since in case of reducing the gases flux at the outlet the engine will become less powerful.

So, it is interesting to involve the REACTOR with an **obconical covering** (Fig 2-8) which maintains the gases balance and to insert a compressor ball of the gases at the outlet of the exhaust pipe of the original engine (Fig. 1-4 and 2-4). With this system it is possible to obtain a constant pressure of the gases without braking the engine.

The REACTOR TO TRANSMUTE THE MATTER must be endowed with a **thick metallic covering, considering the high internal temperatures registered**, made of a material with high thermal conductivity. Also the manifolds that go across this covering (Fig. 2-7) must be made of a material with a good thermal conductivity. Although various types of metals present such required qualities, the **different types of copper, in some cases even an alloy of bronze and brass, proved to better meet the demands of the REACTOR** and to be more economic for construction.





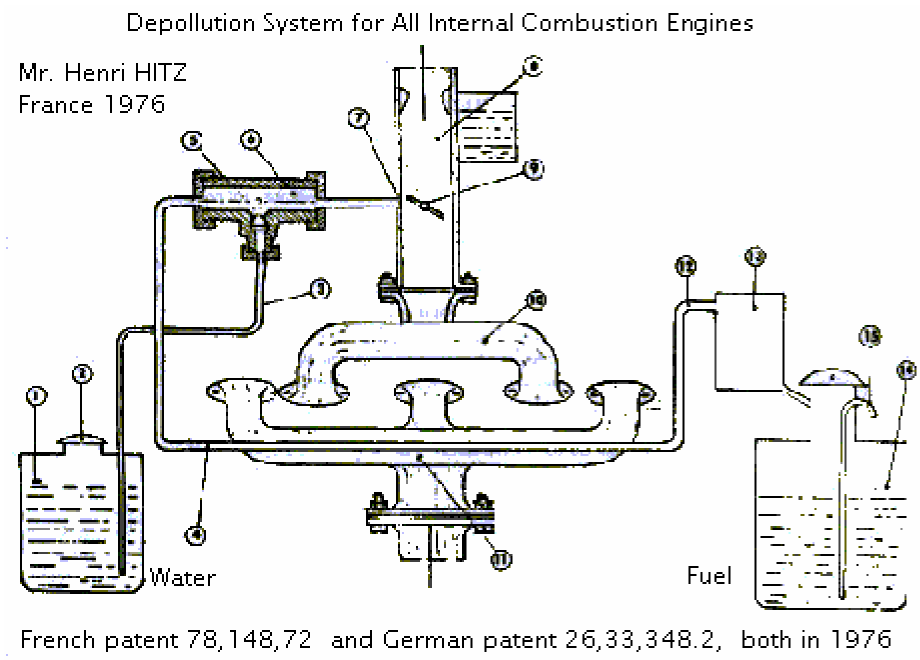
The **results achieved with the REACTOR TO TRANSMUTE THE MATTER are of great importance**. Using a **mixture of dihydrogen oxide and ethylic alcohol**, equally proportioned in weight, as fuel to feed the REACTOR, it was **identified at the outlet of the REACTOR** (before its admission in the engine) 33 (thirty three) **different gases, such as: ARGON, ALUMINIUM, COBALT, MOLYBDENUM, TECHNETIUM, RUTHENIUM, RHODIUM, PALLADIUM, LANTHANUM, THULIUM, ASTATINE, AMERICIUM and CURIUM**. In addition, **<u>at the outlet of the exhaust pipe it was observed 46 (forty six) different gases</u>**. Among these gases it was registered: HYDROGEN, HELIUM, LITHIUM, BERYLLIUM, ALUMINIUM, CHLORINE, TECHNETIUM, RUTHENIUM, RHODIUM, BARIUM, LANTHANUM, POLONIUM, PROTACTINIUM, AMERICIUM, CURIUM, BERKELIUM and HAHNIUM.

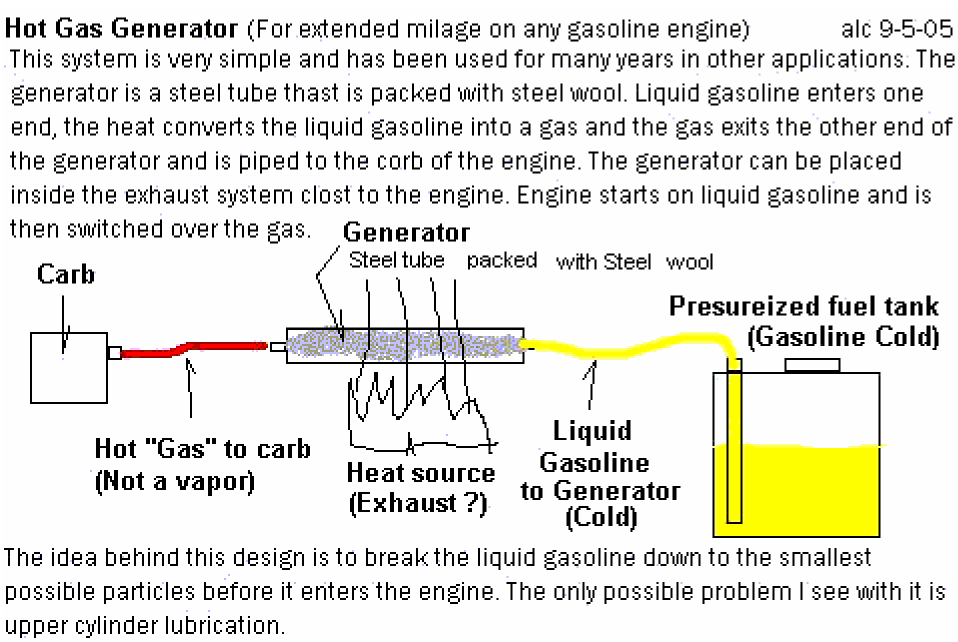
<u>Three other gases which are in the group could not be identified according to the PERIODIC CHART OF THE ELEMENTS; Their numbers are 109, 111 and 131</u>. It is interesting to remember that the PERIODIC CHART OF THE ELEMENTS classifies only till element No.105.

Another innovation of the REACTOR TO TRANSMUTE THE MATTER is the feasibility of storing the exhaust gases and to send them back under a given pressure to the REACTOR, acting in this way as a compressor pipe. If this method is applied, it will have to be injected, for safety's sake, with an electronic injector or any other system, a minimum quantity of alcohol or any other fuel at each revolution of the engine. With this system, it became **possible to reduce substantially the consumption of carburant**. It will be necessary **<u>only one liter of alcohol or any other fuel to cover 60 km (37 miles)</u>**. Or even set a stationary engine into motion with one liter of fuel, at 1,800 rpm (revolutions per minute), during an hour.

All references and information courtesy of http://waterfuel.t35.com/geet_plasma.html

1976, Henri HITZ , French, Patented in France 78,148,72 and in Germany 26,33,348.2, invented during WW2, system to save 20-40% of fuel, by adding water and methanol, and preheating the fuel.





- **2001, RENAULT (French car manufacturer), for a FUEL REFORMER SYSTEM** using Water Steam:

The French patent in pdf http://econologie.com/file/brevets/renault_FR2831532.pdf

SYSTEM AND METHOD FOR HYDROGEN PRODUCTION THROUGH CONVERSION AT HIGH TEMPERATURE WITH WATER STEAM **FR 2 831 532 - A1** **dépôt :** 26.10.01.

Demandeur(s) : *ARMINES ASSOCIATION POUR LA RECHERCHE ET LE DEVELOPPEMENT DES METHODES ET PROCESSUS INDUSTRIELS Association loi de 1901* — FR et *RENAULT* — FR.

Inventeur(s) : GROUSSET DIDIER, MARTY PHILIPPE, FALEMPE MICHEL et BOUDJEMAA FABIEN.

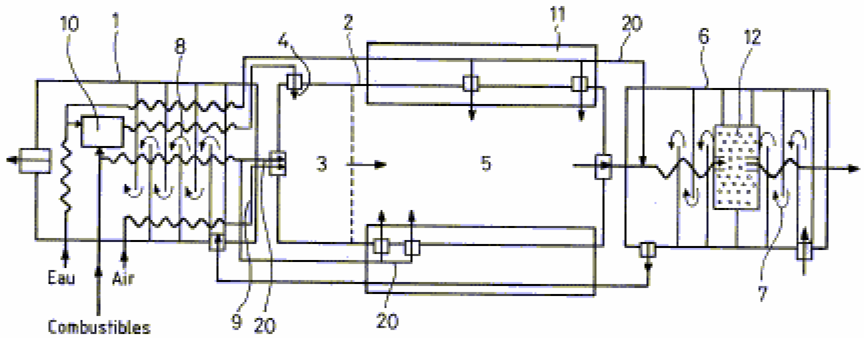
The invention is about a system to generate Hydrogen through the reaction of a fuel with water steam.

the system includes : - (1) a reactor where the water is vaporized and fuel and water steam are preheated - (2) a reactor to convert the fuels in a mix rich in hydrogen and Carbon Monoxide, through high temperature reaction, without catalyst, of the fuel with the water steam.

The reformer reactor (2) includes a first zone (3) where a part of the fuel and water steam are injected (4) at high speed. That the starting of the conversion reaction.

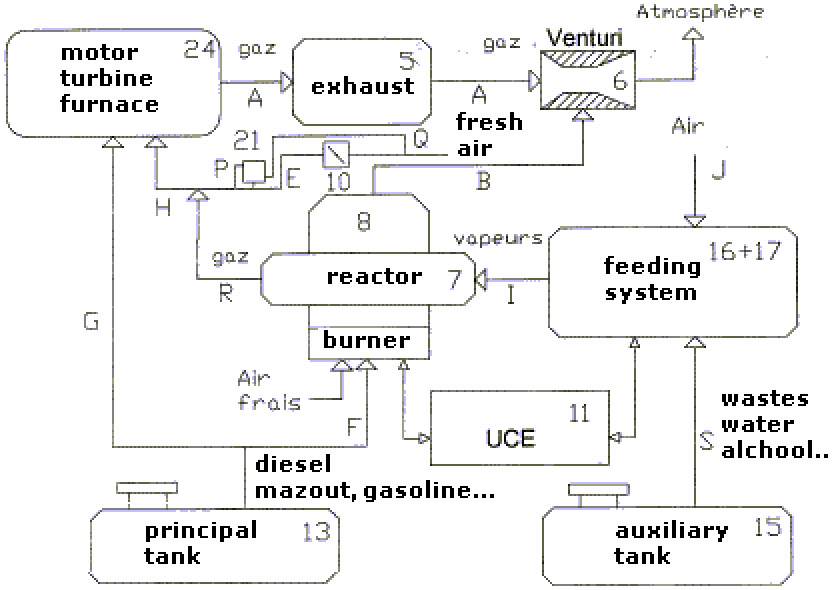
The converter reactor (2) has also a second zone (5) where the fuels and water steam will stay a time long enough for the reaction to reach the thermodynamic equilibrium.

The system has also a colloing area (6).



2003, M. SALELLES and M. MARTZ, French Patent 2,858,364, enhance the GEET PANTONE system:

DISPOSITIF AMELIORANT LE FONCTIONNEMENT DES REACTEURS
SYSTEM TO ENHANCE THE WORKING OF PHYSICO-CHEMICAL REACTORS/CONVERTERS USED ON THE FEEDING LINE OF ENERGY TRANSFORMATION SYSTEMS, AND ESPECIALLY OF ICE ENGINES.
Patent 2,858,364, 2003, M. SALELLES and M. MARTZ ●



The invention is about a system controled through an 'electronic unit control' (11) called UCE.

This system is made of a specific fuel burner (8) that is used to bring up and maintain the temperature of the reactor (7) at a predetermined value.

This system is also made of a doubble evaporator (16/17) feed in fluid that can be combustible, at level, pressure and temperature stable to produce a gas feeding the reactor (7); the reactor is working under an adjustable vacuum, through the clapet (10) and a pneumatic valve (21) fitted on the feeding line (E) of the ICE (24) or other system for energy conversion.

DISPOSITIF AMELIORANT LE FONCTIONNEMENT DES REACTEURS SYSTEM TO ENHANCE THE WORKING OF PHYSICO-CHEMICAL REACTORS/CONVERTERS USED ON THE FEEDING LINE OF ENERGY TRANSFORMATION SYSTEMS, AND ESPECIALLY OF ICE ENGINES.
Patent 2,858,364, 2003, M. SALELLES and M. MARTZ ● ●

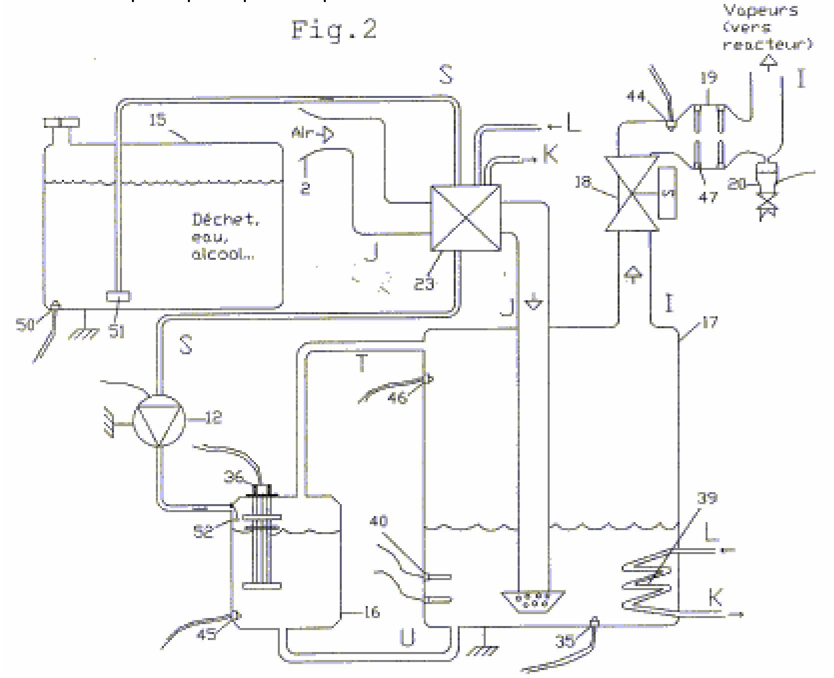


Fig.2

DISPOSITIF AMELIORANT LE FONCTIONNEMENT DES REACTEURS SYSTEM TO ENHANCE THE WORKING OF PHYSICO-CHEMICAL REACTORS/CONVERTERS USED ON THE FEEDING LINE OF ENERGY TRANSFORMATION SYSTEMS, AND ESPECIALLY OF ICE ENGINES.
Patent 2,858,364, 2003, M. SALELLES and M. MARTZ ● ● ●

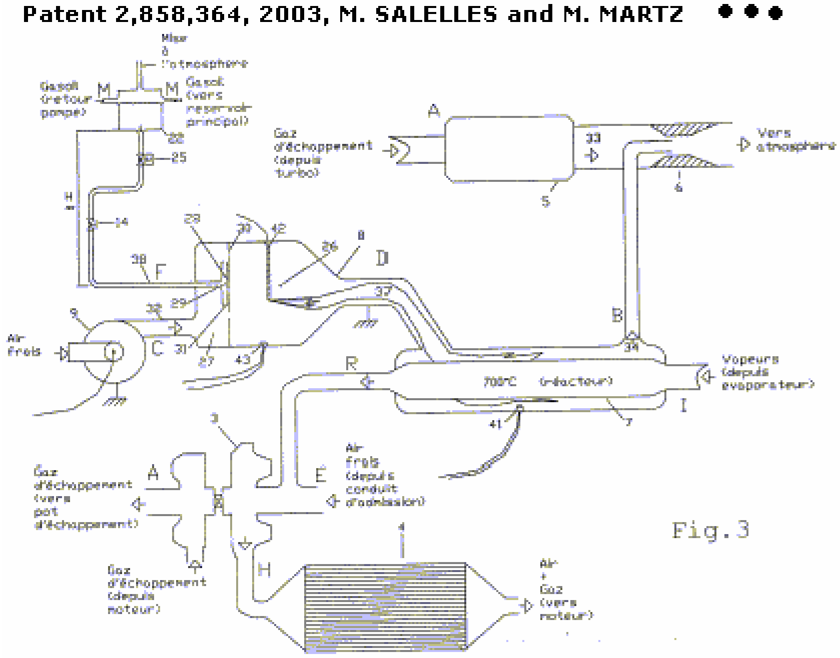
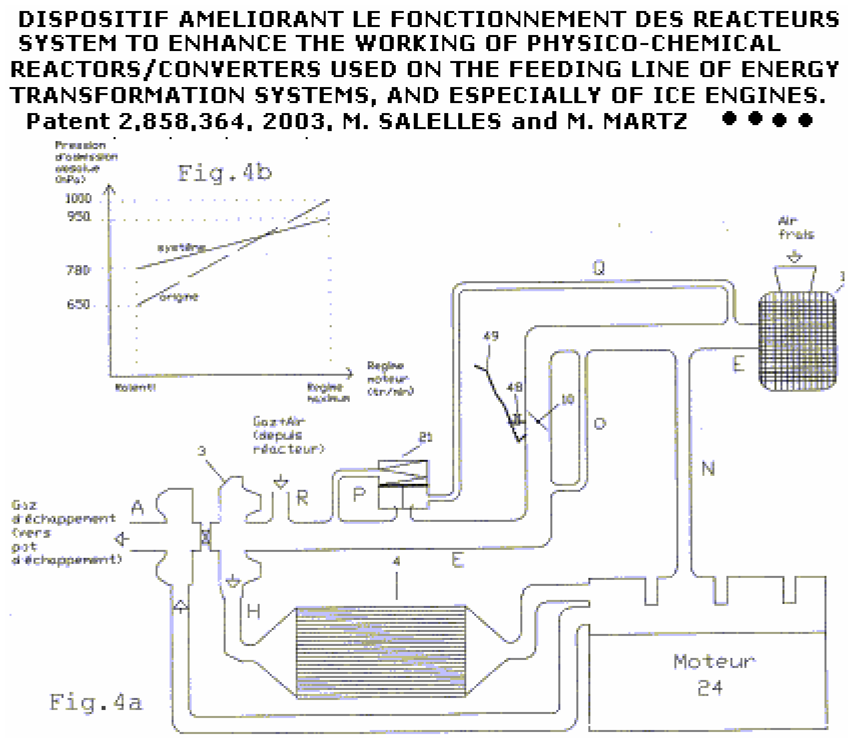


Fig.3



[**GEET Patent on KEELYNET**](http://www.keelynet.com/interact/archive/00000363.htm), by Peter T. Michel, GEETofPA@aol.com, Sun, 20 Jun 1999
http://www.keelynet.com/interact/archive/00000363.htm

As far as the GEET, I decided to build one myself, and it worked great! I ran it on a 4.0 HP Briggs and Stratton on a lawnmower. I was able to run it on **5% gas, 5% oil, and the remaining 90% was a mixture of pickle juice, jalapeño juice, Sprite soda, and water**. I was actually **able to run it in a closed-loop as well! The entire exhaust fed back into the intake and I could shut down any outside air**, and it worked great! On one model, **it ran silently in a closed-loop except for the sound of the valves** opening and closing. That kind of freaked me out, but it was cool as anything to witness!

I also recently testified in Philadelphia at the EPA hearings for Tier 2 emissions standards and spoke of what the GEET has to offer. I had an excellent reception from people, particularly people from various environmental organizations.

At the **Conference on Future Energy (COFE) in Bethesda**, MD, I and a couple of friends demonstrated another GEET mower running on a **mixture of black coffee (delicious hotel made), Mountain Dew, and about 20% gas**. The emissions ran **so clean** that several people held their faces to the exhaust and breathed deeply with big smiles on their faces. And I have it on videotape to prove it!

So, **does it work? I'd have to say, yes, definitely**. I've been experimenting with different designs including a double reaction chamber, but am not getting good results. **I asked**

Paul about this and he said that the greatest reaction occurs when you have the coldest possible fuel vapor, and the hottest possible exhaust, going in opposite directions in the chamber. I'm going to play around with refrigerants that boil at low temperatures and see what happens.

The **clearances also need to be very small in order to have the greatest acceleration**. Essentially, the cold vapors (**best when closest to 32 degrees F** for most fuels) **heat up to around 900 degrees F in an instant** as they accelerate up the rod, effectively **annihilating its molecular (and atomic?) structure**. I couldn't help but notice the similarities of what is happening in the chamber, to the elements necessary for **elemental transmutation as described by Walter Russell**. Indeed, there is a transmutation happening of some type as **mass spec tests performed at Brigham Young Univ. showed 73 elements entering the chamber and only 13 elements coming out**! -- With a new one that they've named **Pantonium!** Most of the fuel that comes out of the chamber is **hydrogen** making it a very clean burning fuel.

The best fuel they found was a mixture of **80% seawater and 20% crude oil**. In an ideal reaction, the reaction is **endothermic** in nature and has actually caused **frost to form on the muffler**. Also, using this "fuel" mixture and having an ideal reaction, they were able to obtain mileage increases that I cannot even mention here. But, let's just say, **<u>it was WELL OVER 25 times the mpg!</u>** (Due to pressures and dangers from the "darker side" the production model will only provide 2-3 times the gas mileage. **Paul already had his brake lines cut, his car blown up, and his house burned down**, because some stupid TV news reporter told the world that he was ready to put the oil companies out of business!)

Some interesting things were noted with the steel rod inside the chamber. Namely, **it takes on magnetic properties** that can actually indicate the latitude where it was last run. Also, **concentric rings form on the rod** and each seems to **have its own magnetic polarity**. Together, these rings seem to comprise a magnetic signature that has to do with where the chamber is situated with respect to the earth's magnetic field, and the type of fuel being used. When first running the chamber, Paul says that you need to "burn in" the rod's signature. You do this by **pointing the end of the reaction chamber** from which the exhaust is entering, **towards magnetic north**. Then, by running the engine for **at least 20 minutes** that signature is burned into the rod. Then, you can run the engine anywhere without any problems.

GEET also has **a new water machine that can create 200,000 gallons of pure water every 24 hours, even in dry desert conditions, using only a 10 HP engine**. Imagine turning dry, barren desert land into green, fertile farmland. Now, using a GEET fuel processor on the generator, you can fill it up once a day and just let it keep running out in the middle of the desert!

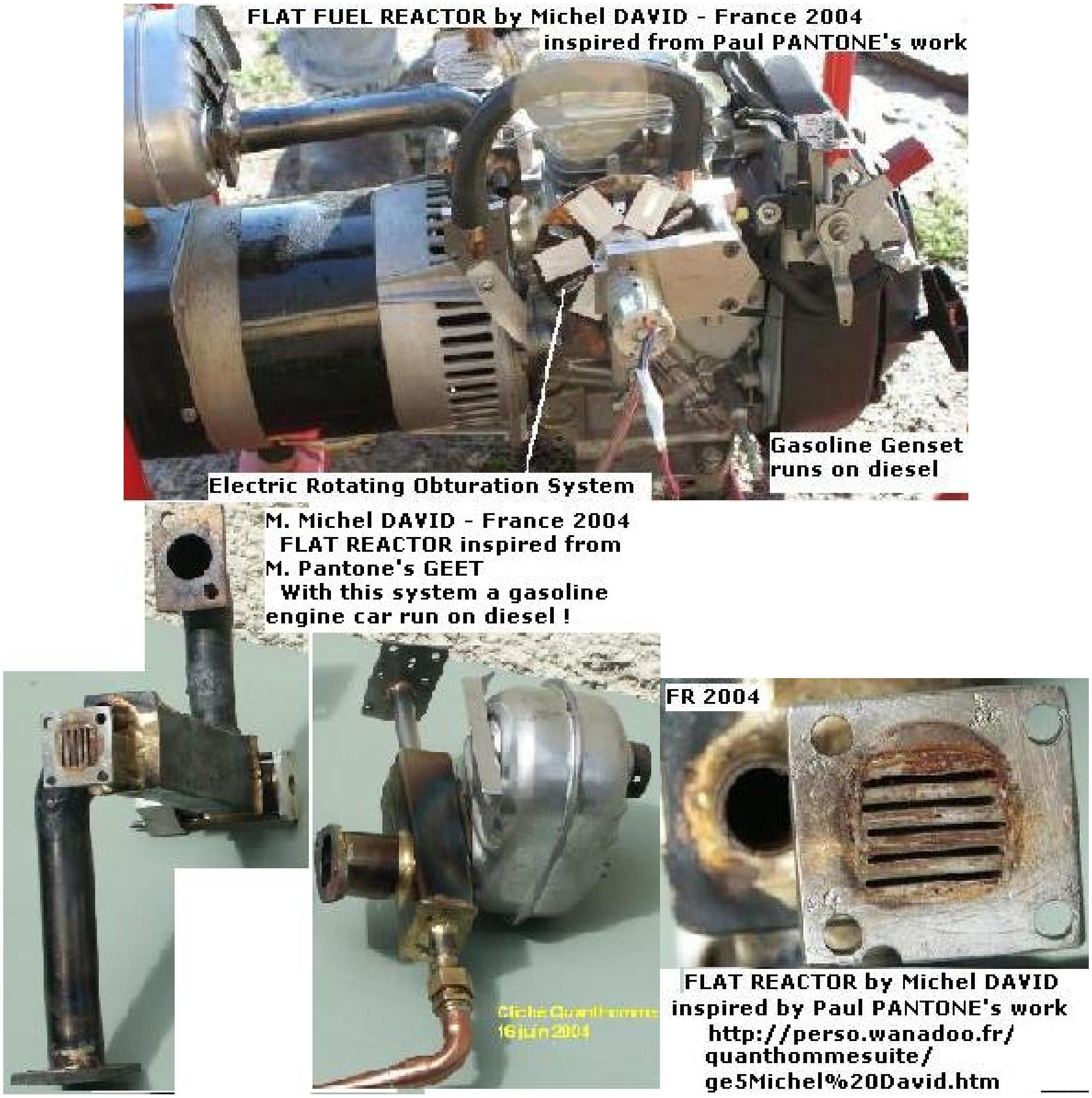
I feel like I hit pay-dirt with GEET. **They have over 400 inventions from inventors all over the world** who just want to keep inventing and want to leave the marketing to GEET management and their worldwide distributors. I asked many people what they thought about GEET and Paul Pantone prior to diving into this and they all had positive things to say.

2004, Michel David, France, a gasoline genset running on diesel, through an original small mechanic 'reactor' inspired by the work of Paul Pantone. http://perso.wanadoo.fr/quanthommesuite/ge5Michel%20David.htm (Link no longer active.)

On the short film to download on the webpage, you'll see that there are no fumes **at the exhaust** when starting the group with gasoline, and also no fumes when it's switched on diesel... absolutely no changes in RPM or noise when the gasoline genset runs with diesel. Michel David, Hervé F., Michel Schmit, Bernadette and Jean Soarès (famous webmasters of www.quanthomme.com) are present on the video

Explication: Starting with a standard Pantone Reactor, M. David had the idea to create a **flat reactor**, where are fitted 5 spaces of 0,75 mm, that **replace the rod and internal tube on a GEET**, what gives a large surface of friction for the gasses accelerated by the rotational obturation system made by M. David at the air intake. M. David used copper pipes that he pressed to make them flat, with a metal saw blade inserted inside to keep the desired space.

Any fuel is passed through a modified carburetor that pulverized it, then through the reactor to the engine, where it arrives as gas. **It's easy to run any genset with diesel,** with or without adding water. It needs a very good vacuum at the intake side, that's why M. David uses a homemade rotating obturator **to increase this vacuum effect**. This system very easy to make by any professional, can be mechanic or with a progressive action for the opening-closing of the holes to obtain a soft feeding following the needs of a vehicle's engine.



Reference

Thermal Water Cracking Devices for making water into Hydrogen & Oxygen by heat alone

1) The conversion unit for furnace was in Texas military motor pool during 1950's. Inventor went to Washington later, and demonstrated smaller boxed unit to Senator, in gov garage.11.-- Oil furnace runs on water - (a modification could be adapted to autos) Read Patent #2863499, Stainless steel tubing coil in fire pot heated with fuel oil initially only, 90 lb pressure water turns to super heated steam at 1500 degrees, goes through T fitting with oil and comes out spray nozzle as hydrogen gas torch. oil shut off, furnace continues to run on water, Patent office allowed only 50 % increase in efficiency with oil to be written, not what manuscript submitted.

Electric motor with dual shafts run two 90lb oil hydraulic fluid pumps. The vibration of flowing steam molecules increases and enters shockwave effect as going into area of low ambient pressure air when exiting nozzle. Smaller unit built with fire brick, vertical tubing coil, valves, outlet ring of pores at bottom. Initial alcohol flame heat to 1500 degrees, then light off gases coming from circular burner ring.

2) http://clients.pppoe.ca/~joso65/info.html Sorrenti Patent CDN patent 2,054,007 basically mixes water and hydrocarbons, heats up to 1200 degrees C using exhaust heat to disassociate the water and adds magnets as an aid and burns in internal combustion engine. He says in the patent, once heated up can increase water up to 100%. Had one brief phone call with above but does not give out phone number and did not respond to emails. The Claim is that a 1978 Chrysler Cordoba was converted & able to run totally on water.

3) http://www.rexresearch.com/celis/celis.htm Philippino Inventor 30% to 50% saving by adding thermo cracked water into engine. Some inventors are the only ones that can get those inventions to work. Also it is possible that water cracking devices use up oxygen that needs to be replaced somehow.
http://v3.espacenet.com/origdoc?DB=EPODOC&IDX=US2863499&QPN=US2863499

www.google.com/patents?id=a8piAAAAEBAJ

Papers and Related Information

The GEET Fuel Processor is a Self-Inducing Plasma Generator by Paul & Molley Pantone

The first working prototype was developed long before the technical analysis was attempted. Plasma research is a fairly new field of acceptable science. At this time most printed text is from foreign Countries, and a majority from Russia.

The technology used in the GEET Fuel Processor is a combination of the most basic scientific principles, most of which falls within the normal rules and of thermodynamics. But some of the 70 simultaneous phenomenon’s are not found in those books, since it is the combination of events, which is the body of this discovery. Put quite simply, the exhaust heat is transferred to the incoming fuel vapor, which must be maintained in a vacuum, and the overall configuration provides a molecular breakdown within the vacuum of all of the heavier elements. Therefore, intensifying the vacuum, the speed of molecular breakdown or reaction is magnified, and less heat is required.

The GEET Plasma unit generates several "electrical" fields at the same time while operating, some of which are in opposite direction and all are affected by the direction of mass movement as well as by the gravitational field of our planet. During lectures from coast to coast Paul and Molley have explained that it is frequency and vibration that determines the amount of plasma or energy being developed. Reseach in private laboratories in Europe is helping to isolate some of the basic field replication of the plasma generator that the Pantones need for visible demonstrations.

Many attempts to use the external electrical mechanical devices to enhance the production of Plasma in the GEET Fuel Processor have all failed to show any promise, such as the Plasmatron. This has occurred because the outside interference has opposed the "natural" order of the energy, which must be self generated to maximize the results, as well as will increase the charge-discharge at specific needs of demand of the Plasma - or GEET GAS. It should also be noted that using outside non-natural heating diminishes the fields which are normally self generated.

The specific movement of vapor within the GEET Fuel Processor is "focused" to exact flow direction and velocity being self created thereby maximizing and intensifying the "field and enhancing the molecular, or atomic disassociation. Without all other natural elements increasing to equal proportions, one cannot expect that merely increasing the Arc-Field will be the main reason for specific success of any given substance to be broken down to its base elements. When the ideal Plasma has been created is the time to begin increasing or decreasing all parameters involved at their respective equal or balanced increments to satisfy an increase or decrease in the Plasma flow. In doing so one can increase the Plasma flow to a viable delivery state for use of all commercial use demands. During tests the over-reving to engines has startled engineers and scientists from around the world, as engines are sped up to over twice the normal rpm, and slowed down to a fraction of their normal idle speed, with no noticeable vibration. Have you ever seen a 350 Chevrolet idle at 80 rpm? We have.

All of the currently studied Plasma generators basically share a design and operational feature in that they attempt to push the Fuel, under pressure, into a reaction, whereby a need for outside energy to force the device to function. The most unique feature of the GEET Plasma Fuel System is that by supplying the fuel into the Plasma chamber in a vacuum and through a longitudinal, natural release, causes a Radial reaction which is self induced, which creates energy as electrons are pulled into the reacting of plasma, instead of consuming energy. Thus the Plasma becomes more "homogenized" with atmospheric air, causing a well-blended fuel for final delivery.

An additional stabilizing feature within our system is the natural circulation of opposing masses as a vorticular motion within the Plasma Field, a condition as described by Molley Pantone as Thermal momentum-or Inertia. Such field is caused in part by the chamber beginning before and after the Field zone. The size of the Field zone must coincide with the fuel and parameters with specific limitations, dependent on the fuel demand. Now we should also explain that a small unit, such as a 10 hp engine can be used as a "servant" to produce fuel to be used by an un-modified larger engine or furnace, by adapting pumps and only modifying the air intake only. Thus a 10 hp engine could make the fuel for a locomotive. The exact length of the Plasma chamber need be adjustable to fully accommodate rapid change of fuels when different blends are being used. This is quite simple but requires some very expensive equipment for analysis of the final exhaust for the average mechanic.

The "balance" point of a perfectly adjusted GEET Plasma reaction chamber, will give the same temperature coming out of the exhaust pipe as the ambient air, as well as the air quality should be the same or a slight increase of oxygen coming out of the tailpipe. So far the inventor has accomplished a 2% increase in oxygen coming out of an internal combustion using crude oil as fuel and a 3.5% increase using Battery acid mixed with 80% saltwater. At the higher than ambient oxygen levels you normally find ice forming on the exhaust pipes as a normal function of this phenomenon.

Paul Pantone had his best results with seawater and crude oil. 1 gallon of seawater has as much energy as 300 gals of gasoline. This is because of the deuterium.
http://library.thinkquest.org/17940/texts/fusion_dt/fusion_dt.html

When the Plasma field chamber is too short or too long for the density of the fuel being used, it overheats the South end and Chills the North end of the reactor, this also causes the field to consume oxygen, instead of creating it. The direction and configuration of the heat source is critical to the proper balance of the reaction to create Plasma. We have now learned that down is the same as South in relationship to using a compass, and therefore North is up.

Other Plasma generators, such as the copy cat from MIT, which they call the Plasmatron, uses outside applied power to create heat to run the units, but have extremely limited use and output, when compared to the GEET system. Since the power output of Plasma is constant and generates power we can only assume that it is of a DC nature and is a constant output which we have not yet attempted to harness. (hopefully coming soon.)

There will be a large number of reports dating back to 1984 that the inventor was not ready to release until he felt the timing was right. He feels the timing is now right and these will be posted as soon as possible.

The Geet fuel processor may soon make it possible for you to own the ultimate home production power plant... one that heats your water, generates electricity, takes care of heating and air conditioning, simply by utilizing waste heat from refrigeration and applying it to storage or hot water, while the generator is giving you all the electricity you want.

In simple definition, the GEET Fuel Processor could be called a new type of carburetor with a miniature refinery built in. With it, There is no need for catalytic converters, smog pumps and many other costly items on cars , as the GEET fuel processor is not just a fuel delivery system it is also a pollution elimination unit! Your mileage will be greatly increased if you are truly consuming ALL of the available energy From whatever fuel you may be using.

I began working on the original concept of better mileage over fifteen years ago, During the fifteen years of testing and research, I was able to achieve the goals of ZERO pollution, while running internal combustion engines on fuel such as crude oil, battery acid, cleaning solvents, even gasoline... some of the tests were done with mixtures with as much as 80% water IMPOSSIBLE??? SEEING BELIEVES!

Having demonstrated the GEET Fuel Processor countless times, I heard over and over " that's impossible." Most of the hundred scientists who have been invited to help in this project have refused to even come out to look at it, claiming it is impossible. Yet after repeated showings, many potential financial backers have depended on the professional opinion of qualified people, who did not even take the time to even look.

One scientist -Jim _ who wanted to help me was employed at a major United States Testing Laboratory. We spent several days reviewing how and why the device worked. Jim claimed he could get all the necessary funding to get through the R&D stages by telling the other scientists at work what he had viewed, Jim told me to call him at work the following Wednesday.

When I , called the number I was informed that Jim was asked to resign. They told me that Jim must have been doing some drugs, if he truly believed that he saw a gasoline engine run on crude oil with no pollution.

This type of response is very normal to this inventor. Many sincere people have turned their backs and walked away, because of the input of knowledge of others who laugh and say it is absolutely not possible.

However, a few years ago , at a smog certification station in California, this fuel system was demonstrated while being monitored and videotaped. While running a gasoline engine on crude oil, the final exhaust was actually cleaner than the air in the establishment --zero pollution. This does not defy physics; it only operates within the most basic laws of physics in a unique manner. Basics of GEET technology

The GEET fuel processor is a self inducing plasma generator. In my case, the working proto type was developed long before the technical analysis was attempted. Plasma research is a fairly new field of science. Most of the available text on this subject is from foreign countries.

The technology used on the GEET fuel processor is a combination of very basic scientific principles which fall within most of the normal rules and laws of thermodynamics.

Put quite simply the exhaust heat is transferred to the incoming fuel, which is in a vacuum, and the overall configuration provides a molecular breakdown within the vacuum , the speed of the molecular reaction, or breakdown, is greatly magnified. The GEET Plasma Generator

The phenomenon which occurs within and around the GEET Fuel Processor can best be described as controlled lightning. As masses of cold and warm air colliding, an electrical discharge occurs. The specific lengths of each colliding mass determine the type and the amount of discharge.

It can be a bolt of lightning, or if the configuration of masses is conductive to a radial type of discharge it may appear as a ball of energy. Many discharges of this nature are so small they are not visible to the human eye. Others are magnified by moisture and radiate in an energy field which is visible as colored light.

When the electromagnetic field is radial as well as longitudinal, and balanced to create the center of the plasma reaction, maximum efficiency of the field is accomplished. This is done within the GEET Fuel Processor, as the plasma is created on demand. Using a steady self generated magnetic field one does not have the problem of random Plasma clusters, as every molecule is held as a constant potential contributor to the demand and the demand controls the field which stabilizes itself within a specific ratio.

The elemental components of the GEET Fuel Processor allow the transfer of virtually all the generated heat into the plasma, which further stabilizes the electromagnetic field, as well as increases the electron flow at any specific need, on demand.

In the GEET device the plasma fields is generated internally. Many attempts to use external electrical mechanical devices to enhance the production of plasmas in the GEET fuel processor, have all failed. This has occurred because the outside interference has opposed the "natural" electromagnetic field, which is self-generated in the GEET fuel processor. Thus the entire magnetic field collapses and entire system shuts down.

In conventional generators, the means of introduction of the magnetic flow is perpendicular or angled to the plasma tube through wave guides; the effectiveness is diminished due the turbulence created. By simply changing the position of the electrode to the center of the plasma field, the turbulence is eliminated, thus more usable energy is created. Furthermore, less extraneous equipment is used to produce and control the plasma.

The movement within the GEET Fuel Processor is "focused" to the specific flow direction of the Plasma being created, thereby maximizing and intensifying the magnetic field and enhancing molecular, or atomic, disassociation.

Without all other elements increasing to equal proportions, one cannot expect that merely increasing the electric arc/magnetic field will be the main reason for specific success of any given test. When the ideal plasma reaction has been created is the time to begin increase or decreasing all parameters involved at their respective equal, or balanced, increments to satisfy an increase or decrease in the plasma flow. In so doing one can increase the Plasma flow to a viable delivery state for commercial use. Plasma Flow

All the current studied Plasma generators basically share a design and operational feature in that they attempt to PUSH the Plasma chamber. One of the unique differences of the GEET Fuel Processor is that reduced pressure (vacuum), PULLS the Plasma, which enhances the homogenization of the newly created fuel.

An additional stabilizing feature within our Plasma unit is the recirculation zone is through and beyond both ends of the magnetic field, thus intensifying and further stabilizing the plasma. The size of the recirculation zone needs to coincide to all other parameters within specific limitations- depending on the fuel source-and demand at any given time.

The exact length of the Plasma generation chamber needs to be fully adjustable, to compensate for changes in the molecular density or massive expansions of the fuel being used for Plasma. An example of this would be when 20% battery acid is mixed with 80% saltwater and used as fuel; it needs a shorter Plasma chamber than the one needed for Alaskan Crude Oil.

If the same or larger unit is chosen for the acid mix, the normal running temperatures are exceeded, and the balance of the plasma field is at its optimum performance when ambient air and the final discharge are at the same temperature, and air quality at both points are equal.

When the plasma field tube is too short or too long for the density of the fuel being used, it overheats the high end or forms ice on the low end, respectively. This characteristic is further evidence by numerous tests. When pollutants are noticeable there is an imbalance.

The direction and configuration of heat applied, was made on many of the prior units to formulate conclusions. The specific natural flow of self generated energy which does create its own fields (outside of lightning, and natural phenomenon).

Other plasma generators using outside applied power seem to have less technological reason and practical use than the GEET fuel processor which requires no outside power. Since the energy field which is radial and longitudinal, as well as self generated and constant, we may assume that the current-voltage characteristic of the GEET plasma field is a pulsating direct current. New Theories Needed

With the proper team of open minded scientists, this technology should be easily understood. since prototypes already exist. A few months ago, when the inventor invited scientists from all over the country, to help in compiling a reasonable theory or formula for why the invention works, he found very few takers.

One scientist, Dr. Andreas Kurt Richter, spent most of a week at the inventor's home as a house guest. There were hours of discussion on physics and unknown phenomenon. In a letter, dated July 3, 1995, Dr. Richter states, I am a consultant to Paul Pantone in the search for the scientific and technical explanations to understand the operation of this energy device. According to my present knowledge it should not work and I would not believe it had I not seen it with my own eyes. It is my opinion that Mr. Paul W. Pantone has invited an amazing energy device or engine with potential as yet unheard of.

Another scientist, Dr. Grant Wood, has similar comments. Dr. Wood has taught automotive science for most of the last 35 years. I am still seeking scientists, doctors, manufacturers, and all other professionals to assist me, not only in this but hundreds of other inventions and products and concepts. Testing

Getting testing done or the interest to get them done at such places as Lawrence Livermore Laboratories, Southwest Research Laboratories Universities, etc., is difficult. First you must convince them it works, and then have a ton of money. These laboratories have expressed that testing would be a waste of money, and their valuable time. Most simply do not understand this device.

To get testing done, the inventor went to numerous companies including Cooper Industries, Briggs and Stratton, Waukesha; (this list is quite long), and in most cases these industries were not interested, even though many sent representative out and can convey that the prototypes did in fact work. At first, most of the tests were accomplished on small internal combustion engines. Combustion studies were done in furnace applications to enable the inventor a better fuel study.

In 1983, I approached the small engine manufacturers in an effort to gain knowledge and technical support. Up to this point I had used old beat up equipment for most of my testing. Briggs and Stratton was the only company willing to discuss such technology which is advanced, they wanted to be the first engine company to go public.

A few years later in 1987, I did go to Wawatosa, Wisconsin and ran this engine, hooked up to their testing dyno. These test were done on crude oils, gasoline, and fuel oils, mixed with water. They knew the engine worked and would be controversial and suggested that I try to market the device the device in third world countries. I still want to market the device in the United States first.

A few test engines have been tested in cars. Now a 240 kW Waukesha Generator (Model #H2475) has also been retrofit with the GEET Fuel Processor and the only thing needed to get this into production is automatic controls and money. A Pollution Solution

Many have asked what the true value of this technology is. To being with, please place a value on what would it be worth, in dollars and cents, if you could just double the mileage/performance on every car, truck, locomotive, ship, furnace, boiler, hot water heater, etc., not to mention reducing pollution, on a world wide application? The truth is that if you only disposed of some forms of toxic waste, it would be invaluable to man. And if you generated energy from raw crude oil, without the need for refineries, this would satisfy many countries all by itself.

Although the automotive field is very large, our global buildup of toxic waste has become my first choice for production. This can be accomplished in a reasonably short time by installing electronic controls to the necessary control components.

Utilities and communities can greatly benefit from the GEET Fuel Processor, while running power plants, desalinization plants, pumping plants, etc., all the while getting paid to take toxic fuel to run the plants. When toxic waste is transported from coast to coast there is always a danger of accents, and by locating toxic disposal units throughout the country this will shorten the risk and distances traveled, providing more safety to the public.

For you,
And the World,
Paul & Molley Pantone

Related Technology

MIT Plasmatron - Principles of the Pantone GEET Device

MIT is/has developed it for ArvinMeritor. Delphi is also into it. Reports given to the organization state that MIT has spent years and millions perfecting and proving the technology. They partnered with Arvin Mentor who worked out the last details of actually implementing in cars delivered by the automakers. Then One Equity Partners ("OEP") bought out Arvin's fuel reformer division and created a new company called EMCON Technologies. Then EMCON Technologies dropped the fuel reformer from their product line and only sell other stuff. After all MIT did to prove its benefits. Bottom line Big Oil wins again”. It know appears that big oil will only let MIT's Plasmatron be used to clean up exhaust as in patent US2005274104, but they can no longer use as originally intended for engine intake gases to increase fuel economy.

The Geet system is alleged to be used in the Pre-Ignition catalytic converter

http://peswiki.com/index.php/Directory:PICC_Pre-Ignition_Catalytic_Converter http://www.preignitioncc.com/better/

The Plasmatron Fuel converter (PDF)

The Microplasmatron Fuel Converter (Plasmatron)

Environmental protection and energy saving using plasma technology by Daniel R. Cohn (PDF)

MIT's plasmatron cuts diesel bus emissions, promises better gas engine efficiency

MIT device could lead to near-term environmental improvements for cars

MIT's Plasmatron Tested on Bus

Device could help cars' gas-burning efficiency

MIT's plasmatron cuts diesel bus emissions (PDF)

Cleaner, Higher Efficiency Vehicles Using Plasmatrons (Power Point)

Le Plasmatron du MIT

Illustration of the Super-Carb process by Himac

Hydrogen yield results from experimental arc pyrolysis of methane by Roseberry, C., Wilson, D. and Lu, F. - AIAA Paper 2005–3401, AIAA/CIRA 13th International Space Planes and Hypersonics Systems and Technologies Conference

Reformation of methane in a supersonic, arc-heated flow, by Lu, F.K., Roseberry, C.M., Meyers, J.M., Wilson, D.R., Lee, Y.-M., Czysz, P., - AIAA Paper 2004–1132, 42nd AIAA Aerospace Sciences Meeting and Exhibit, January 5-8, 2004, Reno, Nevada

Experimental evaluation of methane fuel reformation feasibility by Roseberry, C.M., Meyers, J.M., Lu, F.K., Wilson, D.R., Lee, Y.-M., Czysz, P. - **AIAA Paper 2003–693**7, 12th AIAA International Space Planes and Hypersonic Systems and Technologies, December 15–18, 2003, Norfolk, Virginia.

Faculty information

PAGES OF MICHEL DAVID From Quantum home

CONSTRUCTION AND the INSTALLATION Of a SYSTEM PANTONE ON an ENGINE

(Translated from French)

Running the GEET on 100% water

Paul Pantone was able to get some engines with GEET reactors to run on water only. But I think he did have either some electrolytes in the water else he was mixing in lead acid battery fumes to the water vapors going into the reactor.

If you study Jean Chambrin's patents carefully you will see he also was able to run his engine with fuel reformer on water only also. We have also heard that he went to hide in Brazil in fear of his life from people who didn't want people to know able his invention.

http://v3.espacenet.com/publicationDetails/description;jsessionid=14B1247C59EF48\DC351D8D7F02A3B185.espacenet_levelx_prod_0?CC=WO&NR=8204096A1&KC=A1&FT=D&date=19\821125

http://v3.espacenet.com/publicationDetails/description?CC=WO&NR=8203249A1&KC=A1&\FT=D&date=19820930

In addition, for the GEET type of reactor to work well, the water needs some electrolyte in it, like some ammonia or vinegar or a small amount of battery acid, etc. Something so the water when vaporized will have some ions capable of carrying electric charges of the ions of the vaporized electrolyte. That can help the electromagnetic characteristics to develop that cause the magnetic signature to develop on the rod.

A few things to remember:

1.It will be difficult to run water only so have multiple reactors in parallel and spread out from each other. Have very small gaps for intake and exhaust. In 1 reactor use a hydrocarbon fuel that can keep the engine running while the other reactors have water with electrolyte running through them. They need cool dry vapors of water with electrolyte. If the other reactors' rods develop short magnetic signatures then cut the rod lengths down and then start again on a hydrocarbon fuel. Then after the engine warms up the reactor's slowing turn off the hydrocarbon reactor and see if the engine will keep running on only the cracked vapors of water with electrolytes.

2. When measureing the magnetic signature of where to cut, use a very small compass and turn the rod many diffent directions to get an overall picture in your head because the Earth's magnetic field will add or subtract from the readings and affect how long it looks like between poles.

Some rods materials like steel will hold the signature better than cast iron because the cast iron very easily changes its fields to align with the Earth or even to align with the magnetic field coming from the compass needle. And some rod materials both hold the signature better and give a better reaction than plain cold rolled steel. But GEET never would say what those materials are.

3. Multiple small reactors in parallel with small gaps for both inner vapors and small gaps for exhaust gases will likely be easier than just using 1 reactor to get it to work with only water or almost all water as the fuel. Even with just 1 reactor, a little smaller similar yours with a .4 inch rod may be better for small engines.

Radioactivity in the GEET

Reports given to the nonprofit organization stated the following: The US government nuclear safety people sent a team of people out to shut Paul Pantone down after it was reported that radioactivity was detected. However after testing everything while his unit was running (and with a Geiger counter) they could not identify the radiation as being some form that is classified as dangerous so they said he could keep experimenting.

See this video about it:

http://www.youtube.com/watch?v=nZahzVFpv6U

The man that Paul mentions at the beginning of that video is a physicist who was helping Paul with his research. Paul Pantone has also had ALL emissions from a working GEET analyzed at university physics labs and he has not mentioned poisonous emissions among them. That was when he verified that transmutation was occurring, similar Jean Chambrin's unit. The universities that have helped analyze the emissions have refused to release any official reports about it because they said the results were impossible.

You Tube educational video series.

GEET Water Fuel Plasma Reactor Chamber Explained Part 1

GEET Water Fuel Plasma Reactor Chamber Explained Part 2

At heart of the Multi-Fuels Processor of GEET is a self-inducing plasma generator or a plasma reactor with an endothermic reaction. The endothermic reactor is composed of two coaxial steel cylinders:

- the interior cylinder (threaded at each end), called the pyrolytic chamber (430mm length and 15mm of inner diameter) contains a steel rod of 300mm length and 13mm of diameter (not magnetized before the burning-in). A side of this steel rod is round in order to identify its magnetic polarity after its disassembling. The rod is maintained in the center of the pyrolytic chamber with to 3 small nipples welded at each end.

- the external cylinder (threaded at each end) is a steel tube of 300mm length and 26mm of inner diameter. The two cylinders are placed coaxially with two reducing T (showed on the diagram below) placed at each end. The bubbler is a tank containing a mixture of water and hydrocarbon (gasoline, diesel, kerosene, crude oils and others derived from hydrocarbons...).

The hot gas flow coming from the exhaust of the engine circulates by the outside part of the reactor with a strong kinetic energy, that contributes to bring up to very high temperature the steel rod (being used as heat accumulator) contained in the pyrolytic chamber. The gases cross the engine and penetrate then in the bubbler containing the water/hydrocarbons mixture. The vapor of the mixture is strongly aspired by the vacuum created by the engine intake and is pushed by the pressure coming from the exhaust. The kinetic energy of the vapor is increased considerably by the reduction of the diameter in the pyrolytic chamber (by Venturi effect). The combined effect of the high temperature and the increase of the kinetic energy produces a thermo chemical decomposition (molecular breakdown) of the water/ hydrocarbons mixture.

The endothermic reactor forms an Electro-Plasma-Chemical unit (EPC) and it is now possible to create a high-output fuel coming from the decomposition of the water contained in the water/hydrocarbons mixture. This fact is confirmed by the presence of oxygen gaz (O2) in great amount measured in the exhaust.- Reference End.

Mr. Pantone's invention is to recover energy lost through the exhaust (exhaust cold at fuel processor outlet) and to recover it *into a form that is directly convertible to mechanical energy by internal combustion engines.* The GEET processor produces some sort of highly efficient fuel from various mixtures. In the tractor case, this is an air and water vapor mixture. The GEET processor seems to produce a gaseous fuel using the exhaust heat as energy source. This fact is demonstrated from the following: The exhaust gases at the outlet of the GEET processor are largely colder that at the inlet.

It can't be a simple gas heating effect through the processor. In fact, heating the inlet gases before motor intake dilates these gases, thus reducing the amount of combustible mixture entering the cylinder as the engine always sucks in a constant gaseous volume. Heating the inlet gases thus reduces the motor power. In contradiction, the power or retrofitted engines is augmented for reduced fuel consumption.

Mr. Martz's calculations demonstrate that the water cracking into oxygen and hydrogen in the GEET processor does not produce enough energy to explain the fuel consumption reduction. It is possible that other gases present such as nitrogen are also modified in the GEET.

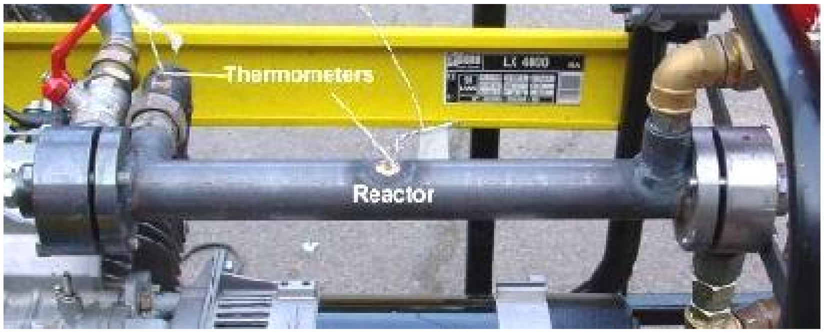
Note that there is no energy generation in the GEET fuel processor: there is only an efficient recovery of the energy normally lost in the exhaust. The recovered energy serves to modify the chemical composition of the gases inside the reactor.

Gases produced by the GEET fuel processor are stable, as the engine intake is somewhat far away – up to more than a meter from the processor outlet to the actual combustion in the engine. If they were unstable composites or molecules, they would decompose or recombine in the piping before reaching the engine. From this reflection, free atoms such as single atoms of hydrogen, oxygen or nitrogen (H, O and N) might not be considered. Or they could be precisely the answer: the GEET fuel processor would produce free atoms which recombination would liberate much more energy than the simple combustion of hydrogen molecules with two atoms (H_2) with oxygen molecules also containing two atoms (O_2). The heat of combustion used by Mr. Martz is the one of the usual hydrogen combustion reaction: $H_2+ O_2= 2 H_2O$

Another important element is that the engine efficiency increase can't be provoked by water vapor's presence. As a matter of fact, water vapor present at the motor intake dilutes the combustible gases (fuel and oxygen), thus reducing temperatures and pressures in the cylinder and therefore the engine power and efficiency. Moreover, the water vapor presence increases the compression work; calculations of Mr. Martz indicate that the compression work is greater than the energy recovered by the expansion work (negative balance). An engine's behavior is normally little affected by the air humidity: dry or humid air does not make much difference. **There is something else going on in the GEET fuel processor, something still to identify. –**End.

"**GEET Pantone study** : Final engineer studying project for ENSAIS (Ecole Nationale Supérieure des Arts et Industries de Strasbourg) Diploma by C. Martz. November 24th, **2001**. http://quanthomme.free.fr/pantone/martz

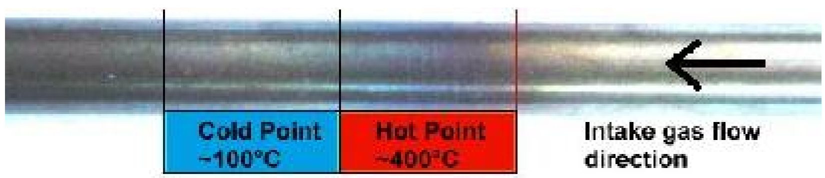
"Design of a testing ground and characterization of P. Pantone's GEET process based on hydrocarbon conversion" This testing ground has been built **to characterize the GEET process** by measuring definite points as, for instance, the specific consumption (i.e. output) ,different flows, temperatures, pressures, H2/O2 gas analysis...



The **fall of pollution on CO and HC (un-burnt) is remarkable**. Furthermore on carbon gas the pollution is falling with the motor's charge (40% to 70% of diminution). Maybe the conversion reaction is **more efficient when the motor is loaded** (that means exhaust gas are hotter).

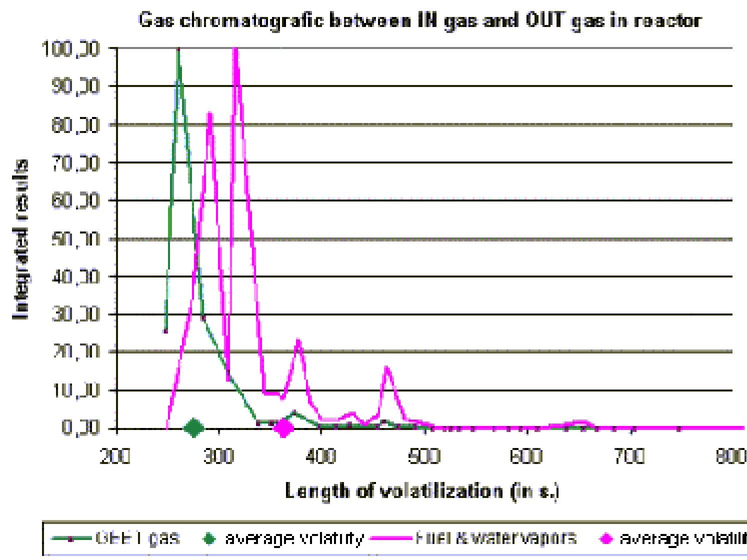
At this time **we don't know where the rest of carbon is:** a classical gas or fuel combustion give 14 to 16% carbon gas (CO+CO2), with GEET we are at 6%maximum (only CO2), which means that **quantity of carbon are missing in exhaust gas**...I hope that further experiences would solve this problem and say where is the carbon.

After a few hours of functioning, we have noted some interesting remarks on some rod (we got 9 rods of different dimensions): Observing a **hot point** on the cold side of the rod. A cold point is just after this hot point which shows that the **reaction has inner-reaction high temperature**. I mean **<u>this hot point does NOT come from exhaust gas high temperature.</u>**




Observing a spiral appearance on one rod after more than 15 h of functioning.

... But if really it is not hydrogen, the GEET gas is **certainly a high hydrogenous gas (smells of ether)** which got the energetic advantages of hydrogen. GEET **gas coming out of reactor is more volatile and simple** than the fuel and water vapors coming in. Then it is **sure than there is a gas conversion in the reactor** but the obtained GEET gas must be exactly definite... **GEET gas is still unknown**, it MUST be analysis by other ways (chemical, spectral...).



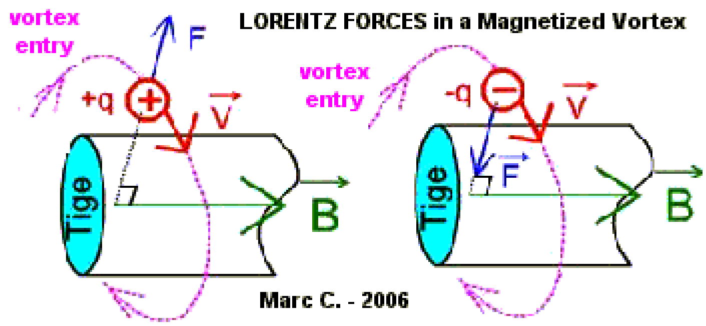
- **SPECIAL DELIVERY from FRANCE**: Le GEET c'est GENIAL ! (GEET it's GREAT !)

Ion Vortex Theory, Marc C., Nov. 2006, original pdf in french:
http://quanthomme.free.fr/qhsuite/imagenews06/theoriechampmarc281106.pdf
and also see http://quanthomme.free.fr/qhsuite/separionfluidmouv.htm

REMARK: need technical translator for this 24 pages document, really very interesting … here I can only put some schematics.

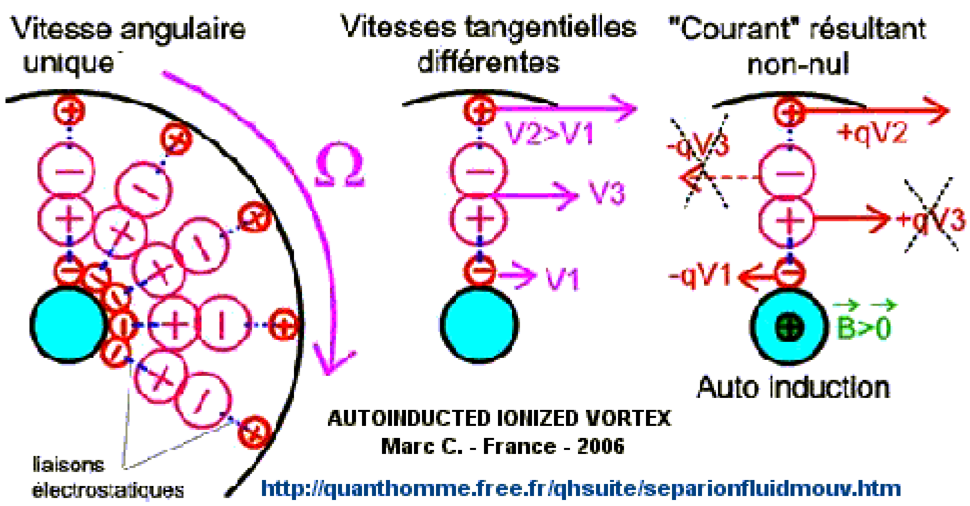
A study on the Pantone Reactor. All info and schematic are given in the Public Domain by their author.
Extracts: The **LORENTZ forces** are alsmost not discussed in high school, they are just to introduce those of **LAPLACE** : we **abandon the 'charged particle', to replace it by 'an electric current in a conductor'**. Big mistake:

- An **Electron moves at only 2.16 kmh** in the copper; we could follow it by walk … with good eyes!
- An Positive or Negative **Ion moves at 800 kmh** … 370 times faster than an electron! The main difficulty is to **control and canalize the Ions** that are just willing to 'fly away' ; that where the Vortex is interesting …

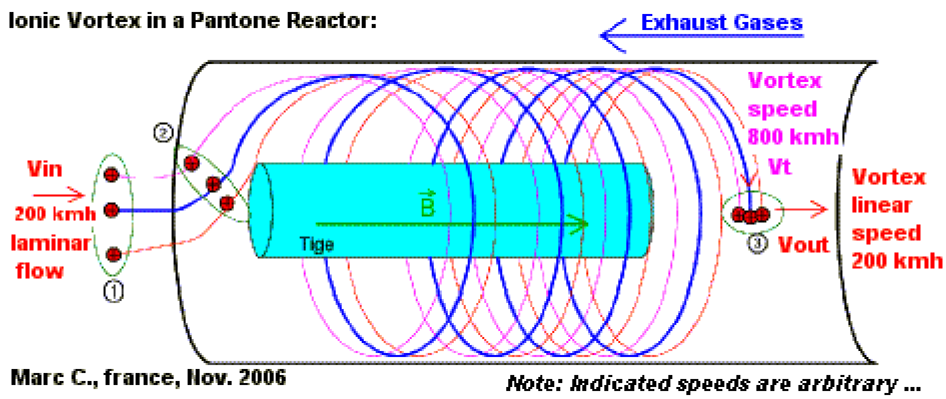


It took me 5 years to take seriously the Pantone's patent. How could I expect that my readers believe immediately in this theory. We must let time for reflex ion and experimentation first .

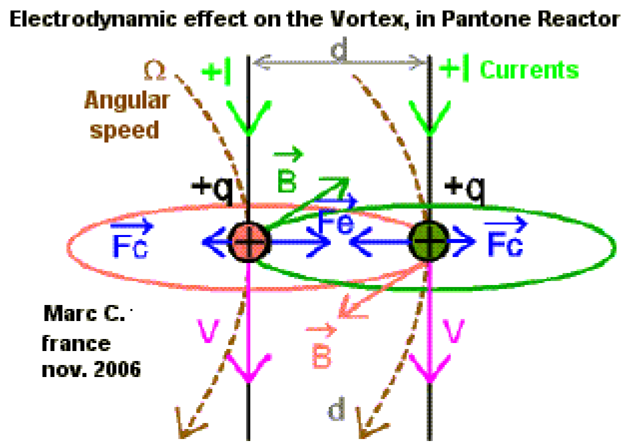
A good example is the **Law of Coulomb** that every good student memorized: " the **opposite signs are attracted, same signs repel**" **How many learned that it is sometime the opposite, the Electrodynamics Forces (Lorentz ones) surpassing those of Coulomb?** Could my theory re-establish at least this notion of DUALITY, ever existing in Physics.



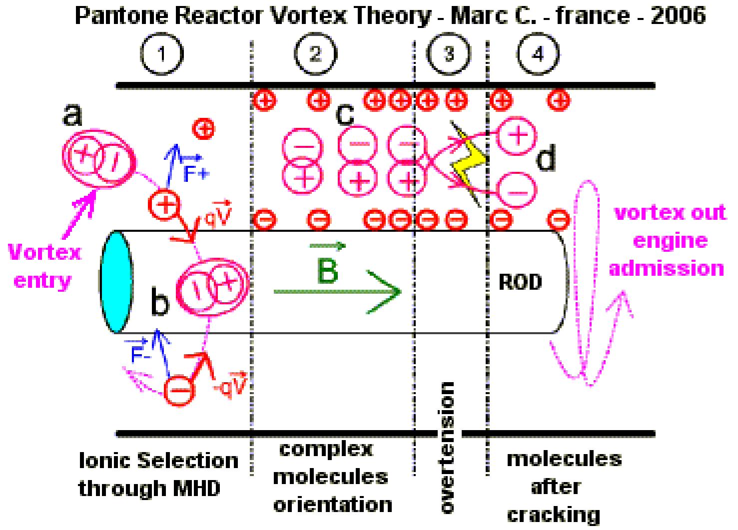
Ionic Vortex in a Pantone Reactor:



Each Ion induces a magnetic field in proportion with its speed. The Ions closed to each other in the vortex tend to come closer when their speed increases.



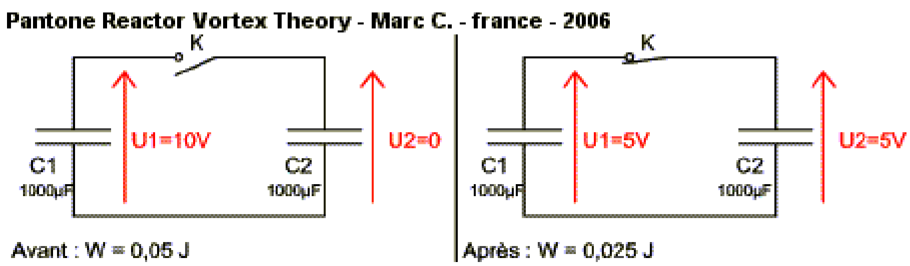
In fact the Electrodynamics Force (Fe) becomes stronger than the Coulomb force (Fe) that was repelling them. Optimal Molecular Breakdown If the reactor can transform the mix of air and water (+ eventual fuel) in a plasma, like Paul Pantone said, then we obtain, theoretically: H2O=H+H+O, CO2=C+O+O, NO2=N+O+O, etc … In other terms, we obtain production of Mono Atomic Hydrogen and Oxygen, from water or from the oxides present in the air ; so the vortex acts like an air purification.



Pantone's Reactor is an electric system that produce an electric field without generator

The Pantone Reactor is **an electric apparatus that produces an electric field without generator!**
The Vortex associates Magnetic and Electric Fields, without limiting the last one. **No electron flow, No energy dissipation**. Efficiency is closed to 1.

The **principle of the Charge Conservation wins on the Energy Conservation one**.
Paradox of the Charge Conservation:



Charge Conservation win on Energy Conservation

By which way this 'half-energy', that became 'un-desirable', has been thrown out of the circuit ? To learn more about this, just put a radio receiver or TV close to the circuit, to see or hear the parasists: joining the 2 condensates liberated some electromagnetic energy as a pulse; from where appear an 'infinity' of radio waves radiated all around .

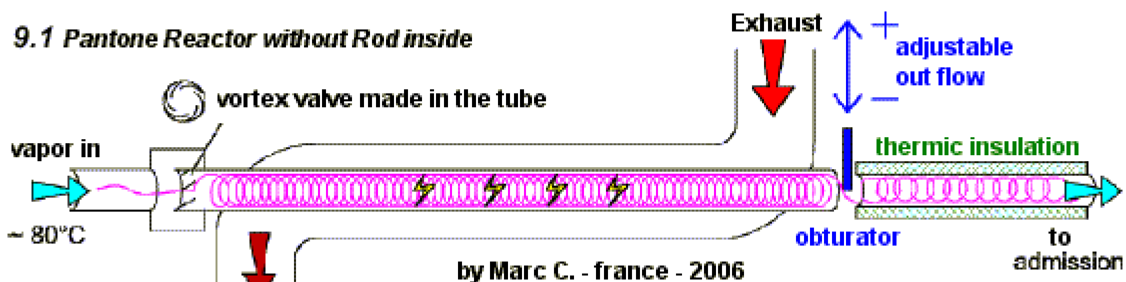
Is it possible to conciliate 'Charge Conservation" and 'Energy Conservation" in a closed system, without ever use external energy ?

About the Law of Charge Conservation : **from : http://jnaudin.free.fr/html/tepcoil.htm**
The Law of charge conservation 1,2,3 has been introduced by Benjamen Franklin (1706-1790) :
The Law of charge conservation can be written "Electricity is never created or destroyed, but only transferred" or like this "In any closed system the sum of all electric charge remains constant ".

Because of the particular links that it creates between the molecules constituting it, an **Ionic Vortex is comparable to a spring that contracts when heat is added, and expands in the opposite case**.

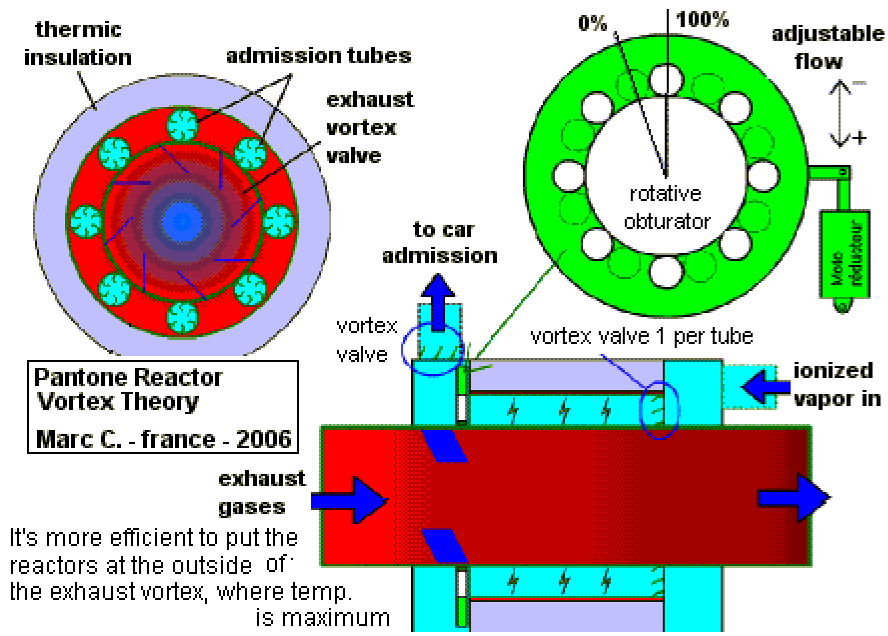
8.3 Water boiling: By which way the boiling **water limits its temperature** at the pot top surface at around 100 Deg.C.? It is said that the water gives away its internal energy in the form of cold; but **where did the water store this energy**? Myself, after reading an article on the web about **'micro vortices'' observed in water vapor** (article that I can't find again), I think that the **liquid to vapor transit** comes with a re-organization of the molecules, what **creates micron sized ionic vortices', converting the heat in a organized molecule movement**.

Pantone Reactor <u>without Rod:</u>

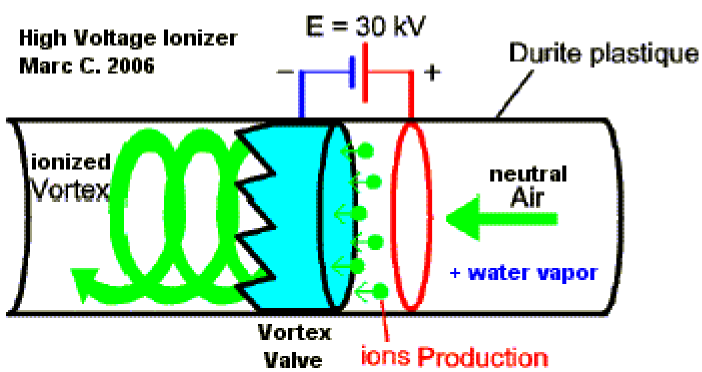


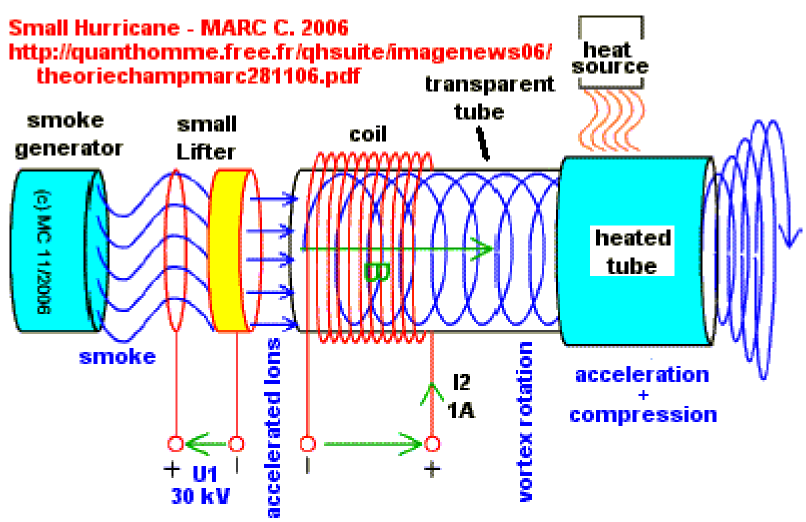
http://quanthomme.free.fr/qhsuite/imagenews06/theoriechampmarc281106.pdf

By **taking out the rod**, the magnetic **induction become less intensive** than before, but the **global efficiency is increased** because the **Ions are no more in friction** with the rod. This Ions at the center of the vortex, rotating almost along the tube axle, have a speed quasi null, then **the magnetic interactions with the external crown of the vortex are increased** ...



http://quanthomme.free.fr/qhsuite/imagenews06/theoriechampmarc281106.pdf

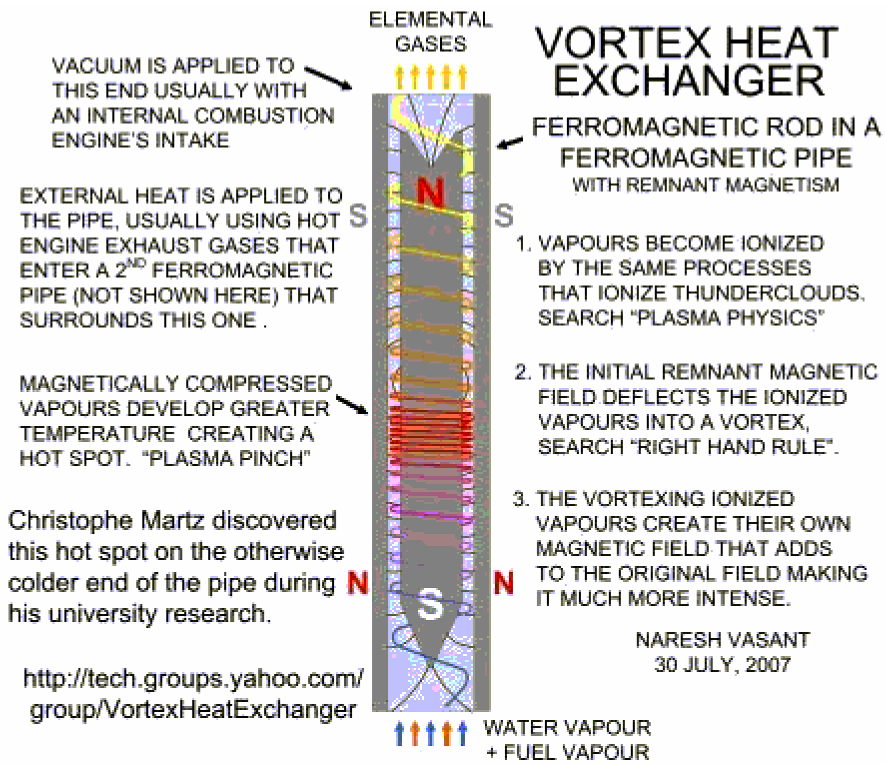




-End

The following is courtesy of open source Engineer Naresh.

Vortex Heat Exchanger Group R and D



Naresh's group is working in this version of the technology, links to his technology group are provided at the end of this document.

This group is for R & D, theory and practical use information about how to build a fuel reformer that uses a vortex heat exchanger for reforming fuels for your car, farm equipment or genset as well as for fuel cells. This fuel reformer is for hydrogen production and will help people to get much better fuel economy. Research on this type of reformer was started by Jean Chambrin and others around the world. Jean Chambrin's system is patented under patent numbers WO8204096 and WO8203249. Later a similar system was patented by Paul Pantone in the USA which he called the GEET (TM) reactor fuel pretreater.

Paul Pantone's U.S. patent number 5.794.601 does not mention vortex action nor unusual electromagnetic phenomena however these were reported by Paul Pantone and other independent researchers investigating his fuel pre treater's performance. The fuel pre treater uses a heat exchanger with a ferromagnetic pipe inside a ferromagnetic pipe with a central ferromagnetic rod in the inner pipe

Adding air causes a partial oxidation reaction which is an exothermic reaction that can provide some of the heat needed for the steam reforming reaction which is an endothermic reaction that releases hydrogen from both the water and the hydrocarbon fuel.

A GEET type of fuel reformer works better with exhaust gases instead of air. The exhaust gases through the reactor with fuel and water vapors and heating the chamber on the outside with exhaust both provide all the heat needed for the endothermic reaction. Sending air through a GEET reactor causes a higher tendency to eventually clog up the reactor with carbon deposits. Engine coolant is no where near as hot as the exhaust gases.

Donato Tommasi's research

The not well understood parameters that made Tommasi's device not work are also what make people's GEET not work well also. But I think if the vapors are well electrified then it works better. And there is something special about the type of electrification that gets intensified by a rapid change in temperature. When water vapors change state rapidly it seems to manifest more electrostatic effects.

Using search word magnetized:

http://books.google.com/books?q=steam+electricity+copper+tube+atmospheres+iron+magnetized

NYPL RESEARCH LIBRARIES, SUPPLEMENT TO "THE ELECTRICAL ENGINEER,"JUNE 29, 1888. THE ELECTRICAL PAGE 266

"The Steam Electro Magnet. — Many of our readers, like us, may have been under the impression that — whatever be the explanation of the fact — a core of iron becomes magnetized, in some degree, when a current of steam is caused to traverse with sufficient velocity a tube of copper, wound helically upon the core. The experimental result in question was first obtained, we understood, by M. Tommasi, and his results were subsequently confirmed by M. Thouvenot. A correspondent of l' Electrician now states that he has repeated the experiment with an entirely negative result, even with higher steam pressures than had been mentioned as necessary. Messrs. Tommasi and Thouvenot have, under these circumstances, been requested to render assistance or explanation; and the Editor of the above-named paper is about to repeat the experiment for the satisfaction of those who are interested in the question."

PAGE 387- "Steam Electro-Magnet. — In a note under this heading (p. 266)

We mentioned that the discovery of M. Tommasi, said to have been confirmed by M. Thouvenot, had been called into question by a correspondent of l ' Electrician. In the current number of that journal, a somewhat unsatisfactory communication from M. Tommasi appears, in which he states that "When a current of steam under a certain pressure is passed through an ' unsoldered copper tube wound about an iron bar the latter becomes magnetized.' " This, M Tommasi affirms, "is an indubitable fact which has been verified by several distinguished persons," amongst whom he mentions MM. Desains, Moigno,and Parville. It is admitted, however, that the experiment does not always succeed, even when it is repeated under conditions apparently identical with those prevailing when a positive result has been obtained. M. Tommasi can suggest no explanation of this curious circumstance, but observes that the necessary conditions are not sufficiently understood. We may presume that this gentleman, those above-mentioned, and M. Thouvenot will make some endeavor to make themselves acquainted with these conditions; for an experiment which cannot be repeated is not scientific evidence, and affords no basis for an alleged discovery. –End

The key to getting the GEET to work correctly is to get the vapors electro statically charged up. Using methods like discussed in these old science publications:

The Edinburgh New Philosophical Journal, page 54

The experiment of Professor Faraday, on the electricity of steam, from which he was led to conclude, that no electrical development could take place from the evaporation of a saline mixture, and that evaporation is not the cause of the electricity' of the atmosphere, tell against the theory ; but the conclusions of Pouillet on the electrical phenomena of evaporation are directly the reverse ;for, although he agrees generally with Dr Faraday on the phenomena of the electricity of steam, he " has demonstrated that the conversion of pure water into vapor, at any temperature, is not attended with any disturbance of the electric equilibrium, but that vapor, rising from solutions, however weak, gives signs of electricity, varying in kind, according to the nature of the substance dissolved. From saline or acid solutions, the vapor carries up a charge of positive electricity, and leaves the solution in a state of negative electricity ;and the rule was verified particularly with regard to solutions of sea-salt." The experiments of Armstrong and Pattison, on the electricity of steam, were with boilers without any arrangement for causing friction, and without regard being had to the purity of the water.

The experiment of Mr Pattison shows the vast quantity of electricity carried off by vapor, as he says, — " I repeated Volta's experiment, by placing a hot cinder upon the cap of a gold-leaf electrometer, and projecting a few drops of water upon it, when the leaves diverged strongly with negative electricity. I observed that when the cinder was very hot, and the production of the steam consequently very rapid, the electricity given out was always most powerful. "I then insulated an iron pan, 12 inches diameter and 2 inches deep, and attached to it a pith-ball electrometer,2 inches deep, and attached to it a pith-ball electrometer, with balls ¡n this of an inch diameter, and threads 5 inches long, and also attached to the pan a metallic wire, the pointed extremity of which was placed about -/„th of an inch distant from the point of another wire connected with the ground.

The iron pan was then filled with cinders, very hot, from a wind-furnace, and on projecting upon them a few ounces of water, steam was evolved with great rapidity, and at the same moment the pith-balls diverged to the distance of an inch, and sparks passed between the metallic wires. This was several times repeated." — (London and Edinburgh Philosophical Magazine, 1840, p. 460.) And the experiment on evaporation from insulated and UN insulated vessels (an account of which I submitted to the Ashmolean Society in 1841)* tends to show that electricity is a necessary agent in evaporation at moderate or low temperatures.

It is of little consequence, as regards the phenomenon in question, whether the vapor carries off the electricity, or whether (as I have endeavored, in former papers, to show) the electricity carries off the vapor ; it is sufficient to know that, during evaporation, positive electricity is carried off, and the water left in a negative state ; Electricity By Robert M. Ferguson

Chapter: FRICTIONAL OR STATICAL ELECTRICITY. page 73

When water is evaporated, it is usually found that the vessel from which it is evaporated is electrified, and that the vapor has the opposite electrification. The electrification apparently depends upon the nature of the other substances present in the water. If it contains free oxides of such metals as potassium, sodium, calcium, the water becomes positively electrified: if there is a soluble acid or a carbonate chloride, the water is negatively electrified.

When the water is perfectly pure, it does not become electrified on evaporation. 64. Sir William Armstrong invented an engine by which electricity can be generated by the friction of steam. It consists of a boiler on insulating supports, which supplies steam to tubes which pass through a condenser, D (fig. 37), filled with cold water. This condenses the steam partially, and it then escapes through nozzles, A, so formed as to cause much friction between the escaping steam and the sides of the nozzles. A comb, P, provided with a series of points is placed in the jets of steam, and collects the electricity and conveys it to the prime conductor, B. In ordinary circumstances the prime conductor was charged positively and the boiler negatively, and large sparks were obtained. Faraday investigated the action of the Hydroelectric Machine, and showed that the small drops of water produced by the partial condensation' were essential to the production of electricity: that electricity was due to the friction between these drops and the sides of the nozzles, for «" changing the material with which they were lined, the amount or the kind of electricity produced was changed:

when the water was made a conductor, by dissolving in it any salts, acids, &c., no electricity was produced : when turpentine or any fatty substance was added to the water, the boiler was charged positively, the prime conductor negatively: the production of electricity increased with the pressure of the steam. A current of moist air driven through the nozzles charged them negatively, but carried positive to the points: there was no electricity produced when perfectly dry air was used.

Elements of Chemistry: Theoretical and Practical By William Allen Miller

ELECTRIC EFFECTS OF CIIKMICAL ACTION, OF VAPORIZATION. P 359

Electricity of Vapor.—The act of evaporation has also been asserted to be one of the sources of electricity, but the truth of this statement is doubtful. It is true that if a few drops of water fall upon a live coal, insulated on the cap of the gold-leaf electroscope, the leaves of the instrument diverge. This, however, is due to the chemical action between the coke and the water, and not to mere evaporation; for by allowing pure water to evaporate in a clean hot platinum dish connected with the electroscope, no signs of electric disturbance occur. Pouillet found that on allowing alkaline solutions to evaporate in the capsule, the electroscope became charged positively; with acid solutions, the charge given to the electroscope was negative: but Peltier states that these electrical effects may nevertheless be due

to friction, as they do not manifest themselves until the liquid is nearly all driven off, and a crepitating of the salt as it detaches itself from the sides of the capsule begins to occur. This is corroborated by Faraday's observation, that if the dish be heated to redness and pure water be dropped in, so long as it evaporates quietly in the spheroid form (198) no electricity is developed; but the moment that it cools down sufficiently to boil violently with friction against the metallic capsule, the leaves diverge powerfully. Electricity is also developed during the process of combustion; carbon, for example, becoming negatively electric, whilst the carbonic acid is positive.

In like manner hydrogen in the act of burning was found by Pouillet to be negative, whilst the vapor produced by it was positive. In accordance with this observation, Faraday has explained the development of electricity by high-pressure steam, which occurs to so remarkable an extent under certain circumstances. This he has traced to the friction of water accompanying the steam against the orifice of the jet through which it escapes into the air. An insulated boiler from which steam is allowed to blow off at high-pressure through long tubes, in which a partial condensation of the steam occurs, furnishes, as in the hydro electric machine of Armstrong, exhibited at the Polytechnic Institution, an admirable source of high electric power. In this experiment, the boiler becomes negative, the escaping steam being positive. It is remarkable that the presence of the smallest quantity of oil or of essence of turpentine in the exit-pipe reverses these electrical states. A solution of acetate of lead produces a similar effect. Indeed the purer the water that is used in the boiler, the better is it for these experiments, and the more uniform re the results. The electric condition of the steam was found by Armstrong to be also influenced by the material of which the exit-pipe was formed ; glass, lead, copper, and tin. -End

In simple definition, the GEET Fuel Processor could be called a new type of carburetor with a miniature refinery built in. With it, There is no need for catalytic converters, smog pumps and many other costly items on cars , as the GEET fuel processor is not just a fuel delivery system it is also a pollution elimination unit! Your car mileage will be greatly increased if you are truly consuming ALL of the available energy from whatever fuel you may be using.

A model suitable for a small two- or four-stroke (lawn-mower or small generator) typically consists of two horizontally-lying, concentric steel or metallic pipes of about 50 cm in length, one inside the other. The outer pipe has an inside diameter of 25.4 mm, the inner pipe an outside diameter of 12.7 mm and an inner diameter of 12.4 mm. Within the latter is a long solid steel or iron bar, whose diameter is 12 mm, that doesn’t touch it, except at three solder points at each of its extremities. Let us call A and B the two ends of the 50 cm long pipes and bar.

The exhaust from the engine travels

* From A along the "outer" concentric space, between the two pipes, to B.
* From there, it is sent bubbling at high pressure to the bottom a jug of water with some fuel that is vaporized by the heat.
* It is then sent along the inner pipe, in the thin space round the central solid steel bar, back from B to A, to near the air intake, where it is mixed with some fresh air.
* The latter mixture is input to the motor

A preliminary analysis of the GEET: Two-strokes are known to be inefficient as only a certain proportion of their fuel is burnt. Their exhaust typically consists of the following:

1 - Air somewhat depleted in oxygen
2 - Carbon dioxide
3 - Carbon and nitrogen monoxide
4 - Water vapor
5 – Un burnt volatile gasoline
6 - Particles of heavier hydrocarbons, oil and soot

In the case of four-strokes, there are less of 5 and 6.

* As the exhaust first travels between the "outer" space, between inner and the outer pipes, it heats their surface to its own temperature. In order that this temperature is as high as possible, the outer pipe should be thermally insulated with a glass wool jacket. Another contribution to higher temperatures at the inner surface of the outer pipe involves the Ranque-Hilsch effect: the exhaust flow should spiral, so that the hotter components in the gas gather against the outer surface where the steam is more thoroughly reduced into hydrogen while the pipe surface is oxidized. In turn, the released hydrogen reacts with the carbon dioxide into carbon monoxide and water ($CO_2 + H_2 >> CO + H_2O$) at high temperatures, while the steam can again be reduced by the hot iron into hydrogen. Provided that the outer surface of the cooler inner tube contains catalyzers such as nickel, already at 200° C, carbon dioxide and hydrogen combine into methane and water ($CO_2 + 4H_2 >> CH_4 + 2H_2O$), the latter of which can again be reduced at the hotter surface of the outer pipe. Therefore, both the water and the carbon dioxide are reduced, the exhaust becomes depleted in carbon dioxide and enriched in fuels such as carbon monoxide, hydrogen and methane.

* This pretreated exhaust bubbles through the jug of water and fuel, the latter remaining at the top when not miscible (gasoline, heavy fuel or miscible glycol alcohol, etc). The depth of the water increases the pressure in the preceding reducing stage. Now, along with some soot, heavy hydrocarbons and unburnt fuel that are recycled, the carbon dioxide dissolves in the water and is removed from the exhaust so long as the water isn't saturated. To increase the amount of carbon dioxide dissolved, the pressure should be maximal and the water circulated. In critical closed-cycle applications, the resulting carbonic acid could react with a metal such as zinc or magnesium to release hydrogen. The resulting carbonate and hydroxide, as well as the reducing metal of the inner surface of the outer pipe could then be recycled later by using solar energy. Another option is using some mix of photosynthetic algae in an adjacent first stage to convert the carbon dioxide into oxygen and biomass, and fermenting anaerobic bacteria in a second stage to generate methane and hydrogen from the latter.

* The fuel as well as some water is vaporized in the bubbler.

* The cooled and enriched exhaust now travels at high speed inside the inner pipe, as the available space is thin, round the solid steel bar. Here, it must be observed that there are heat gradients, as the outer surface of the inner pipe is heated by the exhaust, while the steel bar inside that doesn't touch it is cooled by the cooler flow of the bubbled exhaust. The Ranque-Hilsch effect can again be used to further reduce the temperature round the inner bar. This involves replacing the three extremely solder points by small soldered coiled lines of wire at the B end of the iron bar.

* Some of the previously generated hydrogen may, here again, catalytically combine with the remaining carbon dioxide into methane and water against the outer surface of the nickel inner tube.

* Because steel is magnetic and its Curie temperature is even higher than that of the outer, hotter pipe, all the surfaces inside the GEET are microscopically strongly magnetized, locally, on the level of magnetic domains of about 80nm, even if this magnetism isn't apparent macroscopically. However, only the inner steel bar is in contact with a sufficiently cool flow so it is below the Curie temperature of the Magnegas.

As a result, when the molecules bounce against the surface of the pipes, they experience a strong magnetic field of several Tesla. As R.M. Santilli has shown, diatomic molecules such as H_2 , O_2 and CO can be magnetically polarized, and may assemble into clusters that this researcher calls magnecules. These have a Curie temperature which is at about 150° C for H_2 and CO. The rate of formation of such magnecules will thus be higher on the cooler surface of the steel bar. The corresponding magnetically polarized gas is called a Magnegas $_{(TM)}$. Because most chemical reactions involve polarized molecules while ordinary gases are un polarized, magnegases release far more energy than expected from the combustion of their UN polarized counterparts. Also note that, due to the recycling, the O_2 molecules may pass several times into the magnetically polarizing cavity.

MASER emission might also occur in this cavity, which might accelerate the formation of magnecules.

The recycled and enriched exhaust thus in the end contains:

* CO, NO, O_2 and H_2 molecules, the latter resulting from the reduction of steam on the outer hot steel surface or from biomass recycling.
* Magnecules of the latter.
* Some methane from catalytic conversion of carbon dioxide and hydrogen or from biomass.
* Recycled un burnt fuel.
* Vaporized fuel from the bubbler.
* Less CO_2 than in the original exhaust, at least until the water becomes saturated in the simplest devices. This suggests the importance of increasing the pressure in the bubbler.

The mechanisms involved suggest an improvement in efficiency from:

* Thermally insulating the outer pipe.
* Placing reducing elements at the inner surface of the outer pipe, with high surface area if in the solid state, or as a liquid circulating blanket maintained by centrifugal forces in a rotating configuration.
* Using spiraling vents at the entry of the exhaust into the cylindrical outer space, and coiled elements at the entry of the bubbled exhaust round the inner bar so that the flow spirals and, by the Ranque-Hilsch effect, concentrates its hot components on the outside and its cooler ones on the inside.

* Using a steel or alloy with high magnetic permeability and saturation, or very pure Iron for the inner bar.
* Polarizing the fuel in the bubbler into a Magneliquid, and the fresh air into a Magnegas.
* Increasing the pressure at the bubbler so that a maximal amount of carbon dioxide is dissolved.
* Using a metallic powder of Zinc or Magnesium so that the resulting carbonic acid releases hydrogen and carbonate in critical closed-cycle applications, or a multistage biomass of photosynthetic algae and anaerobic bacteria to convert the carbon dioxide into oxygen and biomass and the latter into methane in less critical or fixed applications.

The central iron bar should be at less than 150° C (the Curie temperature of Magnegas), the surrounding catalytic pipe at about 200° C (that converts carbon dioxide and hydrogen into water and methane), and the outer pipe at yet higher temperatures.

According to the inventor, Mr Pantone, the central steel or iron bar acquires an overall magnetization and must always be oriented in the same way with respect to the magnetic north in devices where it is horizontal, and similarly with respect to the vertical, when vertical.

The energy balance:

On the minus side:

* The vaporized fuel spent (whatever the actual proportion of fuel in the bubbler, which can be as low as 20%)

* The steel or reducing agent oxidized, mainly at the inner surface of the outer pipe

* The metallic powder turned into carbonate.

On the plus side:

* The un-burnt fuel and hydrocarbons recycled, especially for two-strokes

* The un-burnt CO and NO recycled
* The increased energy released by the use of magnecules
* The possibility of using a wide variety of cheap fuels

* Dissolved CO_2 converted to oxygen and biomass and then the latter into methane

and hydrogen in several stages or into carbonates and hydrogen by a metal in the bubbler itself or some adjacent reactor.

Any test of exhaust emissions should take into account the CO_2 retained in the water. Also note that, when this CO_2 is eventually released in the atmosphere or recycled, one is left with a brew consisting of residual, un-volatilized fuel, soot and various heavy hydrocarbons, which would be ideally suited for recycling in a "Hadronic Reactor" into Magnegas. Thus, provided that the overall cycle proves to have a favorable efficiency, there might be a synergy between the GEET and Hadronic reactors, as they both involve Magnegases and the waste from the one may be taken as starting materials for the other.

For most two-strokes, there should be quite a significant improvement in efficiency from the recycling of the UN burnt fuel alone. For other motors in which there is less of the latter, the gain could be lower but still not negligible. Note also that the Magnegas produced in "Hadronic Reactors" is unsuitable for two-strokes, as these require a liquid fuel into which the lubricating oil is mixed.

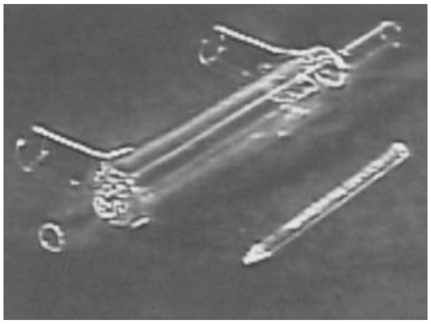
Thus, this system has several positive points. On the other hand, claiming that it runs on 80% of water and 20% of fuel when this is just the proportion that is present in the bubbler where the fuel is preferentially vaporized by the hot exhaust, ignoring the oxidation of the metal in the pipes and their effective lifetime, ignoring the CO_2 retained in the water, especially during the first ten minutes after start-up, as well as the liquid wastes that are produced when measuring the exhaust emissions and not mentioning for how long a specific test was performed can be very misleading, to the point of bordering on fraud.

Suggested improvements involve the use of spiraling aerodynamic flows so as to optimize the temperature gradients at several key locations by the Ranque-Hilsch effect (to minimize the temperature round the central iron bar, and maximize it at the inner surface of the inner and outer pipes), thermally insulating the outer pipe, increasing the pressure so as to maximize the solution of carbon dioxide in the bubbler, and circulating the resulting carbonic acid in adjacent reactors, using a multistage configuration of photosynthetic and anaerobic recycling biomass to convert it to oxygen and methane or using a reactive metal to release hydrogen in certain critical closed-cycle applications. Solar energy can be used at a later stage to release the oxygen taken up by the reducing metal and recycle it.

Glass Reactor

Demonstrations were given in Las Vegas at the Fitzgeralds Hotel and Casino for just a few members and some outsiders of new designs and improvements of the last few months. The components for the "all glass"

GEET made 9 of these glass reactors for research. They saw little sparks along the rod and a white glow 1/2 inch long at the rear end of the rod." It sounds to me like these glass chambers would be perfect for helping us understand what's happening inside the reactor, and determine if it's working, how well it's working, or why it might not be working. For those of us who are using multiple reactors, we could use our DOM tube chambers in conjunction with the glass chamber.



WE have given reports that other have talked about this with Paul. A company tested it and it worked. They wanted to see what was going on inside of it. They were able to take readings and acquire proper information from it. They found electricity inside of the reaction chamber and were able to measure it. It was hooked up the same way everything else is hooked up. Just not to the melting point of glass! be careful with it to not break it. Because that's something that's defiantly needed to be seen in action. That's all I can say on that.

Nickel as a catalyst for the reaction

Have you heard of the Sebatier effect? It's to do with using Nickel as a catalyst basically & it further confirms my idea of getting the Reactor Rods Electroless Nickel plated. Have a look at this & see what you think. http://en.wikipedia.org/wiki/Sabatier_process - http://en.wikipedia.org/wiki/Raney_nickel

Faculty links

Q and A from –the quanthomme web site

Research links

To Help Free Paul Pantone please visit - http://www.geetfriends.net/

http://geet-pantone.com/

http://freeenergynews.com/Directory/Geet/

http://www.rexresearch.com/pantone/pantone.htm http://geetpantone.blogspot.com/

http://waterfuel.t35.com/geet_plasma.html

Technical support groups

http://tech.groups.yahoo.com/group/VortexHeatExchanger/

http://tech.groups.yahoo.com/group/geet-pantone/

French technical support (Translated)

Understanding of the process Pantone

Background on the engine Pantone

In the geographic support around the engine Pantone

Videos

Dual Bubbler GEET

http://www.youtube.com/user/downshydrocustoms

GEET CD Rom Videos

GEET 3.5 demo Dual Reactor Working.

http://www.youtube.com/watch?v=Lk3pM7Q18ss&feature=channel_page

GEET 10hp Engine

http://www.youtube.com/watch?v=G2w24JRQBlU&feature=channel_page

GEET coil

http://www.youtube.com/watch?v=eJ04mwp66Yg&feature=channel_page

GEET geely Scooter

http://www.youtube.com/watch?v=gUGZhu3beng&feature=channel_page

GEET Generator

http://www.youtube.com/watch?v=oR_Sa0bNEvo&feature=channel_page

GEET Generator2 Demo

http://www.youtube.com/watch?v=oVnvIBbc2ok&feature=channel_page

GEET Modified Fuel Injector

http://www.youtube.com/watch?v=nbJDPLgDEoU&feature=channel_page

GEET 1587CC V-Twin Motorcycle

http://www.youtube.com/watch?v=YA33T0qRlH8&feature=channel_page

GEET News Clip Mountin Dew

http://www.youtube.com/watch?v=SQ_HOPawSqM&feature=channel_page

GEET 24 HP Power Washer

http://www.youtube.com/watch?v=MbPW9YVp6wI&feature=channel_page

GEET Wooden 4 Barrel Carb Demo

http://www.youtube.com/watch?v=zGnFKqCiV8s&feature=channel_page

GEET LAWNMOWER WATER POWER HYBRID PART 1

GEET LAWNMOWER WATER POWER and E86 HYBRID PART 2

Paul Pantone interviewed about GEET

Paul Pantone Introduction

MIT Plasmatron - Principles of the Pantone GEET Device

Paul Pantone Plasma Reactor Motor

Panacea-BOCAF GEET Replication

Geet car pantone setup

http://youtube.com/watch?v=uSiTU5Y1b7U

http://youtube.com/watch?v=oq6_8Ll2sco

http://youtube.com/watch?v=VlH_mfS-jrc

French videos

http://nl.youtube.com/watch?v=sq_e0Wqwxn0

http://nl.youtube.com/watch?v=YEfWTZfzDUE

Credits